LOG GLSM AND RELATED TOPICS

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ABSTRACT. These are live-TeXed (for all talks besides my own and those of Shuai Guo, some figures were inserted later using Codex or Claude Code) notes from the log GLSM summer school at Sichuan University in July 2026. Note that all talks (besides mine) were given in Chinese. I apologize for any translation errors as well as for any errors in my lectures. In addition, the notes for the two lectures by Shuai Guo are **experimental** AI-generated transcriptions of his slides. While I did give Claude access to all of my other notes to see what my writing style is, please let me know if there is a mismatch in writing style or if there are any errors.

CONTENTS

1. THETA-STRATIFICATIONS (YAOXIONG WEN AND BIN WANG)

1.1. **Our first moduli problems.** Our first moduli space will be $\mathbb{P}^{N-1}$, which parameterizes 1-dimensional linear subspaces in $\mathbb{C}^n$, or equivalently is $(\mathbb{C}^N \setminus 0)/\mathbb{C}^\times$, which here means the set of closed orbits of $\mathbb{C}^\times$. This construction is our first example of geometric invariant theory. In general, a moduli problem will assign to every scheme T a collection of families over T, and we can think of it as a functor

$$F\colon (\mathsf{Sch}_{/S})^{\mathrm{op}} \to \mathsf{Set}.$$

Example 1.1. Let C be a smooth projective curve. Then consider the assignment

$$F_{r,d}\colon T \mapsto \{E \text{ vector bundle on } C \times T \mid \operatorname{rk} E = r, \deg E = d\}/\sim,$$

which parameterizes vector bundles on C (but not quite correct).

Example 1.2. By the Yoneda Lemma, every scheme Y defines a moduli problem.

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Unfortunately, not every moduli problem is representable by a scheme or algebraic space, for example if it has automorphisms. For example, if we consider the moduli of line bundles on $\mathbb{P}^1$, each object has an entire $\mathbb{C}^\times$ of automorphisms. Therefore, when we consider moduli problems, we will also need to consider isomorphisms between the objects (and therefore the target of the functor should be the ategory of groupoids).

Definition 1.3 (informal)**.** A *stack* $\mathfrak{X}$ is a functor

$$(\mathsf{Sch}_{/S})^{\mathrm{op}} \to \mathsf{Grpd}.$$

Example 1.4. Consider the stack $\mathsf{Bun}_{r,d}(C)$ which sends a scheme T to the groupoid of vector bundles of rank r and degree d on $T \times C$.

Unfortunately, there are far too many stacks to consider, so we will need to consider a subclass of stacks which is amenable to the methods of algebraic geometry.

Definition 1.5. An *algebraic stack* is a stack with representable (by algebraic spaces) diagonal and with a smooth surjective morphism from a scheme U (which is called an atlas).

Note that the first condition means that for any morphism $T \to \mathfrak{X} \times \mathfrak{X}$ from an algebraic space, then $T \times_{\mathfrak{X}\times\mathfrak{X}} \mathfrak{X}$ is an algebraic space. This is equivalent to all Isom sheaves being algebraic spaces and implies that any morphism from a scheme to $\mathfrak{X}$ is representable.

Remark 1.6. In the classical setting, if we have morphisms $f\colon A \to C$ and $g\colon B \to C$, then the fiber product is

$$A \times_C B = \{(a, b) \mid f(a) = g(b)\}.$$

However, in the setting of stacks, we need to replace equality by isomorphism, but this implies that we need to actually record the isomorphism.

1.2. **Quotient stacks.** Let G be an algebraic group an X be a scheme with an action of G.

Definition 1.7. The *quotient stack* $[X/G]$ sends a stack T to the groupoid which consists of diagrams

$$\begin{tikzcd} P \arrow[r, "\varphi"] \arrow[d] & X \\ T & \end{tikzcd}$$

where P is a principal G-bundle over T and φ is an equivariant morphism.

Example 1.8. Any smooth separated DM stack with either generically trivial stabilizer or quasi-projective coarse moduli space is a global quotient. Here, a DM stack is an algebraic stack with unramified diagonal.

Exercise 1.9. Show that the groupoid $[X/G](\mathbb{C})$ is equivalent to the action groupoid of $G(\mathbb{C})$ acting on $X(\mathbb{C})$.

The above exercise implies that as a groupoid (most importantly, not as an algebraic stack), we have

$$[X/G](\mathbb{C}) \simeq \coprod_{[x_i] \in X(\mathbb{C})/G(\mathbb{C})} BG_{x_i}.$$

1.3. **Fine vs coarse vs good moduli spaces.**

Definition 1.10. A *fine moduli space* for a moduli problem is a scheme or algebraic space M which represents the functor.

Unfortunately, it is very hard to find a fine moduli space, so we want to weaken this condition.

Definition 1.11. A *coarse moduli space* M for an algebraic stack $\mathfrak{X}$ is a morphism $\pi\colon \mathfrak{X} \to M$ such that

(1) π is universal for morphisms from $\mathfrak{X}$ to algebraic spaces;
(2) π is bijective on geometric points.

Even these do not always exist, so we consider an even weaker notion.

Definition 1.12. A *good moduli space* $q\colon \mathfrak{X} \to X$ is a *good moduli space* if

(1) The pushforward functor $q_*\colon \mathsf{QCoh}(\mathfrak{X}) \to \mathsf{QCoh}(X)$ is exact;
(2) The natural map $\mathcal{O}_X \to q_*\mathcal{O}_{\mathfrak{X}}$ is an isomorphism.

TABLE 1. Types of moduli spaces

Can handle	finite stabilizer	infinite stabilizer	feature
Fine moduli	No	No	Admits universal family
Coarse moduli	Yes	No	Bijection on points
Good moduli	Yes	Yes	S-equivalence

Theorem 1.13 (Keel-Mori)**.** *Let $\mathfrak{X}$ be an algebraic stack which has finite inertia. Then it has a coarse moduli space.*

Theorem 1.14 (Alper)**.** ......[1]

Unfortunately the stack $\mathsf{Bun}_{r,d}(C)$ is "too big to be good," namely that it is not quasi-compact and non-separated. For example, we can consider a vector bundle $E_r = L_n \oplus M_{d-n} \oplus \mathcal{O}^{\bigoplus r-2}$, then send $n \to \infty$, so this clearly cannot satisfy any finiteness condition. To see why this stack is non-separated, note that if E is a stable bundle, it has automorphisms given by $\mathbb{C}^\times$, which is not proper. To fix this problem, we will study an open substack of semistable objects and then consider S-equivalence classes.

Definition 1.15 (Slope stability)**.** Define the quantity

$$\mu(E) = \frac{\deg E}{\operatorname{rk} E}.$$

E is *(semi)stable* if for any proper subbundle $F \subset E$, we have $\mu(F) \leqslant \mu(E)$ or $\mu(F) < \mu(E)$, respectively.

Armed this definition, we may now consider the substack $\mathsf{Bun}^{ss}_{r,d}(C)$, which is quasi-compact but still not separated (because of the problem of automorphisms).

[1] Yaoxiong literally wrote this on the board.

Definition 1.16 (S-equivalence)**.** Two vector bundles E and E' are *S-equivalent* if they have the same associated graded pieces of their Jordan-Holder filtration. Here, the Jordan-Holder filtration is a filtration

$$0 = J_0 \subset J_1 \subset \cdots \subset J_N = E$$

such that all $J_i/J_{i=1}$ are stable. While the filtration is not necessarily unique, the associated graded is unique.

Now we consider the moduli space $M^{ss}_{r,d}(C)$ be the moduli space of S-equivalence classes of semistable vector bundles on C. The morphism $\mathrm{Bun}^{ss}_{r,d}(C) \to M^{ss}_{r,d}(C)$ is a good moduli space.

Remark 1.17. The original method of proof for this theorem was to construct the moduli space using geometric invariant theory. The goal of this series of lectures will be to discuss how to prove this type of theorem without making reference to GIT.

1.4. **Theta-stratifications.** Let $\mathfrak{X}$ be an algebraic stack and consider $\Theta \coloneqq [\mathbb{A}^1/\mathbb{G}_m]$. Then define the stack $\mathrm{Filt}(\mathfrak{X}) \coloneqq \mathrm{Map}(\Theta, \mathfrak{X})$ of filtered objects and the stack $\mathrm{Grad}(\mathfrak{X}) \coloneqq \mathrm{Map}(B\mathbb{G}_m, \mathfrak{X})$ of graded objects.

Definition 1.18. A Θ-*stratification* of $\mathfrak{X}$ consists of

(1) A totally ordered set I;
(2) For all $\alpha \in I$ an open substack $\mathfrak{X}_{\leqslant \alpha} \subset \mathfrak{X}$;
(3) For all $\alpha \in I$, a Θ-stratum $S_\alpha \subset \mathrm{Filt}(\mathfrak{X}_{\leqslant \alpha})$ such that

$$X_{<\alpha} = \bigcup_{\beta < \alpha} \mathfrak{X}_{\leqslant \beta} = X_{\leqslant \alpha} \setminus \mathrm{ev}_1(S_\alpha).$$

This implies $\mathfrak{X} = \mathfrak{X}^{ss} \sqcup \bigsqcup_\alpha \mathrm{ev}_1(S_\alpha)$.

Here a Θ-stratum is a union of connected components of $\mathrm{Filt}(\mathfrak{X}_{\leqslant \alpha})$ such that $\mathrm{ev}_1(S_\alpha)$ is a closed substack.

Consider the stack $\mathrm{Bun}_G(C) = \mathrm{Map}(C, BG)$. We will consider the stack

$$\mathrm{Map}(\Theta, \mathrm{Bun}_G(C)) = \mathrm{Map}(\Theta \times C, BG).$$

Any such map f is equivalent to a $\mathbb{G}_m$-equivariant principal G-bundle $\mathcal{E}$ over $\mathbb{A}^1 \times \mathbb{C}$, which when $G = \mathrm{GL}_n$ is equivalent to a graded $\mathcal{O}_C$-module. Here, we will take t to have weight 1. By flatness of $\mathcal{E}$, the map $t\colon \mathcal{E}_w \to \mathcal{E}_{w+1}$ is injective.

Remark 1.19. Consider the vector bundle $E = \mathfrak{E}_{|1 \times C}$. Then $\mathcal{E}_{|(\mathbb{G}_m \times C)} = P_C^* E$, where $P_C\colon C \times \mathbb{G}_m \to C$ is the projection to the first factor and thus $E = \mathcal{E}[t^{-1}]_0$. Defining $F^w E = \mathcal{E}_w \cdot t^{-w} \subset E$, we see that $F^w E \subset F^{w+1} E$. This gives us a filtration

$$0 \subset \cdots \subset F^w E \subset F^{w+1} E \subset \cdots \subset E.$$

In the other direction, we will require the Rees construction which sends a vector bundle E with filtration $F^\bullet$ a module

$$\mathrm{Rees}(E, F^\bullet) \sum_w t^{-w} F^w E \subset E \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}].$$

Therefore, a map from Θ to the stack of vector bundles is simply a filtration on a vector bundle.

1.5. **Non-abelian localization.** We will begin with the classical setting. Let X be a smooth projective variety with an action on $\mathbb{G}_m$. Then let $X^{\mathbb{G}_m} = \sqcup_\alpha Z_\alpha$. We can then compute equivariant Euler characteristics as

$$\chi_{\mathbb{G}_m}(X, F) = \sum_\alpha \chi_{\mathbb{G}_m}\left(Z_\alpha, \frac{F_{|Z_\alpha}}{e(N_{Z_\alpha/X})}\right),$$

where

$$e(N_{Z_\alpha/X}) = \lambda_{-1}(N^\vee_{Z_\alpha/Z}) = \sum (-1)^i [\Lambda^i N^\vee_{Z_\alpha/X}].$$

We will now consider the stacky version. Let $\mathfrak{X} = \mathfrak{X}^{ss} \sqcup \bigsqcup S_\alpha$ be a Θ-stratification and $Z_\alpha = \sigma^{-1}(S_\alpha)$, where $\sigma \colon \mathsf{Grad}(\mathfrak{X} \to \mathsf{Filt}(\mathfrak{X}))$ is the projection $\Theta \to B\mathbb{G}_m$.

Theorem 1.20. *Let $\mathfrak{X}$ be a quasi-smooth derived stack and $F \in \mathsf{Perf}(\mathfrak{X})$ satisfy various conditions. Then*

$$\chi(\mathfrak{X}, F) = \chi(\mathfrak{X}^{ss}, F) + \sum_\alpha \chi(Z_\alpha, F_{|Z_\alpha} \otimes \mathbb{E}_\alpha).$$

Here, if we split $\mathbb{L}_{\mathfrak{X},|Z_\alpha} = \mathbb{L}^+_\alpha \oplus \mathbb{L}^-_\alpha$, we have

$$E_\alpha = \mathrm{Sym}(\mathbb{L}^-_\alpha \oplus (\mathbb{L}^+_\alpha)^\vee) \otimes \det(L^+_\alpha)^\vee[-\mathrm{rk}(\mathbb{L}^+_\alpha)].$$

1.6. **Geometric invariant theory.** Let G be a reductive group and X be a projective scheme equipped with a G-equivariant line bundle $\mathcal{L}$. The semistable locus can be defined using the Hilbert-Mumford criterion, where it consists of all points where for all cocharacters $\lambda \colon \mathbb{G}_m \to G$, we can define the point

$$x_0 = \lim_{t \to 0} \lambda(t) x$$

and define the quantity

$$\mu^L(x, \lambda) \coloneqq \frac{-\mathrm{wt}_{x_0}(\mathcal{L})}{\|\lambda\|}.$$

A point is semistable if $\mu^L(x,\lambda) \leqslant 0$ for all cocharacters of G. The stable locus consists of all points where the inequality is strict and the stabilizer is finite.

Theorem 1.21. *The stable and semistable loci are open in X. Moreover, X^{ss}/G is a good quotient and X^s/G is a geometric quotient.*

Example 1.22. Consider the moduli stack $\mathsf{Bun}_{r,d}$ of vector bundles of rank r and degree d on a curve of genus at least @. Then recall that the semistable locus was defined to consist of slope semistable vector bundles. In fact, this can be obtained via GIT.

Fix an ample line bundle $\mathcal{O}(1)$ on C. Then there exists some n_0 such that for all $n \geqslant n_0$ and $E \in \mathsf{Bun}_{r,d}$, $E(n)$ is globally generated. Therefore, we can consider

$$H^0(C, E(n)) \twoheadrightarrow E(n).$$

If we set $V = \mathbb{C}^N$, where $N = h^0(E(n))$, then there is a surjection

$$V \otimes_{\mathcal{O}_C(-n)} \twoheadrightarrow E.$$

Therefore, we can think of E as a member of a Quot scheme. On the Quot scheme, we can consider the universal family

$$\begin{tikzcd} V \otimes \mathcal{O}_C(-n) \arrow[rr, two heads] \arrow[d] & & \mathcal{E} \\ \mathsf{Quot}_{V\otimes\mathcal{O}_C(-n),p_E} \times C \arrow[r, "p"] & \mathsf{Quot}_{V\otimes\mathcal{O}_C(-n)}. \end{tikzcd}$$

Now define the line bundle

$$\mathcal{L} \coloneqq \det R^0 p_*(\mathcal{E}(m))$$

for some $m \gg 0$. We can also consider the action of $\mathrm{SL}(V)$ on the Quot scheme. For some cocharacter λ, we can split

$$V = V_1 \oplus V_2$$

with weights s and $N - s$. Then consider

$$V_1 \otimes \mathcal{O}_C(-n) \subset V \otimes \mathcal{O}_C(-n) \twoheadrightarrow E$$

and consider the filtration

$$0 \to F^1 E \to E \to E/F^1 E \to 0$$

induced by this. Then we see that

$$\lim_{t\to 0} \lambda(t) E = F^1 E \oplus E/F^1 E.$$

Then we compute the restriction $\mathcal{L}_{|\operatorname{gr}_F E}$ and obtain the spaces $H^0(C, F^1 E(m))$ and $H^0(C, E(m)/F^1 E(m))$, and therefore the weight of $\mathcal{L}$ is given by

$$(N - s)p_{F^1 E}(m) - s p_{E/F^1 E}(m),$$

which recovers GIT.

We will now consider the unstable locus. Let E be a vector bundle on C. Then there exists a unique filtration

$$0 \subset F^\ell E \subseteq \cdots \subseteq F^0 E = E$$

such that

(1) All $\operatorname{gr}_F E$ are semistable;
(2) We have $\mu(\operatorname{gr}_F^i E) > \mu(\operatorname{gr}_F^{i-1} E)$ for all i.

Remark 1.23. From the GIT perspective, we should really take μ to be the reduced Hilbert polynomial.

1.7. **Construction of Theta-stratifications.** Let $\mathfrak{X}$ be a reasonable algebraic stack.[2] Now consider some family $\mathcal{E}$ over $\mathbb{A}^1 \times C$. If we restrict to the central fiber, we see that $\mathbb{G}_m$ acts on $\operatorname{gr}_F E$, which means we have a map $\mathbb{G}_m \to \operatorname{Aut} \operatorname{gr}_F E$, and from this we can compute the weight from this.

Note that the morphism $\sigma\colon \mathsf{Grad}(\mathfrak{X}) \to \mathsf{Filt}(\mathfrak{X})$ is a Θ-contraction, and therefore

$$\pi_0(\mathsf{Filt}(\mathfrak{X})) = \pi_0(\mathsf{Grad}(\mathfrak{X})).$$

[2]Here, reasonable means something like locally of finite presentation with qcqs diagonal.

Example 1.24. Let $\mathfrak{X} = BG$. Then we have

$$\mathsf{Filt}(\mathfrak{X}) = \bigsqcup_{[\lambda]} BP_\lambda,$$

where P_λ is the parabolic subgroup corresponding to the cocharacter λ, and we can always choose representatives in a single Weyl chamber. Moreover, we have

$$\mathsf{Grad}(\mathfrak{X}) = \bigsqcup_{\lambda \in \mathbb{X}_*(T)/W} BL_\lambda,$$

where L_λ is the Levi subgroup corresponding to λ.

Definition 1.25. A Θ-*stratum* in $\mathfrak{X}$ is a union of connected components $S \subseteq \mathsf{Filt}(\mathfrak{X})$ such that

$$\mathrm{ev}_1 : S \to \mathfrak{X}$$

is a closed immersion.

Roughly speaking, closed immersions are injective, so an object in S consists of an object in $\mathfrak{X}$ and a "canonical filtration."

Definition 1.26. A Θ-*stratification* is an increasing family of open substacks

$$\mathfrak{X}_{\leqslant c} \subseteq \mathfrak{X}, \qquad c \in \Gamma,$$

where Γ is some totally ordered set, together with Θ-strata

$$S_c \subseteq \mathsf{Filt}(\mathfrak{X}_{\leqslant C}),$$

which are required to satisfy

$$\mathfrak{X}_{\leqslant c} = \mathrm{ev}_1(S_c) \sqcup \bigcup_{c' < c} \mathfrak{X}_{\leqslant c'}.$$

In particular, this implies that

$$\mathfrak{X} = \mathfrak{X}^{ss} \sqcup \bigsqcup_{c>0} \mathrm{ev}_1(S_c).$$

Analogously to the Hilbert-Mumford criterion, we will construct Θ-stratifications using numerical invariants.

Definition 1.27. A *numerical invariant* μ with values in a totally ordered set Γ consists of an assignment of a scaling invariant function

$$\mu\colon \mathbb{R}^q \setminus \{0\} \to \Gamma$$

for any $p \in \mathfrak{X}(\mathbb{C})$ and any nondegenerate homomorphism $\gamma\colon \mathbb{G}_m^q \to \mathrm{Aut}(p)$. These are required to satisfy

(1) The assignment is invariant under field extensions. This point will not be elaborated on because we are working over $\mathbb{C}$;
(2) If we have $\mathbb{G}_m^q \hookrightarrow \mathbb{G}_m^n \hookrightarrow \mathrm{Aut}(p)$, then we have a factorization $\mathbb{R}^q \setminus \{0\} \to \mathbb{R}^n \setminus \{0\} \to \Gamma$;
(3) If we have some $\xi\colon S \to \mathfrak{X}$ and $\gamma\colon \mathbb{G}_{m,S}^q \to \mathrm{Aut}(\xi)$ with finite kernel, the function $\mu_{\kappa(s)}$ is locally constant on S, where s ranges over points of S.

We will give a naive version of a construction starting with a numerical invariant μ. Define $x \in \mathfrak{X}$ to be semistable if $\mu(f) \leqslant 0$ for any nondegenerate $f \in \mathsf{Filt}(\mathfrak{X})$ satisfying $f(1) = x$. If x is unstable, then we define its instability level

$$M^{\mu}(x) := \sup\{\mu(f) \mid f \in \mathsf{Filt}(\mathfrak{X}), \mathrm{ev}_1(f) = x.\}$$

If a filtration f attains this supremum, we call it a "Harder-Narasimhan filtration."

Definition 1.28. A numerical invariant μ is *standard* if

(1) $\mu(x)$ and $\mu(-x)$ are not both possitive for $x \in \mathbb{R}^q \setminus \{0\}$;
(2) Each function μ_γ is strictly quasi-concave, namely for all $\delta \in [0, 1]$ and linearly independent $x_0, x_1 \in \mathbb{R}^q$ such that $\mu(x_0), \mu(x_1) \geqslant 0$, then

$$\mu_\gamma(\delta x_0 + (1-\delta)x_1) \geqslant \max\{\mu_\gamma(x_0), \mu_\gamma(x_1)\}.$$

If either value is strictly positive, the inequality is required to be strict.

The next condition is that we need some kind of rationality. We can see $\mathbb{X}_*(T)_{\mathbb{R}}$, and our goal is to find the rational points. This leads us to condition (R): If for any γ for which μ_γ attains a positive value, then it attains a maximal value at a rational point.

Theorem 1.29. *Let $\mathfrak{X}$ be a stack and μ be a standard numerical invariant satisfying condition (R). Then μ defines a Θ-stratification if and only if it satisfies the following:*

- *For any DVR R with fraction field K and residue field k, then $\mu(f_K^{\mathrm{HN}}) \leqslant M^{\mu}(f_k)$ (HN-specialization);*
- *For any map $B \colon T \to \mathfrak{X}$ from a finite type scheme, there exists a quasi-compact open $\mathfrak{X}' \subseteq \mathfrak{X}$ such that for any point of finite type $p \in T(k)$ and a filtration f of $B(p)$ for which $\mu(f) > 0$, there exists another filtration f' of $B(p)$ with $\mu(f') \geqslant \mu(f)$ satisfying $f'(0) \in \mathfrak{X}'$ (HN-boundedness).*

The point here is that the conditions of being standard and (R) combined with HN-boundedness imply existence and HN-specialization implies uniqueness.

2. Intro to log geometry (Tingting Fang and Zijian Han)

2.1. **Motivation.** In enumerative geometry, we will consider intersection numbers on compactified moduli spaces. This is going to require our objects to degenerate in some way, for example for domain curves to be nodal, maps to land in boundary divisors, or we degenerate the target. In ordinary geometry, we view these as singular, but in log geometry, there is a way to think of these situations as being *log smooth*. In log geometry, we are going to replace functions with the data of functions and how they vanish on the boundary (or namely the morphism $\alpha \colon M_X \to \mathcal{O}_X$).

Example 2.1. Consider a family of curves $(xy = t)$ over $\mathbb{A}^1$, where the central fiber is nodal. In log geometry, if we place the divisorial log struction on $\mathbb{A}^1$, then this family is log smooth.

2.2. **Monoids.**

Definition 2.2. A *monoid* is a commutative semigroup with a unit.

For a monoid P, we may associate a group P^{gp} via the standard Grothendieck construction

$$P^{\mathrm{gp}} = \{(a, b) \mid (a, b) \sim (c, d) \text{ if there exists } s \in P \text{ such that } a + d + s = b + c + s.\}.$$

Definition 2.3. If the natural morphism $P \hookrightarrow P^{gp}$ is injective, we say that P is *integral*.

Example 2.4. If P is has generators a, b, c with relation $a + c = b + c$, then in P^{gp} we have $a = b$, so P is not integral.

Definition 2.5. A monoid P is

- *fine* if P is finitely generated and integral;
- *saturated* if for all $p \in P^{gp}$ and all natural numbers n, if $np \in P$ then $p \in P$;
- *torsion-free* if P^{gp} is;
- *sharp* if the identity is the only invertible element.

2.3. **Log structures.** Let X be a scheme.

Definition 2.6. A *pre-log structure* on X is a sheaf of monoids M_X on the small étale site of X and a morphism $\alpha \colon M_X \to \mathcal{O}_X$. It is a *log structure* if $\alpha^{-1}(\mathcal{O}_X^\times) \simeq \mathcal{O}_X^\times$, where the isomorphism is given by α.

We now consider the exact sequence

$$0 \to \alpha^{-1}(\mathcal{O}_X^\times) \to \bar{M}_X \to 0,$$

and the quotient here is called the *characteristic sheaf* or the *ghost sheaf*.

Example 2.7. Let $M_X = \mathcal{O}_X^\times$ and α be the identity. Then there is no additional information provided by the log structure.

Example 2.8. Let X be a regular variety and D be a closed subscheme. Set $U = X \setminus D$ Then define

$$M_X = \mathcal{O}_X \cap j_* \mathcal{O}_U^\times,$$

where $j \colon U \to X$ is the open immersion. If D is a divisor, then this is called the *divisorial log structure*.

For example, if $X = \mathbb{A}^1$ and $D = \{0\}$, then the characteristic sheaf vanishes everywhere except for the origin. Here, we compute

$$M_X = \left\{ \varphi \cdot t^n \mid \varphi \in \mathcal{O}_X^\times \right\}.$$

The stalk is $\mathcal{O}_X^\times$ everywhere except at the origin, where it is $\mathcal{O}_{\mathbb{A}^1,0}^\times \oplus \mathbb{N}$.

Example 2.9. Now let $X = \mathbb{A}^2$ and D be the union of the coordinate axes. Then

$$\begin{aligned} M_{X,D} &= \{ f \in k[x,y] \mid f \text{ is invertible outside } D \} \\ &= \left\{ \varphi \cdot x^a \cdot y^b \right\}. \end{aligned}$$

Note here that at the general point, we see $M_p = \mathcal{O}_{X,p}^\times$. At the origin, we see $\mathcal{O}_{X,p}^\times \oplus \mathbb{N}^2$. At a general point on the x-axis, we see that x^a is invertible, so we see the stalk $\mathcal{O}_{X,p} \oplus \mathbb{N} \cdot b$. Similarly, we can compute the stalk at a general point on the y-axis.

Just as we can associate a sheaf to a presheaf, we will associate a log structure to a pre-log structure. First, a pushout in a category $\mathcal{C}$ will be denoted by $B \sqcup_A C$ and is the colimit of the diagram

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "g"'] & B \\ C. \end{tikzcd}$$

In the case where $\mathcal{C}$ is commutative monoids, then
$$B \sqcup_A C = (B \oplus C)/\sim, \qquad (f(a)+b, c) \sim (b, g(a)+c).$$

Example 2.10. Consider the morphisms $2\colon \mathbb{N} \to \mathbb{N}$ and $3\colon \mathbb{N} \to \mathbb{N}$. Then the pushout is the monoid
$$P = \frac{\mathbb{N} \oplus \mathbb{N}}{(3,0) \sim (0,2)} \simeq \mathbb{N} \setminus \{1\}.$$

If we replace $\mathcal{C}$ by the category of sheaves of commutative monoids, we can form the pushout on sections and then sheafify.

Definition 2.11. The *associated log structure* of a pre-log structure $\alpha\colon M_X \to \mathcal{O}_X$ is the pushout
$$M_X \sqcup_{\alpha^{-1}(\mathcal{O}_X^\times)} \mathcal{O}_X^\times.$$
The map to $\mathcal{O}_X$ is given by $(a, b) \to \alpha(a)b$.

Definition 2.12. A morphism of log schemes $(X, M_X) \to (Y, M_Y)$ consists of

(1) A morphism $\underline{f}\colon \underline{X} \to \underline{Y}$ of underlying schemes;
(2) A map $f^\flat\colon f^*M_Y \to M_X$, where f^*M_Y is the log structure associated to $f^{-1}M_Y$.

Definition 2.13. A *chart* for a log scheme $(X, \alpha\colon M_X \to \mathcal{O}_X)$ is a morphism from the constant monoid $P_X \to M_X$ such that $M_X \simeq P_X^a \simeq P_X \oplus_{\alpha_P^{-1}(\mathcal{O}_X^\times)} \mathcal{O}_X^\times$, where $\alpha_P\colon P_X \to M_X \to \mathcal{O}_X$.

Charts are useful both in specifying log structures. For example, if we have $P_X \to \mathcal{O}_X$, then P_X^a gives a log structure on X.

Example 2.14. The trivial log structure is given by the chart $P = \{0\}$.

Example 2.15. Let $X = \operatorname{Spec} k$ and give a log structure by $\alpha_{\mathbb{N}}\colon \mathbb{N} \to k$ which sends 0 to 1 and 1 to 0. This is called the *standard log point* and can also be formed by restricting the divisorial log structure on $\mathbb{A}^1$ to the origin.

Example 2.16. Consider the chart $\mathbb{N} \to \mathcal{O}_{\mathbb{A}^1}$ which sends $n \mapsto t^n$. Then this recovers the divisorial log structure on $\mathbb{A}^1$ relative to the origin.

Example 2.17. Consider the chart $\mathbb{N}^2 \to \mathcal{O}_{\mathbb{A}^2}$ which sends $(a, b) \mapsto x^a y^b$. This recovers the divisorial log strucure on $\mathbb{A}^2$ relative to the toric boundary.

Definition 2.18. A *chart* for a log map $f\colon (X, M_X) \to (Y, M_Y)$ is a triple $(P_X \to M_X, Q_Y \to M_Y, Q \to P)$, where $P_X \to M_X$ and $Q_Y \to M_Y$ are charts and the diagram
$$\begin{tikzcd} Q_X \arrow[r] \arrow[d] & P_X \arrow[d] \\ f^*M_Y \arrow[r] & M_X \end{tikzcd}$$
commutes.

Definition 2.19. A log scheme (X, M_X) is *finitely generated, fine, integral, saturated* if there exists an étale local chart $P_U \to M_U$ for a cover $U \to X$ such that P is finitely generated, fine, integral, or saturated, respectively.

2.4. **Log smoothness.**

Definition 2.20. Consider a commutative diagram

$$\begin{tikzcd} T_0 \arrow[r, "\phi"] \arrow[d, "j"] & X \arrow[d, "f"] \\ T_1 \arrow[r, "\psi"] & Y, \end{tikzcd}$$

where j is a strict closed immersion (here, this means that $M_{T_0} = j^*M_{T_1}$). A morphism $f\colon X \to Y$ of fine log schemes is *log smooth (resp. log étale)* if

(1) The underlying map $\underline{f}\colon \underline{X} \to \underline{Y}$ is locally of finite presentation;
(2) For any commutative diagram as above, étale locally on T_1 there exists a (resp. there exists a unique) morphism $g\colon T_1 \to X$ such that $\phi = g \circ j$ and $\psi = f \circ g$.

We will now give a chart local criterion for log smoothness.

Theorem 2.21. *Let $f\colon X \to Y$ be a morphism of fine log schemes. Assume we have a chart $Q \to M_Y$ where Q is a fine monoid. Then f is log smooth if and only if étale locally on X there exists a chart for f extending $Q \to M_Y$ satisfying*

(1) The kernel and torsion part of the cokernel of $Q^{gp} \to P^{gp}$ is zero or finite order invertible on X;
(2) The induced morphism $\underline{X} \to \underline{Y} \times_{\operatorname{Spec}\mathbb{Z}[Q]} \operatorname{Spec}\mathbb{Z}[P]$ is étale in the classical sense.

Example 2.22. Let X be an affine toric variety given by $\operatorname{Spec}\mathbb{C}[P]$ for a monoid P. Note that $P^a \simeq M_{X,X\setminus T}$ induces the divisorial log structure. Then consider the map $f\colon X \to \mathrm{pt}$, where the point has a trivial log structure. We note that the morphism $0 \to P^{gp}$ has no kernel and the cokernel is torsion-free, and the map

$$\underline{X} \to \operatorname{Spec}\mathbb{C} \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec}\mathbb{Z}[P] = \operatorname{Spec}\mathbb{C}[P]$$

is an isomorphism, so any toric variety is log smooth over a point with trivial log structure.

From this example, we can morally think of log smoothness as being locally toric relative to the base.

Example 2.23. Recall the family $xy = t$ with log structure relative to the central fiber over $\mathbb{A}^1$ with the divisorial log structure relative to 0. The map of charts we will consider is in fact $\Delta\colon \mathbb{N} \to \mathbb{N}^2$. Taking the groupification, the map becomes $\Delta\colon \mathbb{Z} \to \mathbb{Z}^2$. The kernel is trivial and the cokernel is torsion-free, so we have verified the first condition. Then we consider the map

$$\underline{X} \to \operatorname{Spec}\mathbb{C}[t] \times_{\operatorname{Spec}\mathbb{Z}[\mathbb{N}]} \operatorname{Spec}\mathbb{Z}[\mathbb{N}^2] = \operatorname{Spec}\frac{\mathbb{C}[u,v,t]}{uv - t},$$

which is an isomorphism, so this family is log smooth.

2.5. **Stack of log structures.**

Let S be a fine log scheme. We will consider the category Log_S whose objects are fine log schemes $(T, M_T) \to S$ and whose morphisms are strict morphisms $(T, M_T) \to (T', M_{T'})$ over S.

(1) This category Log_S is a category fibered in groupoids over $\mathsf{Sch}_{/S}$ classifying all fine log schemes mapping to S;
(2) Log_S is an algebraic stack locally of finite type over S;

(3) A morphism $f\colon X \to S$ is log smooth if and only if $X \to \mathsf{Log}_S$ is smooth. The same holds with smooth replaced by étale.

2.6. **Tropicalization.** Let X be a fine log scheme. Then the tropicalization of X is defined to be the colimit

$$\Sigma(X) := \bigsqcup_{x \in X} \operatorname{Hom}(\bar{M}_{X,x}, \mathbb{R}_{\geqslant 0}) \Big/ \sim,$$

where the equivalence relation is given by if a point x specializes to y, then the induced map $\sigma_x \to \sigma_y$ induces an equivalence from σ_x to its image.

Now suppose there is a stratification of X by X_i such that $\bar{M}_{|X_i}$ is constant. For any X_i, all points $x \in \overline{\{\eta_{X_i}\}}$, so we only need to consider the generic points of the strata and then glue various faces.

Example 2.24. Consider X with the trivial log structure, Note that for any point $x \in X$, we have $\bar{M}_{X,x} = 0$, so $\sigma_X = \{0\}$. Then $\Sigma(X) = \{0\}$.

Example 2.25. Consider $\mathbb{A}^1$ relative to the origin (with divisorial log structure). It is easy to see that $\sigma_0 = \operatorname{Hom}(\mathbb{N}, \mathbb{R}_{\geqslant 0}) = \mathbb{R}_{\geqslant 0}$ and $\sigma_\eta = \{0\}$, so the tropicalization is $\mathbb{R}_{\geqslant 0}$.

Example 2.26. Consider $\mathbb{A}^2$ with the toric log structure (divisorial log structure relative to $xy = 0$). Then $\sigma_{(0,0)} = \mathbb{R}^2_{\geqslant 0}$, $\sigma_{(x,0)} = \mathbb{R}_{\geqslant 0}$, $\sigma_{(0,y)} = \mathbb{R}_{\geqslant 0}$, and $\sigma_{(x,y)} = \{0\}$, so we obtain the fan of $\mathbb{A}^2$ as a toric variety treated as a cone complex.

Example 2.27. Let X_Σ be a toric variety with the divisorial log structure relative to $X_\Sigma \setminus T$. Then

$$\Sigma(X_\Sigma) = \Sigma,$$

so a possible slogan is that tropicalizing a toric variety recovers the structure of ordinary toric geometry (minus the embedding of the fan in some real vector space).

Example 2.28. Let $X = \operatorname{Spec} k[x,y,t]/(xy - t^\ell)$ with divisorial log structure over $xy = 0$. Then there is a chart given by

$$Q = \langle e_x, e_y, e_t \mid e_x + e_y = \ell e_t \rangle.$$

Note that $M_X = Q^a = Q \oplus_{\alpha^{-1}(\mathcal{O}_X^\times)} \mathcal{O}_X^\times$ and $\bar{M}_X = Q$. Therefore, we have

$$\begin{aligned}
\sigma_q &= \operatorname{Hom}(Q, \mathbb{R}_{\geqslant 0}) \\
&= \left\{ (\rho_x, \rho_y, \rho_t) \in \mathbb{R}^3_{\geqslant 0} \mid \rho_x + \rho_y = \ell \rho_t \right\} \\
&= \left\{ (\rho_t, \rho_x) \in \mathbb{R}^2_{\geqslant 0} \mid 0 \leqslant \rho_x \leqslant \ell \rho_t \right\}
\end{aligned}$$

at the singular point q, which recovers the cone which gives X as a toric variety. At the generic point η_x of $x = 0$, we see that $\bar{M}_{X,\eta_x} = \mathbb{N} e_t$, and therefore $\sigma_{\eta_x} = \mathbb{R}_{\geqslant 0}$. Similarly, we see that $\sigma_{\eta_y} = \mathbb{R}_{\geqslant 0}$. The remaining problem is gluing.

Note that $\bar{M}_{X,q} \to \bar{M}_{X,\eta_x}$ is given by $e_t \mapsto e_t$, $e_x \mapsto \ell e_t$, and $e_y \mapsto 0$, and therefore the map $\sigma_{\eta_x} \to \sigma_q$ sends ρ_t to (ρ_t, ρ_x), where $\rho_x = \ell \rho_t$.

The morphism $X \to \mathbb{A}^1_t$ giving a family of curves with nodal central fiber has tropicalization given by projecting the cone onto $\mathbb{R}_{\geqslant 0}$.

2.7. **Log curves.** For now, all monoids and log schemes will be fine (or even fine saturated). There are two reasons for this:

(1) If P is a fine monoid, its dual is fs and sharp;
(2) If $X = (\underline{X}, M_X)$ is a fine log scheme, for any geometric point $\bar{x} \to X$, there exists an étale neighborhood of $\bar{x}$ which has a chart.

We will now discuss various geometric properties of maps between monoids.

Definition 2.29. A map $\theta\colon P \to Q$ is *integral* if the map $\operatorname{Spec}\mathbb{C}[Q] \to \operatorname{Spec}\mathbb{C}[P]$ is flat.

For example, any nonzero map $\mathbb{N} \to \mathbb{N}$ and the diagonal $\Delta\colon \mathbb{N} \to \mathbb{N}^2$ are integral. If a map is integral, then for any $P \to P'$, the monoid $P' \oplus_P Q$ is fine.

Example 2.30. Pushouts may not be fine, for example consider

$$\langle a, b, c \mid a + c = b + c\rangle = \mathbb{N}^3 \oplus_{\mathbb{N}^2} \mathbb{N}.$$

Definition 2.31. A map $f\colon X \to Y$ between fine log schemes is integral if at every geometric point of X, the map

$$(f^*\bar{M}_Y)_{\bar{x}} \to \bar{M}_{X,\bar{x}}$$

is integral.

Definition 2.32. A map between fs monoids $P \to Q$ is *saturated* if for all $P \to P'$, the monoid $P' \oplus_P Q$ is also fs.

Definition 2.33. The map $\Delta\colon \mathbb{N} \to \mathbb{N}^2$ is saturated, but $\ell\colon \mathrm{N} \to \mathbb{N}$ is not.

Definition 2.34. A *log curve* over an fs log scheme S is a log smooth and integral map $\pi\colon \mathcal{C} \to S$ between fs log schemes such that the underlying geometric fibers are reduced connected curves.

Proposition 2.35. *Let $f\colon X \to Y$ be a log smooth and integral map between fs log schemes. Then $\underline{f}$ is flat.*

Theorem 2.36. *The underlying family of a log curve is a family of nodal curves.*

Example 2.37. Let $P = \mathbb{N} \setminus \{1\}$. Then $\operatorname{Spec}(P \to k[P])$ is log smooth over k, but $\operatorname{Spec} k[P]$ is a cusp.

Definition 2.38. An n-*pointed log curve* over an fs base S consists of the data $(\pi\colon \mathcal{C} \to S), \vec{p} = \{p_i\}_{i=1}^n$ such that

(1) The underlying family is a prestable curve;
(2) π is a log smooth map between fs log schemes;
(3) On the smooth locus U of $\underline{\pi}$, we have

$$\bar{M}_{C,|U} \simeq \pi^*\bar{M}_S \oplus \bigoplus_{i=1}^n p_i\underline{\mathbb{N}}.$$

2.8. **Canonical log structures on prestable curves.** Let $\mathfrak{M}_{g,n}$ be the moduli of n-pointed prestable curves. Then the singular locus is a normal crossings divisor, so there is a canonical log structure

$$M^{\mathrm{can}}_{\mathfrak{M}_{g,n}} = M_{\partial\mathfrak{M}_{g,n}}.$$

If we allow the universal family $\mathfrak{C}_{g,n}$, we can consider $\pi^{-1}(\partial\mathfrak{M}_{g,n}) \subseteq \mathfrak{C}_{g,n}$. Therefore, we have a map

$$\pi^* M_{\partial\mathfrak{M}_{g,n}} \to M_{\pi^{-1}(\partial\mathfrak{M}_{g,n})}.$$

We can also consider the divisors $p_i(\mathfrak{M}_{g,n}) \subset \mathfrak{C}_{g,n}$, which also define log structures

$$P_i \coloneqq M_{p_i(\mathfrak{M}_{g,n})}.$$

Therefore, we can define a log structure

$$M^{\mathrm{can}}_{\mathfrak{C}_{g,n}} \coloneqq M_{\pi^{-1}(\partial\mathfrak{M}_{g,n})} \oplus_{\mathcal{O}^*} \bigoplus_{i=1}^{n} P_i.$$

Then if we consider

$$\pi^\flat \colon \underline{\pi}^{-1} M^{\mathrm{can}}_{\mathfrak{M}_{g,n}} \to M_{\pi^{-1}(\partial\mathfrak{M}_{g,n})} \to M^{\mathrm{can}}_{\mathfrak{C}_{g,n}},$$

we can lift π to a map of log stacks.

Theorem 2.39. *The map π is a log smooth integral map.*

Definition 2.40. For a family $(\underline{C}/\underline{S}, \vec{p})$ of prestable curves, we can pull back along the diagram

$$\begin{tikzcd} \underline{C} \arrow[r] \arrow[d] & \mathfrak{C}_{g,n} \arrow[d] \\ S \arrow[r] & \mathfrak{M}_{g,n} \end{tikzcd}$$

and endow the left edge with a canonical log structure.

Remark 2.41. Let $\underline{S} = \operatorname{Spec} k$. If $S^{\mathrm{can}} = \operatorname{Spec}(\mathbb{N}^r \to k)$, then r is the number of nodes on $\underline{C}$. Working étale locally on a node q, the log structure looks like

$$\begin{tikzcd} \mathbb{N}^{r-1} \oplus \mathbb{N}^2 \arrow[r] & k[x,y]/xy \\ \mathbb{N}^{r-1} \oplus \mathbb{N} \arrow[r] \arrow[u, "\mathrm{id} \oplus \Delta"] & k. \arrow[u] \end{tikzcd}$$

Theorem 2.42. *Let $(\mathcal{C}/S, \vec{p})$ be an n-pointed log curve on an fs base S. Then there exists a unique $M^{\mathrm{can}}_S \to M_S$ such that*

$$\begin{tikzcd} \mathcal{C} \arrow[r] \arrow[d] & \mathcal{C}^{\mathrm{can}} \arrow[d] \\ S \arrow[r, "a"] & S^{\mathrm{can}} \end{tikzcd}$$

is a Cartesian diagram in the category of fs log schemes.

A slogan is that a log curve is a prestable curve together with $M^{\mathrm{can}}_S \to M_S$.

Example 2.43. Let $\underline{S} = \operatorname{Spec} k[s]$ and $\underline{T} = \operatorname{Spec} k[t]$. Also choose $\ell \in \mathbb{N}$. Then consider the diagram

$$\begin{tikzcd} \operatorname{Spec} \frac{k[x,y,s]}{xy-s^\ell} \arrow[r] \arrow[d] & \operatorname{Spec} \frac{k[x,y,s]}{xy-s^\ell} \arrow[r] \arrow[d] & \operatorname{Spec} \frac{k[x,y,t]}{xy-t} \arrow[r] \arrow[d] & \mathfrak{C}_{g,n} \arrow[d] \\ \operatorname{Spec} k[s] \arrow[r, "\underline{\mathrm{id}}"] & \operatorname{Spec} k[s] \arrow[r] & \operatorname{Spec} k[t] \arrow[r] & \mathfrak{M}_{g,n}. \end{tikzcd}$$

On log structures, consider the case where the bottom left arrow corresponds to $1 \mapsto \ell$ on $\mathbb{N}$. Then at the top left corner, the monoid becomes $\mathbb{N}^2 \oplus_{\mathbb{N},\ell} \mathbb{N}$. If we call this P, then the map $k[P] \to k[\mathbb{N}]$ is not a family of log curves with the canonical log structure, but is a family of log curves.

2.9. **Moduli of log curves.**

Definition 2.44. A map between n-pointed log curves $(\mathcal{C}, S) \to (\mathcal{C}', S')$ is a pair $(\ell\colon \mathcal{C} \to \mathcal{C}', \Theta\colon S \to S')$ such that Θ is strict and the diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, "\ell"] \arrow[d] & \mathcal{C}' \arrow[d] \\ S \arrow[r, "\Theta"] & S' \end{tikzcd}$$

is Cartesian. Denote by $\mathfrak{M}^{\log}_{g,n}$ the fibered category parameterizing n-pointed log curves.

Proposition 2.45. *We have* $\mathfrak{M}^{\log}_{g,n} \simeq \mathsf{Log}_{(\mathfrak{M}_{g,n}, M^{\mathrm{can}}_{\mathfrak{M}_{g,n}})}$.

In particular, for a scheme $\underline{T}$, the groupoid $\mathfrak{M}^{\log}_{g,n}(\underline{T})$ parameterizes log structures M_T together with log curves $\mathcal{C}/T$.

Consider a log curve $\pi\colon \mathcal{C} \to S, \vec{p}$ over $S = \mathrm{Spec}(Q \to k)$. Denote $\bar{\pi}^\flat\colon \bar{\pi}^*\bar{M}_S \to \bar{M}_{\mathcal{C}}$. At a generic point or a general smooth point η, the map is

$$\bar{\pi}^\flat_\eta\colon (\pi^{-1}\bar{M}_S)_\eta = Q \simeq \bar{M}_{\mathcal{C},\eta}.$$

At a marked point $p_i \in \overline{\{\eta\}}$, we have $\bar{M}_{\mathcal{C},p_i} = Q \oplus \mathbb{N}$. In this case we have

$$\chi_\eta\colon \bar{M}_{\mathcal{C},p_i} \xrightarrow{\mathrm{pr}_1} \bar{M}_{\mathfrak{C},\eta}$$

the specialization map and the pullback

$$\bar{\pi}^\flat_{p_i}\colon Q \hookrightarrow Q \oplus \mathbb{N}.$$

At a node q, we have $\bar{M}_{\mathcal{C},q} = Q \oplus_{\mathbb{N}} \mathbb{N}^2$. Let ρ_q denote the image of 1 in Q, which is the edge length or smoothing parameter at q. Denote by $\bar{\eta}_1$ and $\bar{\eta}_2$ denote the generic points of the two branches of q. Then the generalization maps are

$$\begin{aligned} \chi_{\eta_1}\colon Q \oplus_{\mathbb{N}} \mathbb{N}^2 \to Q \qquad & (q,(a,b)) \mapsto q + a\rho_q; \\ \chi_{\eta_2}\colon Q \oplus_{\mathbb{N}} \mathbb{N}^2 \to Q \qquad & (q,(a,b)) \mapsto q + b\rho_q. \end{aligned}$$

2.10. **Tropicalization of log curves.** We will only consider the situation over a log point $S = \mathrm{Spec}(Q \to k)$. We will consider the category of cones Cone, whose objects are pairs (w, N_w) such that $w \subseteq N_w \otimes_{\mathbb{Z}} \mathbb{R}$, where w is a full-dimensional strongly convex rational polyhedral cone. The morphisms are maps $(w_1, N_{w_1}) \to (w_2, N_{w_2})$, which consist of

$$\chi\colon N_{w_1} \to N_{w_2}$$

such that $\chi_{\mathbb{R}}(w_1) \subseteq w_2$.

Example 2.46. Let X be an fs log scheme and consider $((\bar{M}^\vee_{X,x})_{\mathbb{R}}, (\bar{M}^\vee_{X,x})_{\mathbb{Z}})$.

We will now consider the category GCC of generalized cone complexes. Its objects are topological spaces $|\Sigma|$ together with a presentation as the colimit of a diagram in Cone with all arrows being inclusions of faces. Its morphisms are continuous map $|\Sigma| \to |\Sigma'|$ such that for each $(w, N_w) \in \Sigma$, the map $w \to |\Sigma'|$ factors through $(w, N_w) \to (w', N_{w'})$.

Example 2.47. If $f\colon X \to Y$ is a morphism of fs log schemes, the map $\bar{f}^\flat\colon f^*\bar{M}_Y \to \bar{M}_X$ induces $\Sigma(X) \to \Sigma(Y)$ on tropicalizations.

Definition 2.48. A *cone complex* is a generalized cone complex Σ such that for each $(\sigma, N_\sigma) \in \Sigma$, the map $\sigma \to |\Sigma|$ is injective.

Theorem 2.49. *If X is fs and log smooth over $\underline{\mathbb{C}}$ with Zariski log structure, then $\Sigma(X)$ is a cone complex.*

For a monoid Q, we will denote $w_Q = Q_{\mathbb{R}}^\vee = \mathrm{Hom}(Q, \mathbb{R}_{\geqslant 0})$ and the lattice $\mathrm{Hom}(Q, \mathbb{Z}) = Q_{\mathbb{Z}}^\vee$. At a smooth point, we have

$$w_\eta = (Q_{\mathbb{R}_{\geqslant 0}}^\vee, Q_{\mathbb{Z}}^\vee).$$

At a marked point, we see that

$$w_p = \left(\mathrm{Hom}(Q \oplus \mathbb{N}, \mathbb{R}_{\geqslant 0}) = Q_{\mathbb{R}}^\vee \oplus \mathbb{R}_{\geqslant 0}, Q_{\mathbb{Z}}^\vee \oplus \mathbb{Z}\right).$$

At a node, we have

$$w_q = \left(\mathrm{Hom}(Q \oplus_{\mathbb{N}} \mathbb{N}^2, \mathbb{R}_{\geqslant 0}), (\bar{M}_{\mathcal{C},q})_{\mathbb{Z}}^\vee\right).$$

Note that the cone is in fact equal to

$$\{(m, m_x, m_y) \mid m = m_x + m_y\} \simeq \{(m, e_x) \mid e_x \leqslant m(\rho_q)\} \subseteq Q_{\mathbb{R}}^\vee \times \mathbb{R}_{\geqslant 0}.$$

Now that we understand the cones associated to every point, we now need to discuss the face maps. The generalization maps corresponding to marked points are

$$Q \oplus \mathbb{N} \xrightarrow{\mathrm{pr}_1} Q,$$

and the associated face maps are

$$w_\eta \hookrightarrow w_p \qquad m \mapsto (m, 0).$$

At a node, the two face maps are

$$\chi_{\eta_1}^\vee\colon Q_{\mathbb{R}}^\vee \to w_q \qquad m \mapsto (m, m(\rho_q))$$

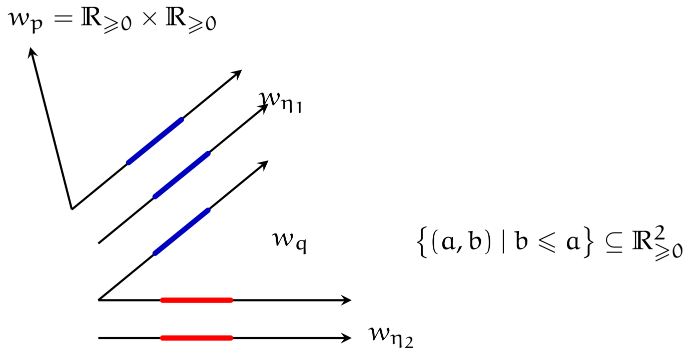
and

$$\chi_{\eta_2}^\vee\colon Q_{\mathbb{R}}^\vee \to w_q \qquad m \mapsto (m, 0).$$

Example 2.50. Consider a curve which is a union of two rational curves with a marked point p on one branch. Then let $\underline{S} = \operatorname{Spec} \mathbb{C}$. Then we see that

$$S^{\mathrm{can}} = \mathrm{Spec}(\mathbb{N} \to \mathbb{C}),$$

$Q = \mathbb{N}$, and $Q_{\mathbb{R}}^\vee = \mathbb{R}_{\geqslant 0}$. We also see that $\bar{M}_{C,q} = \mathbb{N} \oplus_{\mathbb{N}} \mathbb{N}^2$ and $\bar{M}_{C,p} = \mathbb{N} \oplus \mathbb{N}$.



2.11. **Tropical curves.**

Definition 2.51. A family of *tropical curves* $(G, \vec{g}, \vec{\ell})$ over a cone w, N_w consists of

(1) A connected graph $G = V(G) \cup E(G) \cup L(G)$;
(2) A leg ordering $\vec{m}\colon L(G) \xrightarrow{\sim} \{1, \dots, n\}$;
(3) A genus function $\vec{g}\colon V(G) \to \mathbb{N}$;
(4) An edge length function $\ell\colon E(G) \to \mathrm{Hom}(w \cap N_w, \mathbb{N}) \setminus \{0\}$.

Now we will construct $\Sigma(G, \ell)$. For $v \in V(G)$, we set $w_v = w$. For $e \in E(G)$, we set

$$w_e = \left\{(s, \lambda) \mid \lambda \leqslant \vec{\ell}(e)(s)\right\} \subseteq w \times \mathbb{R}_{\geqslant 0}.$$

For legs $\ell \in L(G)$, we set $w_\ell = w \times \mathbb{R}_{\geqslant 0}$. We now need to define the face maps. If ℓ is incident to v, then the face map is

$$w_v \hookrightarrow w_\ell \mid w \mapsto (s, 0).$$

If e is incident to two vertices v_1, v_2, then the face maps are

$$w_{v_1} \hookrightarrow w_e\colon s \mapsto (s, \vec{\ell}(e)(s))$$
$$w_{v_2} \hookrightarrow w_e\colon s \mapsto (s, 0).$$

Remark 2.52. The choice of v_1 and v_2 does not change the cone complex up to isomorphism, but it does matter when we want to define contact orders for log maps. However, in general there is no canonical way to make this choice.

2.12. **Punctured curves.** We consider a nodal curve

$$\mathrm{Spec}\, k[x, y]/xy \to \mathrm{Spec}\, k$$

with log structures given by $\mathbb{N}^2$ with chart $(a, b) \mapsto x^a y^b$ and $\mathbb{N} \to k$ being the standard log point. Now consider $\underline{C}_1 = Z(x) = \mathrm{Spec}\, k[y]$ and mark the node by $p\colon \mathrm{Spec}\, k \to \mathrm{Spec}\, k[y]$. Note that

$$C^{\mathrm{can}} = (\underline{C}, M_C) \to S^{\mathrm{can}} = \mathrm{Spec}(\mathbb{N} \to k).$$

Denote $M^\circ_{C_1} = M_C|_{C_1}$. Then we see that on log structures we have the diagram

$$\begin{tikzcd}
k[y] & \mathbb{N}^2 \arrow[l, "(a,b) \mapsto 0^a y^b"'] \\
k \arrow[u] & \mathbb{N}. \arrow[l] \arrow[u, "\Delta"']
\end{tikzcd}$$

In particular, we can compute the stalk $\bar{M}^{\circ}_{C_1}|_p = \mathbb{N}^2$, and therefore we have

$$\bar{\pi}^{\flat}_{C_1^{\circ}}|_p \colon \mathbb{N} \xrightarrow{\Delta} \mathbb{N}^2.$$

However, we should have the map

$$\bar{\pi}^{\flat}_{C_1}|_p \colon \mathbb{N} \xrightarrow{i_1} \mathbb{N} \oplus \mathbb{N},$$

and therefore C_1 is **not** a log curve. We can then consider the diagram

$$\begin{tikzcd}
\mathbb{N}\oplus\mathbb{Z} & & \\
\mathbb{N}^2 \arrow[u, hook] \arrow[rr, "{(0,1)\mapsto(0,1)}", "{(1,0)\mapsto(1,1)}"'] & & \mathbb{N}^2 \arrow[llu, "{(1,0)\mapsto(1,-1)}"', "{(0,1)\mapsto(0,1)}"] \\
& \mathbb{N}. \arrow[lu, "i_1"] \arrow[ru, "\Delta"'] &
\end{tikzcd}$$

Therefore, we will consider extending the log structures.

Recall that for a log curve, we have $M_{\mathcal{C}} = M'_{\mathcal{C}} \oplus_{\mathcal{O}^{\times}_{\mathcal{C}}} P_{\mathcal{C}}$, where $P_{\mathcal{C}}$ is the divisorial log structure by the markings.

Definition 2.53. A *punctured curve* over an fs log scheme S consists of $(\mathcal{C}^{\circ} \xrightarrow{p} \mathcal{C} \xrightarrow{\pi} S, \{p_i\}_{i=1}^{n})$, where $(\mathcal{C} \xrightarrow{\pi} S, \vec{p})$ is a log curve and the map $p \colon \mathcal{C}^{\circ} \to \mathcal{C}$ is a map of fine log schemes satisfying the following conditions:

(1) The underlying map $\underline{p}$ is an isomorphism;
(2) On log structures, the map $p^{\flat} \colon M_{\mathcal{C}} \hookrightarrow M_{\mathcal{C}^{\circ}}$ is an isomorphism away from the p_i and induces
$$M_{\mathcal{C}} \xrightarrow{p^{\flat}} M_{C^{\circ}} \hookrightarrow M_{\mathcal{C}} \oplus P^{gp}.$$
(3) For any marking p_i and any $s \in M_{\mathcal{C}^{\circ}, p_i} \setminus M_{\mathcal{C}, p_i}$, we have $\alpha_{\mathcal{C}^{\circ}}(s) = 0$.

Roughly speaking, a punctured curve is a log curve plus the data of a commutative diagram

$$\begin{tikzcd}
M_{\mathcal{C}^{\circ}} \arrow[r, hook] \arrow[dr] & M'_{\mathcal{C}} \oplus_{\mathcal{O}^{\times}_{\mathcal{C}}} P^{gp} \\
 & \mathcal{O}_{\mathcal{C}} \\
\mathcal{M}_{\mathcal{C}}. \arrow[uu, hook] \arrow[ur] &
\end{tikzcd}$$

Example 2.54. Let $(\underline{\mathcal{C}} \to \underline{S} = \operatorname{Spec} k, p)$ be a smooth 1-pointed curve.

(1) Let $(\mathcal{C}/S, p)$ be endowed with the canonical log structure (so S has no log structure). Let $M_{\mathcal{C}} = P$. Then any puncturing $p^{\flat} \colon M_{\mathcal{C}} \hookrightarrow M_{\mathcal{C}^{\circ}} \subset P^{gp}$ must be trivial. The reason is that $M_{\mathcal{C}^{\circ}}$ cannot have a section s such that $\alpha_{\mathcal{C}^{\circ}} = 0$. If $S \in P = M_{\mathcal{C}}$, then $\alpha_{\mathcal{C}^{\circ}}(s) = \alpha_{\mathcal{C}}(s)$. Otherwise, if $s \notin P$, then because $\alpha_{\mathcal{C}^{\circ}}(-s)$ is nonzero, so is $\alpha_{\mathcal{C}^{\circ}}(s)$.

(2) Let $M_{\mathcal{C}^\circ} \subseteq (\mathbb{N} \oplus \mathcal{O}_{\mathcal{C}}^\times) \oplus P^{gp}$ be a puncturing. Choose a local generator x of $\mathcal{O}(-p)$. Then any $s \in M_{\mathcal{C}^\circ,p}$ has the form $s = ((n, \varphi), x^m)$ where $n \in \mathbb{N}$ and $\varphi \in \mathcal{O}_{\mathcal{C},p}^\times$. If $m < 0$, then we require that $\alpha_{\mathcal{C}^\circ}(s) = 0$. This implies that $\alpha_{\mathcal{C}}((n, \varphi), 0) = 0$, and therefore

$$\bar{M}_{\mathcal{C}^\circ,p} \subseteq \{(n, m) \in \mathbb{N} \oplus \mathbb{Z} \mid m \geqslant 0 \text{ if } n = 0\}.$$

In fact, any fine submonoid of the right hand side containing $\mathbb{N} \oplus \mathbb{N}$ can be realized by a puncturing.

(3) Let $S = \operatorname{Spec}(\mathbb{N} \to k)$. Consider $M_{\mathcal{C}^\circ}$ such that $\bar{M}_{\mathcal{C}^\circ,p} = \mathbb{N}^2 + \mathbb{N}(2, -2)$. Because $(1, -1) \notin \bar{M}_{\mathcal{C}^\circ,p}$, we see that $M_{\mathcal{C}^\circ}$ is not saturated.

Lemma 2.55. *Let Q be a sharp fs monoid. Let $w := \operatorname{Hom}(Q, \mathbb{R}_{\geqslant 0}) = Q_{\mathbb{R}}^\vee$. Let $Q^\circ \subseteq Q \oplus \mathbb{Z}$ be finitely generated with $Q \oplus \mathbb{N} \subsetneq Q^\circ$ and $Q^\circ \cap \{0\} \times \mathbb{Z}_{<0} = \emptyset$. Then there exists a nonzero, concave, piecewise linear function*

$$\ell \colon w \to \mathbb{R}_{\geqslant 0}$$

with rational slopes such that

$$(Q^\circ)_{\mathbb{R}}^\vee = \left\{(s, \lambda) \in w \times \mathbb{R}_{\geqslant 0} \mid 0 \leqslant \lambda \leqslant \ell(s)\right\}.$$

Example 2.56. Let $Q = \mathbb{N}^2$ and set

$$\begin{aligned} Q^\circ &= \langle((1, 0), -1), ((0, 1), -1), ((0, 0), 1)\rangle \subset \mathbb{N}^2 \oplus \mathbb{Z} \\ &= \{((a, b), m) \mid a + b + m \geqslant 0, a, b \geqslant 0\}. \end{aligned}$$

Then

$$(Q^\circ)_{\mathbb{R}}^\vee = \{(e_x, e_y, e_t) \mid 0 \leqslant e_t \leqslant \min(e_x, e_y)\}.$$

3. Intro to GW theory with a view to higher genus (Patrick Lei)

3.1. **Introduction to GW theory.** The goal of these lectures is to discuss Gromov-Witten theory in a way that is amenable to doing calculations in higher genus. Gromov-Witten theory concerns itself with integrals on the moduli space $\overline{\mathcal{M}}_{g,n}(X, \beta)$, where X is a smooth projective algebraic variety and $\beta \in H_2(X, \mathbb{Z})$ is an effective curve class. There is a diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, "f"] \arrow[d, shift right, "\pi"'] & X \\ \overline{\mathcal{M}}_{g,n}(X, \beta) \arrow[u, shift right, "\sigma_i"'] \arrow[ur, "ev_i"'] & \end{tikzcd}$$

where $\mathcal{C}$ is the universal curve, σ_i are the sections of π which give the marked points, f is the universal stable map, and $ev_i = f \circ \sigma_i$ are evaluation morphisms. While $\overline{\mathcal{M}}_{g,n}(X, \beta)$ is generally singular, not equidimensional, and reducible, there is a special class

$$[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\mathrm{vir}} \in A_{(c_1(X),\beta)+(\dim X-3)(1-g)+n}(\overline{\mathcal{M}}_{g,n}(X, \beta), \mathbb{Q}),$$

which is known as the *virtual fundamental class*.

Exercise 3.1. Use standard arguments in deformation theory to verify that the "expected dimension" of $\overline{\mathcal{M}}_{g,n}(X, \beta)$ the number given above.

Under the assumption that X is smooth and projective, $\overline{\mathcal{M}}_{g,n}(X,\beta)$ is proper, so we may define *GW invariants*

$$\left\langle \tau_{a_1}(\alpha_1),\cdots,\tau_{a_n}(\alpha_n)\right\rangle_{g,n,\beta} \coloneqq \left(\prod_{i=1}^{n} \mathrm{ev}_i^*\,\alpha_i\cdot\psi_i^{a_i}\right)\cap[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}$$

for $\alpha_i \in H^*(X)$. Here, the ψ_i are defined by $\psi_i = c_1(\sigma_i^*(\omega_\pi))$.

Unfortunately, calculating individual Gromov-Witten invariants is very difficult, so we will consider the even more difficult problem of studying generating series of GW invariants, for example

$$F_g = \sum_{\beta\geqslant 0} q^\beta \langle\rangle_{g,0,\beta}$$

when X is a Calabi-Yau threefold. Here, q is a formal variable which keeps track of the curve class and we will denote the ring $\mathbb{Q}[\![q^\beta \mid \beta\in H_2(X,\mathbb{Z})_{\mathrm{eff}}]\!]$ by Λ. There are some other structures we will consider. First, we may consider the *quantum cohomology ring*, whose product $a\star_{\mathbf{t}} b$ is given by the formula

$$(a\star_{\mathbf{t}} b, c) = \sum_{\substack{\beta\geqslant 0\\ n\geqslant 0}} \langle a,b,c,\mathbf{t},\ldots,\mathbf{t}\rangle_{0,3+n,\beta}\frac{q^\beta}{n!}.$$

Here $a,b,c,\mathbf{t}\in H^*(X)$. We will also consider the *quantum connection*. Let $\{e_i\}$ be a basis for $H^*(X)$. Then the quantum connection is the connection on the trivial bundle with fiber $H^*(X)$ over $H^*(X)\times\mathbb{P}^1_z$ given by the formula

$$\nabla_{t^i} = \partial_{t^i} + \frac{1}{z}e_i\star_{\mathbf{t}}.$$

Sometimes, it is useful to also consider the connection in the z-direction given by

$$\nabla_{z\frac{d}{dz}} \coloneqq z\frac{d}{dz} - \frac{1}{z}E_X\star_{\mathbf{t}} + \mu_X,$$

where $\mu_X(\alpha) = \frac{\deg(\alpha)-\dim X}{2}\alpha$ is the grading operator and

$$E_X = c_1(X) + \sum_i\left(1-\frac{\deg(e_i)}{2}\right)t^i e_i$$

is the *Euler vector field*. Because we will be mostly interested in Calabi-Yau varieties, we will not consider this.

3.2. **Enumerative mirror symmetry in genus zero.** The impetus for mathematicians to develop and study Gromov-Witten theory came in the early 1990s, when a group of physicists predicted the number of rational curves of low degree on a general quintic threefold by converting the problem to one of computing period integrals on a different Calabi-Yau threefold called the *mirror quintic* (equivalently, studying its variation of Hodge structure). In genus zero, all invariants can be recovered from the *J-function* $J_{\mathbf{t}}(q,z)\in H^*(X,\Lambda)[z]$, which is defined by

$$(J_{\mathbf{t}}(q,z),\alpha) \coloneqq (z+\mathbf{t},\alpha) + \sum_{n,\beta}\frac{q^\beta}{n!}\left\langle\frac{\alpha}{z-\psi},\mathbf{t},\ldots,\mathbf{t}\right\rangle_{0,1+n,\beta}$$

for any $\mathbf{t}\in H^*(X,\Lambda)[z]$ $\alpha\in H^*(X)$, where $(-,-)$ is the Poincaré pairing. Therefore, the problem of computing all genus-zero Gromov-Witten invariants with descendants (these are the ψ-classes) reduces to the problem of computing the J-function.

Remark 3.2. We expand $\frac{1}{z-\psi}$ as $\sum_{k=0}^{\infty} \psi^k z^{-k-1}$.

We should note that the J-function can be obtained via a fundamental solution to the quantum differential equation, which is given by

$$(S_{\mathbf{t}}(z)\phi, \alpha) = (\phi, \alpha) + \sum_{n,\beta} \frac{q^\beta}{n!} \left\langle \frac{\alpha}{z-\psi}, \phi, \mathbf{t}, \ldots, \mathbf{t} \right\rangle_{0,2+n,\beta}.$$

This has the property that $S_{\mathbf{t}}^*(-z)S_{\mathbf{t}}(z) = \mathrm{Id}$, which is usually known as the *symplectic property*. From this, we can recover the J-function by $zS_{\mathbf{t}}^*(z) \cdot 1 = J_{\mathbf{t}}(q, z)$.

In physics, mirror symmetry is a duality between two different theories called the A-model and B-model, so now we will need to introduce the B-model object. In the interest of time (and notation), we will assume that X is a toric variety and that $-K_X$ is nef. Let $D_1, \ldots, D_r$ be the complete set of torus-invariant divisors in X. Then define the I-function

$$I_{\mathbf{t}}(q, z) := ze^{\mathbf{t}/z} \sum_{\beta \geqslant 0} q^\beta \frac{\prod_{i=1}^r \prod_{m=-\infty}^0 (D_i + mz)}{\prod_{i=1}^r \prod_{m=-\infty}^{\beta \cdot D_j} (D_i + mz)}.$$

Note here that we expand any $(D_i + mz)^{-1}$ which appears as $\frac{1}{mz}\left(1 - \frac{D_i}{mz} + \cdots\right)$, or namely, as a geometric series in $\frac{D_i}{mz}$. In addition, the I-function is annihilated by a differential equation known as the *Picard-Fuchs equation*, which is some hypergeometric type equation.

Example 3.3. If $\mathbf{t} = 0$ and $X = \mathbb{P}^1$, we obtain

$$\begin{aligned} I(q, z) &= z \sum_{d \geqslant 0} q^d \frac{1}{\prod_{m=0}^d (H + mz)^2} \\ &= z \sum_{d \geqslant 0} \left(\frac{q}{z^2}\right)^d \frac{1}{(d!^2)} + O(H), \end{aligned}$$

so in fact we see the Bessel function of the first kind as the leading term.

If X is a complete intersection in a toric variety Y cut out by line bundles $L_1, \ldots, L_k$, all of which satisfy $H^1(f^*L_i) = 0$ for any genus zero stable map to Y (this is called being *convex*), then the quantum Lefschetz theorem allows us to replace $[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\mathrm{vir}}$ with $e(\bigoplus \pi_* f^* L_i) \cap [\overline{\mathcal{M}}_{g,n}(Y, \beta)]^{\mathrm{vir}}$ and perform all computations on Y. In this case, the I-function is defined to be

$$I_{\mathbf{t}}(q, z) := ze^{\mathbf{t}/z} \sum_{\beta \geqslant 0} q^\beta \frac{\prod_{i=1}^k \prod_{m=-\infty}^{L_i \cdot \beta} (L_i + mz)}{\prod_{i=1}^k \prod_{m=-\infty}^0 (L_i + mz)} \frac{\prod_{i=1}^r \prod_{m=-\infty}^0 (D_i + mz)}{\prod_{i=1}^r \prod_{m=-\infty}^{\beta \cdot D_j} (D_i + mz)},$$

where we abuse notation and conflate the L_i with their first Chern classes.

Example 3.4. For the quintic threefold, which is cut out by a section of $\mathcal{O}_{\mathbb{P}^4}(5)$ the I-function at $\mathbf{t} = 0$ is given by

$$I(q, z) = z \sum_{d \geqslant 0} q^d \frac{\prod_{m=1}^{5d} (5H + mz)}{\prod_{m=1}^d (H + mz)^5}.$$

The content of the genus zero mirror theorem is a correspondence between these two objects. More precisely, in the case where $-K_X$ is nef, they are related by a change of variables.

Theorem 3.5. *Write the I-function with* $\mathbf{t} = 0$ *as* $I_0(q) \cdot z + I_1(q) + O(z^{-1})$. *Then we have*

$$J_{\tau(\mathbf{t})}(q, z) = \frac{I_{\mathbf{t}}(q, z)}{I_0(q)},$$

where $\tau(\mathbf{t}) = \frac{\mathbf{t} + I_1(q)}{I_0(q)}$.

Example 3.6. When $X = \mathbb{P}^1$, note that $I_0(q) = 0$ and $I_1(q) = 0$, so in fact, we have the equality

$$J_{\mathbf{t}}(q, z) = I_{\mathbf{t}}(q, z).$$

Example 3.7. When X is the quintic threefold, we will rewrite the components of the I-function as $I_0(q) \cdot z + I_1(q)H + O(z^{-1})$ for convenience. We have

$$I_0(q) = \sum_{d \geqslant 0} \frac{(5d)!}{(d!)^5} q^d, \qquad I_1(q) = H \sum_{d \geqslant 0} \frac{(5d)!}{(d!)^5} \left(5 \sum_{m=d+1}^{5d} \frac{1}{m} \right),$$

so there is a nontrivial mirror map in this case. Specializing to $\mathbf{t} = 0$, we see that $\tau = \frac{I_1(q)H}{I_0(q)}$, so we obtain

$$J_\tau(q, z) = \frac{I(q, z)}{I_0(q)}.$$

By the divisor equation, the left hand side also equals $J(qe^{I_1/I_0}, z)$, and we will also refer to this as the mirror map. Therefore, we can use the mirror theorem to compute $\star_\tau$. The quantum connection in the direction of H^2 becomes the ODE

$$(H + zD)S_\tau(z)^* = S_\tau(z)^* \cdot I_{11} H \star_\tau,$$

where $D \coloneqq q\frac{d}{dq}$ (here, the coordinate is $\log q$) and $I_{11} \coloneqq 1 + D\left(\frac{I_1(q)}{I_0(q)}\right)$. An explicit computation using the mirror theorem and the results of Zagier-Zinger yields

$$\begin{aligned} I_{11} H + \cdots &= S_\tau(z)^* (I_{11} H \star_\tau \mathbf{1}) \\ \left(1 + D \left(\frac{\frac{I_1}{I_0} + D\left(\frac{I_2}{I_0}\right)}{I_{11}} \right) \right) H^2 + \cdots &= S_\tau(z)^* (I_{11} H \star_\tau H) \\ I_{11} H^3 &= S_\tau(z)^* (I_{11} H \star_\tau H^2). \end{aligned}$$

Because S_τ begins with the identity, this computes $H \star_\tau$.

3.2.1. *Oscillating integrals.* We will now reinterpret the genus zero mirror theorem, and we will return to this interpretation when we discuss higher genus invariants. In particular, we will interpret the I-function as an integral on some mirror family to X, which for us will be a toric variety. Let

$$0 \to \Lambda \to \mathbb{Z}^r \to N \to 0$$

be the fan sequence of X. Applying $\mathrm{Hom}(-, \mathbb{C}^\times)$, we obtain the exact sequence

$$1 \to (\mathbb{C}^\times)^{\dim X} \to (\mathbb{C}^\times)^r \to \mathrm{Hom}(\Lambda, \mathbb{C}^\times) \to 1.$$

Suppose that the map $\Lambda \to \mathbb{Z}^r$ is given by the matrix $(m_{ij})_{i=1,\dots,r}^{j=1,\dots,\rho(X)}$. Then we consider the variety

$$E_q = \left(\prod_{i=1}^{r} x_j^{m_{ij}} = q_i \middle| i = 1, \dots, \rho(X) \right)$$

which is mirror to X and consider the superpotential $f = x_1 + \cdots + x_r$. If we wish to consider the equivariant theory with equivariant parameters $\lambda_1, \dots, \lambda_r$ acting on each homogeneous coordinate of X, we will consider $f_\lambda = \sum_{i=1}^{r}(x_i - \lambda_i \log x_i)$.

Example 3.8. When $X = \mathbb{P}^n$, the mirror to X is the superpotential $\sum_{i=1}^{n+1}(x_i - \lambda_i \log x_i)$ on the variety $E_q = (x_1 \cdots x_{n+1} = q)$.

We now choose an appropriate Lefschetz thimble Γ_α near each critical point (x_α) of the (equivariant) superpotential and consider the integral

$$I_\alpha = \int_{\Gamma_\alpha} e^{f_\lambda/z} \prod_{i=1}^{r} \frac{dx_i}{x_i}.$$

This corresponds in the I-function to replacing each D_i in the expression by $D_i - \lambda_i$ (turning on the equivariant parameters) and in the J-function to considering the fully equivariant GW theory of X. In genus zero, the limit as $\lambda_i \to 0$ exists, so we can safely recover the original mirror theorem from the following:

Theorem 3.9. *This integral is annihilated by the same Picard-Fuchs equation as the I-function, and up to various prefactors, its asymptotic expansion as $z \to 0^-$ along the negative real axis coincides with the restriction of the I-function corresponding to the T-fixed point on X which corresponds to the critical point (x_α).*

3.2.2. *The Lagrangian cone.* The quantum cohomology of X is an example of a *Frobenius manifold*, which we will see later can be produced from any cohomological field theory. Write $V = H^*(X)$ and let (e_μ) be a basis for V, so we can write $\mathbf{v} = \sum_{\mu,n} t_n^\mu e_\mu z^n \in V[\![z]\!]$. The structure of a Frobenius manifold comes from the function

$$\mathcal{F}_0(\mathbf{v}) = \sum_{n,\beta,\mu} \frac{q^\beta}{n!} \langle \mathbf{v}(\psi), \dots, \mathbf{v}(\psi) \rangle_{0,n,\beta},$$

which is also called the *genus zero descendant potential* of X. It satisfies the string equation, dilaton equation, and topological recursion relations:

$$\frac{\partial}{\partial t_0^1}\mathcal{F}_0 = \frac{1}{2}(\mathbf{v}_0, \mathbf{v}_0) + \sum_{n=0}^{\infty}\sum_{\mu} t_{n+1}^\mu \frac{\partial}{\partial t_n^\mu}\mathcal{F}_0;$$

$$\frac{\partial}{\partial t_1^1}\mathcal{F}_0 = \sum_{n=0}^{\infty}\sum_{\mu} t_n^\mu \frac{\partial}{\partial t_n^\mu}\mathcal{F}_0 - 2\mathcal{F}_0;$$

$$\frac{\partial^3}{\partial t_{k+1}^\alpha\, \partial t_m^\beta\, \partial t_n^\gamma}\mathcal{F}_0 = \sum_{\mu,\nu} \frac{\partial^2}{\partial t_k^\alpha\, \partial t_0^\mu}\mathcal{F}_0 \cdot \eta^{\mu\nu} \cdot \frac{\partial^3}{\partial t_0^\nu\, \partial t_m^\beta\, \partial t_n^\gamma}\mathcal{F}_0.$$

To state the following result, we consider the infinite-dimensional vector space $V(\!(z^{-1})\!)$ (or a completion of this) with the symplectic form

$$\omega(\mathbf{f}(z), \mathbf{g}(z)) \coloneqq \mathrm{Res}_{z=0}(\mathbf{f}(-z), \mathbf{g}(z)).$$

We also consider the new variable[3] $\mathbf{q} = \mathbf{v} - z$, so $\mathcal{F}_0$ is now a function near $\mathbf{q} = -z$.

Theorem 3.10 (Givental). *$\mathcal{F}$ satisfies the above PDEs if and only if the graph $\mathcal{L}$ of $d\mathcal{F}$ is a Lagrangian cone with vertex at $\mathbf{q} = 0$ such that its tangent spaces L are tangent to $\mathcal{L}$ exactly along $zL \subset L$.*

We may recover the Lagrangian cone $\mathcal{L}$ by the following procedure. We find a fundamental solution

$$S = \mathrm{Id} + S_1 z^{-1} + S_2 z^{-2} + \cdots$$

to the quantum connection, which satisfies the equation

$$z\frac{\partial}{\partial t^\mu} S = e_\mu \star S.$$

Then, setting $J := zS^*(z)\mathbf{1}$, we can then recover the Lagrangian cone by the following procedure:

- The derivatives $\frac{\partial}{\partial t^\mu} J$ form a basis of $L \cap V[\![z^{-1}]\!]$;
- This implies that $z\frac{\partial^2}{\partial t^\mu \partial t^\nu} \in L \cap V[\![z^{-1}]\!]$. Writing these in terms of the first derivatives and using the fact that J solves the quantum connection, we recover the Frobenius structure and hence the Lagrangian cone.

Using this, we can interpret the genus zero mirror theorem as:

Theorem 3.11. *The I-function $I_{\mathbf{t}}(q, z)$ lies on the Lagrangian cone of X.*

3.3. **Moduli of curves and CohFTs.** We will denote by $\overline{\mathcal{M}}_{g,n}$ the moduli space of stable curves of genus g with n marked points. This is nonempty if and only if $2g - 2 + n > 0$, in which case it is a Deligne-Mumford stack of dimension $3g - 3 + n$. There is a combinatorial structure to the collection of $\overline{\mathcal{M}}_{g,n}$, which is given by a collection of morphisms.

- There is the gluing morphism
$$q \colon \overline{\mathcal{M}}_{g,n+1} \times \overline{\mathcal{M}}_{h,m+1} \to \overline{\mathcal{M}}_{g+h,n+m},$$
which takes two curves and glues them along the last marked point to form a node;
- There is the self-gluing morphism
$$s \colon \overline{\mathcal{M}}_{g-1,n+2} \to \overline{\mathcal{M}}_{g,n},$$
which glues the last two marked points.

There is of course another interesting map, which is the forgetful map

$$\mathrm{pr} \colon \overline{\mathcal{M}}_{g,n+1} \to \overline{\mathcal{M}}_{g,n}$$

given by deleting the last marked point and then stabilizing. We are now able to define our enumerative theories of interest, which include Gromov-Witten theory.

Definition 3.12. Given a vector space V with a nondegenerate symmetric bilinear form η and a unit element $\mathbf{1} \in V$, a *Cohomological Field Theory (CohFT)* Ω on V is a collection $\Omega_{g,n}$ of S_n-equivariant linear maps

$$\Omega_{g,n} \colon V^{\otimes n} \to H^*(\overline{\mathcal{M}}_{g,n})$$

[3]This is called the *dilaton shift*.

which satisfy the basic identity

$$\Omega_{0,3}(\mathbf{1}, u, v) = \eta(u, v)$$

and the following combinatorial identities (we will let e_μ be a basis of V such that $e_1 = \mathbf{1}$ and e^μ be the dual basis):

$$q^*\Omega_{g+h,n+m}(\mathbf{v}_1, \mathbf{v}_2) = \sum_\mu \Omega_{g,n+1}(\mathbf{v}_1, e_\mu) \cdot \Omega_{h,m+1}(\mathbf{v}_2, e^\mu);$$

$$s^*\Omega_{g,n}(\mathbf{v}) = \sum_\mu \Omega_{g,n+2}(\mathbf{v}, e_\mu, e^\mu).$$

If in addition the identity

$$\mathrm{pr}^*\,\Omega_{g,n}(\mathbf{v}) = \Omega_{g,n+1}(\mathbf{v}, \mathbf{1})$$

is satisfied, then we say the CohFT has a *flat unit* (or satisfies the *string equation*).

All of this can be generalized to super vector spaces, but for simplicity we will not deal with this case.

Example 3.13. The most important example of a CohFT is the Gromov-Witten theory of a smooth projective variety X. Here, recall that the source of a stable map is a **prestable** curve, so there is a stabilization morphism

$$\mathrm{st}\colon \overline{\mathcal{M}}_{g,n}(X, \beta) \to \overline{\mathcal{M}}_{g,n}$$

which forgets the map and stabilizes the curve. Then, working over the Novikov ring, we will set $V = H^*(X)$, η to be the Poincaré pairing, and $\mathbf{1}$ to be the fundamental class of X. The linear maps $\Omega^X_{g,n}$ are given by the formulae

$$\Omega^X_{g,n}(\boldsymbol{\tau}) \coloneqq \sum_\beta q^\beta\, \mathrm{st}_*\left(\prod_{i=1}^n \mathrm{ev}_i(\tau_i) \cap [\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\mathrm{vir}}\right),$$

where $\mathrm{ev}_i\colon \overline{\mathcal{M}}_{g,n}(X, \beta) \to X$ is the i-th evaluation map.

Example 3.14. Let $\pi\colon \mathcal{C}_{g,n} \to \overline{\mathcal{M}}_{g,n}$ be the universal curve. Then the *Hodge bundle* is the vector bundle

$$\mathbb{E}_{g,n} \coloneqq \pi_*\omega_\pi.$$

We then consider the vector space $V = \mathbb{C}$ with the usual pairing and define the CohFT $\Omega^{\mathbb{E}}$ by the formula

$$\Omega^{\mathbb{E}}_{g,n} \coloneqq c(\mathbb{E}_{g,n}).$$

The gluing axioms are satisfied by the equation $q^*\mathbb{E}_{g+h} = \mathbb{E}_g \oplus \mathbb{E}_h$ and the exact sequence

$$0 \to \mathbb{E}_{g-1} \to \mathbb{E}_g \to \mathcal{O} \to 0.$$

Here, because genus zero components do not contribute to global sections of ω and $\mathbb{E}$ does not depend on the marked points, this CohFT has a flat unit.

Remark 3.15. We will see later that this is related to the GW CohFT of a point by the quantum Riemann-Roch theorem.

Given a CohFT Ω, we may produce invariants by pairing the classes $\Omega_{g,n}(\mathbf{v})$ with various cohomology classes on $\overline{\mathcal{M}}_{g,n}$. The most important ones to consider are the classes

$$\bar{\psi}_i \coloneqq c_1(s_i^*\omega_\pi),$$

where $s_i\colon \overline{\mathcal{M}}_{g,n} \to \mathcal{C}_{g,n}$ is the i-th marked point.

Remark 3.16. In Gromov-Witten theory, these are called *ancestors*. There are also descendants, which are given by the same formula except using the moduli space $\mathfrak{M}_{g,n}$ of prestable curves instead of $\overline{\mathcal{M}}_{g,n}$, where we factor the morphism st defined above as

$$\overline{\mathcal{M}}_{g,n}(X,\beta) \xrightarrow{\text{forget map}} \mathfrak{M}_{g,n} \xrightarrow{\text{stabilize}} \overline{\mathcal{M}}_{g,n}.$$

The descendant classes are denoted by ψ_i.

Example 3.17. We will calculate an invariant which will appear in the GW theory of Calabi-Yau threefolds. For any CohFT Ω, consider the invariant

$$\langle \mathbf{1}\bar{\psi}_1 \rangle_{1,1}^{\Omega} := \int_{\overline{\mathcal{M}}_{1,1}} \Omega_{1,1}(\mathbf{1})\bar{\psi}_1.$$

Using the second gluing equation, the degree zero part of $\Omega_{1,1}(\mathbf{1})$ is equal to

$$\sum_{\mu} \Omega_{g,n+2}(\mathbf{1}, e_\mu, e^\mu) = \sum_{\mu} \eta(e_\mu, e^\mu),$$

which yields the (graded) dimension $\chi(V)$ of V (for example, if $V = H^*(X)$ for a Calabi-Yau threefold X, this is the topologial Euler characteristic of X). Therefore, we obtain

$$\langle \mathbf{1}\bar{\psi}_1 \rangle_{1,1}^{\Omega} = \frac{\dim V}{24}.$$

Example 3.18. Consider the GW CohFT of a point. This is given by $V = \mathbb{C}$ with the usual pairing and the formula

$$\Omega_{g,n} = 1.$$

Then we define the invariants

$$\langle \bar{\psi}_1^{a_1} \cdots \bar{\psi}_n^{a_n} \rangle_{g,n} := \int_{\overline{\mathcal{M}}_{g,n}} \Omega_{g,n} \cdot \bar{\psi}_1^{a_1} \cdots \bar{\psi}_n^{a_n}.$$

Theorem 3.19 (Kontsevich)**.** *The function*

$$\mathcal{D} := \exp\left(\sum_{g,n} \frac{\hbar^{g-1}}{n!} \sum_{a_1+\cdots+a_n=n} \langle \bar{\psi}_1^{a_1} \cdots \bar{\psi}_n^{a_n} \rangle_{g,n} t_{a_1} \cdots t_{a_n} \right)$$

is annihilated by the operators

$$\begin{aligned} L_{-1} &:= -\frac{\partial}{\partial t_0} + \frac{\hbar^{-1}}{2} t_0^2 + \sum_{i=0}^{\infty} t_{i+1} \frac{\partial}{\partial t_i}; \\ L_0 &:= -\frac{3}{2}\frac{\partial}{\partial t_1} + \sum_{i=0}^{\infty} \frac{2i+1}{2} t_i \frac{\partial}{\partial t_i} + \frac{1}{16}; \\ L_n &:= -\frac{(2n+3)!!}{2^{n+1}} \frac{\partial}{\partial t_{n+1}} + \sum_{i=0}^{\infty} \frac{(2i+2n+1)!!}{(2i-1)!!2^{n+1}} t_i \frac{\partial}{\partial t_{i+n}} \\ &\quad + \frac{\hbar}{2} \sum_{i=0}^{n-1} \frac{(2i+1)(2i-1)\cdots(2i+1-2n)}{2^{n+1}} \frac{\partial^2}{\partial t_i \, \partial t_{n-1-i}}. \end{aligned}$$

Remark 3.20. This result (Virasoro constraints for a point) is equivalent to $\mathcal{D}$ being a tau-function for the KdV hierarchy and has been generalized by various authors.

3.4. R**-matrix action.** In the early 2000s, Givental discovered a remarkable property of axiomatic enumerative theories, namely that one can transform CohFTs by the action of matrices $R(z) = R_0 + R_1 z + R_2 z^2 + \cdots \in \mathrm{Hom}(V, V')[\![z]\!]$ which satisfy the property

$$R^*(-z)R(z) = \mathrm{Id}_V.$$

Traditionally, the literature requires that $V = V'$ and $R_0 = \mathrm{Id}$, but the MSP group has removed this restriction and also allows $\dim V < \dim V'$.

In order to define the action of R on a CohFT Ω defined on V, we need to review some of the combinatorial structure of $\overline{\mathcal{M}}_{g,n}$. For a curve $C \in \overline{\mathcal{M}}_{g,n}(\mathbb{C})$, we may consider the dual graph of C, which has vertices, edges, and legs, which are defined as follows:

- The vertices correspond to irreducible components of C and are labelled by a non-negative integer, which is the genus;
- The edges correspond to nodes of C (in particular, we allow loops);
- The legs correspond to marked points.

Any graph which appears as the dual graph of a stable curve is called a *stable graph*.

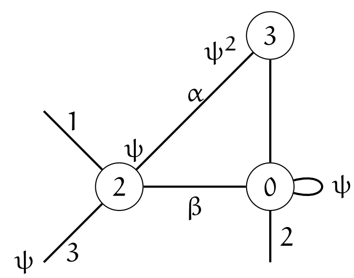


FIGURE 1. Example of a stable graph in $\overline{\mathcal{M}}_{7,3}$ and associated tautological class. This stable graph describes the image of a map $\overline{\mathcal{M}}_{2,4} \times \overline{\mathcal{M}}_{3,2} \times \overline{\mathcal{M}}_{0,5} \to \overline{\mathcal{M}}_{7,3}$.

The first type of action on CohFTs is a translation action. Let $T \in V[\![z]\!]$. For a CohFT Ω, we define

$$(T\Omega)_{g,n}(\mathbf{v}) := \sum_m \frac{1}{m!} p_* \Omega_{g,n+m}(\mathbf{v}, T, \ldots, T)$$

whenever the infinite sum makes sense.

Example 3.21. If Ω^X is the GW CohFT of a smooth projective variety X and $\tau \in H^2(X)$ is a divisor class, then the infinite sum makes sense and we can define the shifted GW CohFT $\Omega^{X,\tau}$ of X. For example, if X is the quintic threefold and $\tau = \frac{I_1}{I_0} H$, then shifting to τ is the same as the mirror map $q \mapsto Q(q) = q e^{\frac{I_1}{I_0}}$.

The second type of action is the action of a matrix R as in the beginning of this section. Let $G_{g,n}$ define the set of all stable graphs of genus g with n legs. Then we define

$$(R\Omega)_{g,n} := \sum_{\Gamma \in G_{g,n}} \frac{1}{|\mathrm{Aut}\,\Gamma|} \mathrm{Cont}_\Gamma,$$

where Cont_Γ is defined by the following construction:

- At the i-th leg, we place the map
$$R(-\bar\psi_i)^* \in \mathrm{Hom}(V', V)[\![\bar\psi_i]\!];$$
- At every edge, we place the bivector
$$V(\bar\psi_1, \bar\psi_2) \coloneqq \sum_\mu \frac{e_\mu \otimes e^\mu - R(-\bar\psi_1)^* e_\mu \otimes R(-\bar\psi_2)^* e^\mu}{\bar\psi_1 + \bar\psi_2};$$
- At every vertex, we place the linear map
$$\Omega_{g_v, n_v} \colon V^{\otimes n_v} \to H^*(\overline{\mathcal{M}}_{g_v, n_v});$$
- Finally, we consider the pushforward in cohomology along the gluing morphism $\prod_v \overline{\mathcal{M}}_{g_v,n_v} \to \overline{\mathcal{M}}_{g,n}$.

Definition 3.22. Let R be as above. Then we define the translation
$$T_R \coloneqq z(\mathbf{1} - R(-z)^* \mathbf{1}') \in zV'[\![z]\!].$$
Whenever it makes sense, we define
$$R.\Omega \coloneqq RT\Omega.$$

Theorem 3.23 (Chang-Guo-Li). *Suppose we work with coefficients in* $\mathbb{C}[\![q]\!]$. *Then if*
$$T_R \in z^2 V[\![z]\!] + zqV[\![z]\!],$$
$R.\Omega$ *is a well-defined CohFT. Moreover, if* $\dim V = \dim V'$, $\mathbf{1}'$ *is a unit for* $R.\Omega$.

We would like to remark a bit more about the translation action when $R_0 \neq \mathrm{Id}$. For simplicity, we will assume that $V = V'$ and $R_0 \mathbf{1} = c \cdot \mathbf{1}$ for some constant c. Then we set $\tilde{T}_R \coloneqq z(\mathbf{1} - cR(z)^{-1}\mathbf{1})$ and use the dilaton equation to compute
$$\begin{aligned}
T_R \Omega_{g,n}(\mathbf{v}) &= \sum_{m=0}^\infty \frac{1}{m!} p_* \Omega_{g,n+m}(\mathbf{v}, T_R, \ldots, T_R) \\
&= \sum_{k,\ell \geqslant 0} \frac{1}{k!\ell!} p_* \Omega_{g,n+k+\ell}(\mathbf{v}, ((1-c^{-1})\mathbf{1} \cdot \psi)^{\otimes k}, (c^{-1}\tilde{T}_R)^{\otimes \ell}) \\
&= \sum_{k,\ell \geqslant 0} \frac{(1-c^{-1})^k \cdot c^{-\ell}}{\ell!} \binom{2g-2+n+k+\ell-1}{k} p_* \Omega_{g,n+\ell}(\mathbf{v}, \tilde{T}_R^{\otimes \ell}) \\
&= \sum_{m=0}^\infty \frac{c^{2g-2+n}}{m!} p_* \Omega_{g,n+m}(\mathbf{v}, \tilde{T}_R, \ldots, \tilde{T}_R)
\end{aligned}$$
whenever this makes sense.

3.5. **Reconstruction theorem.** Recall that every CohFT defines a Frobenius algebra. We will call a CohFT *semisimple* if the corresponding Frobenius algebra is semisimple.

Theorem 3.24 (Teleman). *Let* Ω *be a semisimple CohFT with flat unit and* ω *be its topological part. If* Ω *is semisimple, there exists a unique*
$$R = \mathrm{Id} + R_1 z + \cdots \in \mathrm{End}(V)[\![z]\!]$$
such that
$$\Omega = R.\omega.$$

Example 3.25. Recall the Hodge bundle CohFT from before. Recall that it is given by the formula

$$\Omega^{\mathbb{E}}_{g,n} = c(\mathbb{E}) = 1 + \lambda_1 + \cdots + \lambda_g.$$

Taking the degree zero part, we see that $\omega^{\mathbb{E}}$ is the GW CohFT of a point. Using Mumford's computation

$$\mathrm{ch}(\mathbb{E}) = g + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!}\left(\kappa_{2k-1} + \frac{1}{2}\iota_* \sum_{i=0}^{2k-2} \bar{\psi}_1^i \bar{\psi}_2^{2g-2+i}\right),$$

where $\kappa_m = p_* \bar{\psi}_{n+1}^{m+1}$, B_{2k} are the Bernoulli numbers, and ι is the inclusion of the boundary up to a $2:1$ étale cover, and the formula

$$c(E) = \exp\left(-\sum_k (-1)^k (k-1)!\, \mathrm{ch}_k(E)\right)$$

for any vector bundle E (or by just using the quantum Riemann-Roch theorem[4] directly), we obtain the R-matrix

$$R(z) = \exp\left(\sum_{k=1}^{\infty} \frac{B_{2k}}{2k(2k-1)} z^{2k-1}\right) = 1 + \frac{1}{12}z + \cdots.$$

As a sanity check, we will compute $\Omega^{\mathbb{E}}_{1,1}$ using the R-matrix. First, we consider the stable graphs in Figure 2. The first graph Γ_1 gives us the contribution

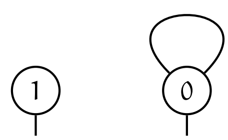


FIGURE 2. Stable graphs for $g = 1, n = 1$

$$\begin{aligned}\mathrm{Cont}_{\Gamma_1} &= T\omega^{\mathbb{E}}_{1,1}(R(z)^{-1}\mathbf{1}) \\ &= \omega^{\mathbb{E}}_{1,1}(R(z)^{-1}\mathbf{1}) + p_*\omega^{\mathbb{E}}_{1,1}(R(z)^{-1}\mathbf{1}, T_R(z)).\end{aligned}$$

Using the formula $B_2 = \frac{1}{6}$ and the fact that $\dim \overline{\mathcal{M}}_{1,1} = 1$, this becomes

$$1 - \frac{1}{12}\psi_1 + \frac{1}{12}\kappa_1.$$

The second graph does not receive tail contributions because $\dim \overline{\mathcal{M}}_{0,3} = 0$. The constant term of the edge contribution is $\frac{1}{12}$, so considering the automorphism and pushing forward to $\overline{\mathcal{M}}_{1,1}$ gives us

$$\mathrm{Cont}_{\Gamma_2} = \frac{1}{12}\Delta,$$

where Δ denotes the boundary divisor (with the correct stack structure). Because $\psi_1 = \kappa_1$ in this case, we obtain

$$1 + \lambda_1 = 1 + \frac{1}{12}(\kappa_1 - \psi_1 + \Delta) = 1 + \frac{1}{12}\Delta.$$

[4] There is a sign error in Coates-Givental, which has propagated to the rest of the literature.

3.6. **Operator formalism and geometric quantization.** We will return to the symplectic formalism of Givental. This is more convenient for certain computations, but it is in fact equal to what we have before, at least when we want to calculate generating functions. Recall that we had a Frobenius manifold structure on V and we considered the vector space $\mathcal{V} := V(\!(z^{-1})\!)$ with symplectic form

$$(\mathbf{f}(z), \mathbf{g}(z)) := \operatorname{Res}_{z=0} \eta(\mathbf{f}(-z), \mathbf{g}(z)).$$

If we consider the polarization given by $\mathcal{V}_+ = V[z]$ and $\mathcal{V}_- = z^{-1}V[\![z^{-1}]\!]$, then letting $\mathbf{q}$ be as before (with the dilaton shift), let $\mathbf{p}$ be coordinates on $\mathcal{V}_-$ such that $(\mathbf{q}, \mathbf{p})$ form a system of Darboux coordinates for $\mathcal{V}$.

Definition 3.26. We will call any formal function of the form

$$\mathcal{D} = \exp\left(\sum_{g=0}^{\infty} \hbar^{g-1}\mathcal{F}_g\right)$$

an *asymptotic element of the Fock space*.

Then given such an asymptotic element of the Fock space, we will quantize (infinitesimal) symplectic transformations on $\mathcal{V}$ by the following formulae:

$$\widehat{p_a p_b} := \hbar\, \partial_{q_a} \partial_{q_b}, \qquad \widehat{p_a q_b} := q_b\, \partial_{q_a}, \qquad \widehat{q_a q_b} = \frac{q_a q_b}{\hbar}.$$

In particular, this will allow us to understand expressions like $\hat{R}\mathcal{D}$. However, we need to be careful because our formulae will involve both the fundamental solution S_τ and the R-matrix, and S_τ is a power series in z^{-1}.[5]

Theorem 3.27 (Givental)**.** *An operator of the form* $S(z^{-1}) = \mathrm{Id} + S_1 z^{-1} + \cdots$ *acts on (asymptotic) elements of the Fock space by the formula*

$$\hat{S}^{-1}\mathcal{D}(\mathbf{q}) = e^{\frac{1}{2\hbar}W(\mathbf{q},\mathbf{q})}\mathcal{D}([S\mathbf{q}]_+),$$

where $W = \sum \eta(W_{mn} q_m, q_n)$ *is defined by the formula*

$$\frac{S(w^{-1})^* S(z^{-1}) - \mathrm{Id}}{w^{-1} + z^{-1}} = \sum \frac{W_{mn}}{w^m z^n}.$$

Operators of the form $R(z) = \mathrm{Id} + R_1 z + \cdots$ *act by the formula*

$$\hat{R}\mathcal{D}(\mathbf{q}) = e^{\frac{\hbar}{2}V(\partial_\mathbf{q}, \partial_\mathbf{q})}\mathcal{D}(R^{-1}\mathbf{q}),$$

where $V = \sum \eta(p_m, V_{mn} p_n)$ *is defined by*

$$\frac{R(w)^* R(z) - \mathrm{Id}}{w + z} = \sum V_{mn} w^m z^n.$$

For a semisimple CohFT, there is a system of *canonical coordinates* u_α such that the 1-form du is a homomorphism of algebras $T_v V \to \mathbb{C}$. Then near a semisimple point, there is a asymptotic solution to the quantum connection of the form

$$\Psi \cdot R \cdot e^{\frac{U}{z}},$$

[5]This is because we expand $\frac{1}{z-\psi} = \sum_{n=0}^{\infty} \psi^n z^{-n-1}$.

where Ψ switches from flat coordinates to canonical coordinates and $U = \operatorname{diag}(u_\alpha)$ is the matrix of canonical coordinates. Finally, define

$$C = \frac{1}{2}\int^{u} \sum R_1^{\alpha\alpha}\, du_\alpha.$$

A corollary of Teleman's reconstruction theorem is the formula

$$\mathcal{D}^X = e^{C(u)}\hat{S}_\tau^{-1}\hat{\Psi}\hat{R}\widehat{e^{\frac{U}{z}}}\prod_{i=1}^{\dim V}\mathcal{D}^{pt}$$

for the Gromov-Witten theory of any smooth projective variety with semisimple quantum cohomology.

Example 3.28. As a final example, we will apply Teleman's theorem to compute F_1 of any semisimple CohFT. Let e_μ denote an idempotent basis for the quantum product. We will compute

$$\int_{\overline{\mathcal{M}}_{1,1}} \Omega_{1,1}(e_\beta).$$

Using the reconstruction theorem, we obtain

$$\begin{aligned}
\int_{\overline{\mathcal{M}}_{1,1}} \Omega_{1,1}(e_\beta) &= \int_{\overline{\mathcal{M}}_{1,1}} T_R\omega_{1,1}(R(\bar\psi)^{-1}e_\beta) + \frac{1}{2}T_R\omega_{0,3}(R(\bar\psi)^{-1}e_\beta, V(\bar\psi_2,\bar\psi_3))\\
&= \int_{\overline{\mathcal{M}}_{1,1}} T_R\omega_{1,1}(e_\beta) - \int_{\overline{\mathcal{M}}_{1,1}} T_R\omega_{1,1}(R_1 e_\beta)\bar\psi\\
&\quad + \frac{1}{2}\sum_\mu \omega_{0,3}(e_\beta, R_1 e^\mu, e_\mu)\\
&= \int_{\overline{\mathcal{M}}_{1,1}} (\omega_{1,1}(e_\beta) + p_*(\omega_{1,2}(e_\beta, R_1\mathbf{1})\bar\psi_2^2))\\
&\quad - \int_{\overline{\mathcal{M}}_{1,1}} (\omega_{1,1}(R_1 e_\beta)\bar\psi_1 + p_*(\omega_{1,2}(R_1 e_\beta)\bar\psi_2^2)\bar\psi_1)\\
&\quad + \frac{1}{2}\sum_\mu \eta(e_\beta \star e_\mu, R_1 e^\mu)\\
&= \frac{1}{24}\sum_\mu (\omega_{0,4}(e_\mu, e^\mu, e_\beta, R_1\mathbf{1}) - \omega_{0,3}(e_\mu, e^\mu, R_1 e_\beta))\\
&\quad + \frac{1}{2}\eta(e_\beta, R_1 e^\beta)\\
&= \frac{1}{24}\sum_\mu\sum_\nu \omega_{0,3}(e_\mu, e^\mu, e_\nu)\omega_{0,3}(e^\nu, e_\beta, R_1\mathbf{1})\\
&\quad - \frac{1}{24}\sum_\mu \omega_{0,3}(e_\mu, e^\mu, R_1 e_\beta) + \frac{1}{2}(R_1)_{\beta\beta}\\
&= \frac{1}{24}\left(\eta(e^\beta, R_1\mathbf{1}) - \sum_\mu \eta(e^\mu, R_1 e_\beta)\right) + \frac{1}{2}(R_1)_{\beta\beta}.
\end{aligned}$$

Using the identities

$$\begin{aligned}\eta(R_1\mathbf{1}, e^\beta) - \sum_\mu \eta(R_1 e_\beta, e^\mu) &= \sum_\mu \Delta_\mu \int_{\overline{\mathcal{M}}_{0,4}} \Omega_{0,4}(e_\mu, e_\mu, e_\mu, e_\beta) \\ &= -\frac{1}{2}\sum_\mu \Delta_\mu\, \partial_{u_\beta} \Delta_\mu^{-1} \\ &= \frac{1}{2}\sum_\mu \partial_{u_\beta} \log \Delta_\mu,\end{aligned}$$

where $\Delta_\mu \coloneqq \eta(e_\mu, e_\mu)^{-1}$, we obtain the result

$$\int_{\overline{\mathcal{M}}_{1,1}} \Omega_{1,1} e_\beta = \frac{1}{48}\sum_\mu \partial_{u_\mu} \log \Delta_\mu + \frac{1}{2}(R_1)_{\beta\beta},$$

which can be placed in the suggestive form

$$dF_1^\Omega = \sum_\mu \frac{d\log \Delta_\mu}{48} + \frac{1}{2}(R_1)_{\mu\mu}\, du_\alpha.$$

This recovers a result which was already known to Givental.

3.7. **How to compute the R-matrix.** We will conclude by explaining how to compute the R-matrix, starting in the setting where the Frobenius manifold is conformal (for example the Gromov-Witten theory of a smooth projective variety, r-spin curves, etc). In this example, we will consider the Frobenius manifold for 3-spin curves. Let V be 2-dimensional with coordinates x and y, where x corresponds to the unit. The metric is given by

$$\eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

and the primary genus zero potential is $F(x,y) = \frac{1}{2}x^2y + \frac{1}{72}y^4$. There is an *Euler vector field*

$$E = x\frac{\partial}{\partial x} + \frac{2}{3}y\frac{\partial}{\partial y},$$

which acts with weights -1 and $-\frac{2}{3}$ on ∂_x and ∂_y, respectively. The conformal dimension is $\delta = \frac{1}{3}$, and we will define the degree operator

$$\mu(v) = [E, v] + \left(1 - \frac{\delta}{2}\right)v.$$

Here, ∂_x has degree $-\frac{1}{6}$ and ∂_y has degree $\frac{1}{6}$.

To make our lives easier, we will set $\phi = \frac{y}{3}$ and define

$$\hat{\partial}_x = \phi^{1/4}\,\partial_x, \qquad \hat{\partial}_y = \phi^{-1/4}\,\partial_y,$$

and then compute in this frame. In this setting, quantum multiplication is given by

$$\hat{\partial}_x \star \hat{\partial}_x = \phi^{1/4}\hat{\partial}_x, \qquad \hat{\partial}_x \star \hat{\partial}_y = \phi^{1/4}\hat{\partial}_y, \qquad \hat{\partial}_y \star \hat{\partial}_y = \phi^{1/4}\hat{\partial}_x.$$

The last thing we need is that quantum multiplication by E is given by

$$E \star - = \begin{pmatrix} x & 2\phi^{3/2} \\ 2\phi^{3/2} & x \end{pmatrix}.$$

We will now compute the R-matrix $R = 1 + R_1 z + \cdots$ by using the recursive equation

$$[R_{k+1}, E \star -] = (k + \mu) R_k.$$

Setting $R_k = \left(\begin{smallmatrix} a_k & b_k \\ c_j & d_k \end{smallmatrix}\right)$, we obtain the equation

$$\left[\begin{pmatrix} a_{k+1} & b_{k+1} \\ c_{k+1} & d_{k+1} \end{pmatrix}, \begin{pmatrix} x & 2\phi^{3/2} \\ 2\phi^{3/2} & x \end{pmatrix}\right] = \frac{1}{6}\begin{pmatrix} 6m-1 & \\ & 6m+1 \end{pmatrix}\begin{pmatrix} a_k & b_k \\ c_k & d_k \end{pmatrix}.$$

The unique solution of this is in fact

$$\begin{aligned} a_k &= \frac{1}{1728^k \phi^{3k/2}} \frac{1+6k}{1-6k} \frac{(6k)!}{(3k)!(2k)!} \delta_{k,\mathrm{even}}, \\ b_k &= \frac{1}{1728^k \phi^{3k/2}} \frac{1+6k}{1-6k} \frac{(6k)!}{(3k)!(2k)!} \delta_{k,\mathrm{odd}}, \\ c_k &= \frac{1}{1728^k \phi^{3k/2}} \frac{(6k)!}{(3k)!(2k)!} \delta_{k,\mathrm{odd}}, \\ d_k &= \frac{1}{1728^k \phi^{3k/2}} \frac{(6k)!}{(3k)!(2k)!} \delta_{k,\mathrm{even}}. \end{aligned}$$

If you are familiar with tautological relations on the moduli space of curves, you will see that the series $a_k + b_k$ and $c_k + d_k$ enter in the formulation of the Faber-Zagier relations and the formulation of Pixton's relations.

In the case where the Frobenius manifold is not conformal (for example in equivariant Gromov-Witten theory), we will need to use an oscillating integral or some other technique to fix initial values in the R-matrix. We will give an example computation here, which is due to Givental. In this setting, $X = \mathbb{P}^n$, but the argument can be made to work for any Fano toric variety. Recall that the equivariant mirror of $\mathbb{P}^n$ is the superpotential

$$W = x_1 + \cdots + x_{n+1} + \sum_{i=1}^{n+1} \lambda_i \log x_i$$

(here, we change the sign of the equivariant parameters for convenience). Therefore, the oscillating integrals will look like

$$I = \int_{\Gamma \subset E_q = (x_1 \cdots x_{n+1} = q)} e^{W/z} \prod_{i=1}^{n+1} \frac{dx_i}{x_i}.$$

This is annihilated by the operator

$$\prod_{i=1}^{n+1} \left(zq \frac{d}{dq} - \lambda_i \right) - q,$$

which is exactly the Picard-Fuchs equation satisfied by the I-function of projective space.

To expand this integral near the critical points, we first need to compute the critical points. Here, using coordinates $t = \log q$ and $T_i = \log x_i$, we obtain the equation

$$x_i = L - \lambda_i,$$

where L is the Lagrange multiplier. These satisfy the equation

$$\prod_{i=1}^{n+1}(L-\lambda_i)=q,$$

and therefore for generic values of λ, near $q=0$, the possible values L_i of the Lagrange multiplier are distinguished by which of the λ_i they begin with. Note that the function W, the values of the Lagrange multipliers, and therefore the critical points, all depend continuously on λ, so they admit non-equivariant limits at $\lambda=0$ when $q\neq 0$.

When $q=0$, the matrix Ψ, transitions from the flat basis H^i to the basis

$$L_i(H)=\prod_j \frac{H-\lambda_j}{(\lambda_i-\lambda_j)^{1/2}},$$

which are the leading terms of the canonical coordinates (simply compute the fixed points of the torus action via equivariant localization, and we will obtain this Lagrange interpolation). Therefore, we need to compute $L_j\left(zq\frac{d}{dq}\right)I$ in the $q=0$ limit. A direct computation shows that applying $L_i\left(zq\frac{d}{dq}\right)$ to the integral introduces an additional factor proportional to $\frac{q}{x_i}$ to the integrand of the oscillating integral. Because we can permute the indices, we will compute only the case where $i=n+1$. Another direct computation shows that the integral vanishes at $q=0$ unless $j=n+1$, so in fact we see that R is diagonal in the classical limit. When $j=n+1$, we have

$$\frac{q}{x_{n+1}}=x_1\cdots x_n,$$

and the integral becomes

$$\int_\Gamma e^{\left(\sum_{i=1}^n x_i+(\lambda_i-\lambda_{n+1})\log x_i\right)/z}\,dx_1\cdots dx_n=\prod_{i=1}^n\int_{\mathbb{R}} e^{x_i/z}x_i^{(\lambda_i-\lambda_0)/z}\,dx_i,$$

which is simply a product of gamma functions.

Lemma 3.29. *We have the asymptotics*

$$\log\Gamma(1+s)\sim s\log\left(\frac{s}{e}\right)+\frac{1}{2}\log(2\pi s)+\sum_{k=1}^\infty \frac{B_{2k}}{2k}\frac{s^{1-2k}}{2k-1}.$$

Using this, we see that at $q=0$, the diagonal matrix element of the R-matrix is given by

$$\log R=\sum_{k=1}^\infty\frac{B_{2k}}{2k}\frac{z^{2k-1}}{2k-1}\sum_{i=1}^n(\lambda_i-\lambda_{n+1})^{1-2k}.$$

By a result of Givental, this implies that this R-matrix recovers the equivariant GW theory of $\mathbb{P}^n$, and as a corollary, because the non-equivariant limit exists, we obtain Virasoro constraints for $\mathbb{P}^n$.

4. Decomposition theorem for semi-small maps (Xueqing Wen)

Question 4.1. *Let X be a (manifold, variety, analytic space, etc). How can we compute $H^*(X, \mathbb{C})$?*

Unfortunately, this question is too hard, so we will replace it with a new question.

Question 4.2. *How can we detect $H^*(X, \mathbb{C})$ using smaller spaces?*

Theorem 4.3 (Künneth formula)**.** *We have $H^*(X, \times Y, \mathbb{C}) = H^*(X, \mathbb{C}) \otimes H^*(Y, \mathbb{C})$.*

Unfortunately, most spaces are not products, so we consider the following question.

Question 4.4. *Let $f: X \to Y$ be a morphism. Can we detect the cohomology of X using the cohomology of Y and the cohomology of the fibers of f?*

This question is still very difficult, but at least we can study it. We will begin with the case where f is a fiber bundle between compact manifolds. The answer is of course the Leray or Serre spectral sequences, but before we get there, we need some preliminaries.

Definition 4.5. A $\underline{\mathbb{C}}_Y$-local system on Y is a locally constant sheaf $\mathcal{L}$ of vector spaces on Y.

For example, if Y is connected, then local systems correspond to representations of $\pi_1(Y)$. Also, note that if Y is an algebraic variety, we will need to work with its analytification.

Example 4.6. Let f be a fiber bundle with fiber F as above. Then $R^i f_* \underline{\mathbb{C}}_X$ is a local system on Y with fiber $H^i(F, \mathbb{C})$.

Proposition 4.7. *We have*

$$H^*(X, \mathbb{C}) = H^*(X, \underline{\mathbb{C}}_X) = H^*(\mathrm{pt}, Ra_{X*}\underline{\mathbb{C}}_X).$$

Theorem 4.8 (Leray/Serre spectral sequence)**.** *We have a spectral sequence*

$$E_2^{p,q} = H^p(Y, R^q f_* \underline{\mathbb{C}}_X) \Rightarrow H^{p+q}(X, \mathbb{C}).$$

We will now consider the setting where $f: X \to Y$ is a smooth projective map between smooth projective varieties.

Theorem 4.9 (Deligne)**.** *We have the non-canonical isomorphism*

$$Rf_* \underline{\mathbb{C}}_X = \bigoplus_{q \geqslant 0} R^q f_* \underline{\mathbb{C}}[-q].$$

At the level of cohomology, we have

$$H^*(X, \mathbb{C}) = \bigoplus_{q \geqslant 0} H^{*-q}(Y, R^q f_* \underline{\mathbb{C}}).$$

This is a consequence of the hard Lefschetz theorem for projective varieties and also states that the Serre spectral sequence degenerates at the E_2 page. In topology, this fails for the Hopf fibration.

We will now consider the setting where $f\colon X \to Y$ is a proper map between algebraic varieties. Then we have a *Whitney stratification*[6] $Y = \bigsqcup_\lambda Y_\lambda$ such that $f^{-1}(Y_\lambda) \to Y_\lambda$ is a C^∞ fibration in possibly singular varieties.

Definition 4.10. A $\underline{\mathbb{C}}_Y$-module $\mathcal{F}$ on Y is called a *constructible sheaf* if there exists a Whitney stratification $Y = \bigsqcup Y_\lambda$ such that $\mathfrak{F}_{|Y_\lambda}$ is a local system for all λ.

With this definition, we may consider the derived category $\mathcal{D}^b_c(Y)$ of constructible sheaves on Y. This assignment admits a six functor formalism, where the functors are $f_*, f^*, f_!, f^!, \otimes, \mathcal{H}om$, and satisfies Verdier duality via the functor

$$D_Y\colon \mathcal{D}^b_c(Y) \to \mathcal{D}^b_c(Y).$$

Definition 4.11. A complex $\mathfrak{C}^\bullet \in \mathcal{D}^b_c(Y)$ is a *perverse sheaf* if it satisfies the following conditions:

(1) (Support condition) We have $\dim \operatorname{supp} \mathcal{H}^{-i}(\mathcal{F}) \leqslant i$ for all i;
(2) (cosupport condition) We have $\dim \operatorname{supp} \mathcal{H}^{-i}(D_Y \mathcal{F}^\bullet) \leqslant i$ for all i.

Remarks 4.12.

(1) Perverse sheaves are not sheaves, but are complexes in degrees $[-\dim X, 0]$.
(2) They have two origins. One is intersection homology, and the other is the Riemann-Hilbert correspondence.

We will need to use the following fact: the category $\mathsf{Perv}(Y)$ of perverse sheaves on Y is an abelian category and is moreover the heart of t-structure on $\mathcal{D}^b_c(Y)$. In particular, for any $\mathcal{G} \in \mathcal{D}^b_c(Y)$, we can consider the perverse cohomology ${}^p\mathcal{H}^i(\mathcal{G}) \in \mathsf{Perv}(Y)$.

Definition 4.13 (Beilinson-Bernstein-Deligne-Gabber)**.** Let $f\colon X \to Y$ be a proper morphism between algebraic varieties. Assume that X is smooth and connected.[7] Then

$$Rf_* \underline{\mathbb{C}}_X[\dim X] = \bigoplus {}^p\mathcal{H}^i(Rf_*\underline{\mathbb{C}}_X)[-i].$$

Moreover, each ${}^p\mathcal{H}^i(Rf^*\underline{\mathbb{C}}_X)$ is semi-simple (meaning each subobject is a direct summand).

Definition 4.14. Let $f\colon X \to Y$ be a proper surjective morphism with X smooth and irreducible. Let $Y = \bigsqcup Y_\lambda$ be a stratification such that $f^{-1}(Y_\lambda) \to Y_\lambda$ is a C^∞-fibration. We say that f is *semi-small* if for all $y_\lambda \in Y_\lambda$, we have $\operatorname{codim}_Y Y_\lambda \geqslant 2\dim f^{-1}(y_\lambda)$. If equality holds, we say that Y_λ is *relevant*. If the only relevant stratum is the dense one, then we say that f is *small*.

Theorem 4.15. *If* $f\colon X \to Y$ *is semi-small, then* $Rf_*\underline{\mathbb{C}}_X[\dim X]$ *is a perverse sheaf on* Y.

Proof. For $y_\lambda \in Y_\lambda \subseteq Y$, we see that

$$Rf_*\underline{\mathbb{C}}_X|_{y_\lambda} = \bigoplus H^i(f^{-1}(y_\lambda), \mathbb{C}).$$

Note that this vanishes if $i > 2\dim f^{-1}(y_\lambda) \leqslant \operatorname{codim}_Y(Y_\lambda)$. If we vary over all strata Y_λ, we obtain the support condition. Because f is proper, we have $Rf_* = f_!$, and by using $D_Y Rf_* = f_! D_X$, we obtain the cosupport condition. ☐

[6] We explicitly refuse to define this.

[7] This is not necessary, but if we remove the smoothness assumption we need to consider the intersection complex.

Examples 4.16. We now give some examples of semi-small maps.

(1) Any generically finite proper map between surfaces are semi-small.
(2) Let $Y = \{xy = t^{\ell+1}\} \subseteq \mathbb{C}^3$. Then there is a crepant resolution $f\colon \tilde{Y} \to Y$ of singularities, and the fiber over the singular point is a chain of $\mathbb{P}^1$s, so it is semi-small (see also the previous example).

Theorem 4.17 (Kaledin)**.** *A projective birational map from a nonsingular holomorphic symplectic variety is semi-small.*

Example 4.18. For example, the Springer resolution and the Grothendieck-Springer resolution are semi-small. In addition, the Hilbert-Chow morphism for the Hilbert scheme of points on a surface is semi-small.

Let $f\colon X \to Y$ be semi-small and consider a stratification $Y = \bigsqcup Y_\lambda$. If Y_λ is relevant, consider

$$\mathcal{L}_\lambda = R^{d_\lambda} f_* \underline{\mathbb{C}}_{f^{-1}(Y_\lambda)},$$

where $d_\lambda = 2\dim f^{-1}(y_\lambda)$. Then $\mathcal{L}_\lambda$ is semi-simple. Here, note that $\mathcal{L}_{|\lambda} = H^{d_\lambda}(f^{-1}(y_\lambda), \mathbb{C})$. Now let $\mathrm{IC}(\mathcal{L}_\lambda)$ be the intermediate extension of $\mathcal{L}_\lambda[\dim Y_\lambda]$ on $\overline{Y}_\lambda$, which is a perverse sheaf.

Theorem 4.19. *Let $f\colon X \to Y = \bigsqcup Y_\lambda$ be semi-small. Then there is a canonical decomposition*

$$Rf_* \underline{\mathbb{C}}_X[\dim X] = \bigoplus_{Y_\lambda \text{ relevant}} \mathrm{IC}(\mathcal{L}_\lambda).$$

Example 4.20. Let G be a reductive group with Lie algebra $\mathfrak{g}$ and choose a Borel B. Then

$$G \times_B \mathfrak{b} = \tilde{\mathfrak{g}} \xrightarrow{\pi} \mathfrak{g}$$

is small. Therefore, we have

$$R\pi_* \underline{\mathbb{C}}_{\tilde{\mathfrak{g}}}[\dim] = \mathrm{IC}(\mathcal{L}).$$

Recall that $\tilde{\mathfrak{g}}^{rs} \to \mathfrak{g}^{rs}$ is a W-torsor. Then for $a \in \mathfrak{g}^{rs}$, we have $\mathcal{L}_{|a} \cong \mathbb{C}[W]$, and therefore from $\mathcal{L} = R^0\pi_* \underline{\mathbb{C}}_{\mathfrak{g}^{rs}}$. This implies that W acts on $\mathcal{L}$, and this action can be extended to $\mathrm{IC}(\mathcal{L})$ and also on $R\pi_*\mathbb{C}_{\tilde{\mathfrak{g}}}$, and in particular acts on the fiber $R\pi_*\mathbb{C}_{\tilde{\mathfrak{g}}}|_a$ for any $a \in \mathfrak{g}$.

Example 4.21. Consider the Springer resolution $\mu\colon T^*\mathrm{GL}_3/G \to \mathcal{N} = \mathbb{O}_{(1^3)} \sqcup \mathbb{O}_{(2,1)} \sqcup \mathbb{O}_{(3)}$. This map is semi-small, but unfortunately all nilpotent orbits are relevant. Note that the fiber above the $(2,1)$ orbit is a union of two $\mathbb{P}^1$s glued at a node, the fiber above the (3) orbit is a point, and the fiber above the (1^3) orbit is GL_3/B.

We see that $\mathcal{L}_{(3)}$ must be trivial because μ is an isomorphism above this orbit. Then $\mathcal{L}_{(2,1)} = R^2f_*(X, \mathbb{C})$ is a rank 2 local system on $\mathbb{O}_{(2,1)}$. Finally, $\mathcal{L}_{(1^3)} = R^6 f_*(\mathrm{GL}_3/B, \mathbb{C})$ is a rank 1 local system on $\mathbb{O}_{(1^3)} = \mathrm{pt}$. Therefore, we have

$$R\mu_* \underline{\mathbb{C}}_{T^*\mathrm{GL}_3/B}[6] = \mathrm{IC}(\mathcal{L}_{(3)}) \oplus \mathrm{IC}(\mathcal{L}_{(2,1)}) \oplus \mathrm{IC}(\mathcal{L}_{(1^3)}).$$

Restricting to $b \in \mathbb{O}_{(3)}$, we obtain $\mathcal{L}_{(3)} = H^0(\mathrm{pt})$. Restricting to $a \in \mathbb{O}_{(2,1)}$, we obtain

$$H^0(X, \mathbb{C}) \oplus H^2(X, \mathbb{C}) = \mathrm{IC}(\mathcal{L}_{(3)})|_{(a)} \oplus \mathrm{IC}(\mathcal{L}_{(2,1)})|_{(a)}.$$

Restricting to $0 = \mathbb{O}_{(1^3)}$, we obtain

$$\begin{aligned} H^*(GL_3/B, \mathbb{C}) = \mathbb{C}[W] &= IC(\mathcal{L}_{(3)})|_0 \oplus IC(\mathcal{L}_{(2,1)})|_0 \oplus \mathcal{L}_{(1^3)}|_0 \\ &= H^0(GL_3/B, \mathbb{C}) \oplus \text{has dimension } 4 \oplus H^6(GL_3/B, \mathbb{C}). \end{aligned}$$

As representations of $W = S_3$, these correspond to the trivial representation, two copies of the standard representation, and the sign representation.

5. Quantum K-theory (Xiaohan Yan and You-Cheng Chou)

5.1. **Overview.** Gromov-Witten invariants are some way of "counting" curves in a smooth projective variety X. GW theory produces numbers of the form

$$\langle \alpha_1, \dots, \alpha_n \rangle_{g,n,\beta} \in \mathbb{Q}$$

for various classes $\alpha_i \in H^*(X)$, genus g, number n of marked points, and curve class $\beta \in H_2(X, \mathbb{Z})$. This has been used to study mirror symmetry, and outside of mirror symmetry has produced birational invariants. Our goal now is to lift this entire story to K-theory. Quantum K-theory produces numbers

$$\langle V_1, \dots, V_n \rangle_{g,n,\beta} \in \mathbb{Z}$$

for $V_1, \dots, V_n \in K(X)$. These satisfy various difference equations[8] and have found various applications in 3D mirror symmetry and representation theory.

5.2. **K-theory.**

Definition 5.1. Let X be a smooth projective variety. Then the *K-theory* $K(X)$ of X is the Grothendieck group of vector bundles on X with the relation that if

$$0 \to V_1 \to V_2 \to V_3 \to 0$$

is a short exact sequence, then $V_2 = V_1 + V_3$.

Remark 5.2.

(1) $K(X)$ is a ring with the operations of direct sum and tensor product;
(2) The Chern character gives an isomorphism $K(X) \to A^*(X)$, at least working over $\mathbb{Q}$.

Example 5.3. Let $X = \mathbb{P}^1$ and set $P = \mathcal{O}(-1)$. Then we have

$$K(X) = \mathbb{C}[P^{\pm 1}]/(P-1)^{n+1} \xrightarrow{\text{ch}} A^*(X) = \mathbb{C}[H]/H^{n+1}.$$

For a Deligne-Mumford stack which is not assumed to be smooth, then there are two versions

$$K_0(X) \supset K^0(X),$$

where K_0 means coherent sheaves and K^0 means perfect complexes. These coincide on a smooth projective variety, so from now on $K(X)$ will mean $K_0(X)$. There is a tensor product

$$K^0(X) \otimes K_0(X) \to K_0(X)$$

and a pushforward

$$f_* \colon K(X) \to K(Y) \qquad V \mapsto \sum (-1)^i R^i f_* V$$

[8] Compare to differential equations in cohomology.

along proper morphisms. If f is a perfect morphism (for example flat, regular embedding, etc), then there is a pullback

$$f^*\colon K(Y) \to K(X) \qquad W \mapsto \sum (-1)^i \operatorname{Tor}_i^{f^{-1}\mathcal{O}_Y}(\mathcal{O}_X, f^{-1}W).$$

5.3. **Moduli of curves.** Recall the moduli space

$$\bar{M}_{g,n}(X,\beta) = \{f\colon (\Sigma, p_1, \dots, p_n) \to X \mid g(\Sigma) = g, f_*[\Sigma] = \beta, |\mathrm{Aut}\, f| < \infty\}$$

of stable maps to X. This has evaluation maps

$$\mathrm{ev}_i\colon \bar{M}_{g,n}(X,\beta) \to X \qquad (f, \Sigma, \underline{p}) \mapsto f(p_i).$$

There are also the universal cotangent lines $L_i = \sigma_i^* \omega_{\mathcal{C}/\bar{M}}$.

The moduli space is a proper DM stack which has a perfect obstruction theory, which produces a virtual structure sheaf $\mathcal{O}^{\mathrm{vir}}_{g,n,\beta}$. This comes from the structure of a *virtual tangent bundle*, where

$$\mathbb{T}_{(f,\Sigma,\underline{p})}\bar{M}_{g,n}(X,\beta) = \chi(\Sigma, f^*T_X) - \chi(\Sigma, T_\Sigma(-p_1 - \cdots - p_n)).$$

Roughly, if $\bar{M}$ is the zero locus of a section of a vector bundle E on a variety Y, then $\mathcal{O}^{\mathrm{vir}} = \mathcal{O}_Y \otimes \mathrm{Eu}(E)$, where the K-theoretic Euler class is $\mathrm{Eu}(E) = \sum (-1)^i \Lambda^i E^\vee$.

5.4. **KGW invariants.**

Definition 5.4. K-theoretic GW invariants are defined by

$$\langle V_1, \dots, V_n \rangle_{g,n,\beta} \coloneqq \chi\left(\bar{M}_{g,n}(X,\beta); \mathcal{O}^{\mathrm{vir}} \otimes \bigotimes_{i=1}^n \mathrm{ev}_i^* V_i\right).$$

If $V_1, \dots, V_n$ are actual vector bundles, then the invariant is an integer, but generally we will consider K_0 with coefficients in a field of characteristic zero. We may also define gravitational descendents

$$\left\langle V_1 L_1^{k_1}, \dots, V_n L_n^{k_n} \right\rangle_{g,n,\beta} \coloneqq \chi\left(\bar{M}_{g,n}(X,\beta); \mathcal{O}^{\mathrm{vir}} \otimes \bigotimes_{i=1}^n (\mathrm{ev}_i^* V_i) L_i^{k_i}\right).$$

Using the action of S_n on $\bar{M}_{g,n}(X,\beta)$ given by permuting the marked points, then the objects $\bigotimes_{i=1}^n \mathrm{ev}_i^* V$ and $\bigotimes_{i=1}^n L_i^k$ are S_n-equivariant.

Definition 5.5. We define *permutation-invariant K-theoretic GW invariants* by

$$\begin{aligned}\left\langle V L_1^k, \dots, V L_n^k \right\rangle_{g,n,\beta}^{S_n} &\coloneqq \chi\left([\bar{M}_{g,n}(X,\beta)/S_n]; \mathcal{O}^{\mathrm{vir}} \otimes \bigotimes_{i=1}^n (\mathrm{ev}_i^* V) L_i^k\right) \\ &= \dim \chi_{S_n}\left(\bar{M}_{g,n}(X,\beta); \mathcal{O}^{\mathrm{vir}} \otimes \bigotimes_{i=1}^n (\mathrm{ev}_i^* V) L_i^k\right)^{S_n}.\end{aligned}$$

Given $t(q) = \sum_{n\in\mathbb{Z}} t_n q^n \in K(X)[q, q^{-1}]$, we can define

$$\langle t(L_1), \dots, t(L_n) \rangle_{g,n,\beta}^{S_n} \coloneqq \chi([\bar{M}_{g,n}(X,\beta)/S_n], \cdots).$$

More generally, if $H = S_{a_1} \times \cdots \times S_{a_m} \subset S_n$, we can define

$$\left\langle t_1(L_1), \dots, t_1(L_{a_1}), t_2(L_{a_1+1}), \dots, t_2(L_{a_1+a_2}), \dots \right\rangle_{g,n,\beta}^{H}.$$

Remark 5.6. We would like to discuss some relation to ordinary GW invariants. Recall that Hirzebruch-Riemann-Roch states that

$$\chi(M;V) = \int_M \mathrm{ch}(V)\,\mathrm{td}(T_M).$$

In the case where M is a smooth DM stack, Töen proved the formula

$$\chi(M;V) = \int_{IM} \widetilde{\mathrm{ch}}(V)\widetilde{\mathrm{td}}(T_M),$$

where $IM = M \times_{\Delta, M\times M, \Delta} M$ is the inertia stack of M and the tildes mean taking the orbifold Chern character and Todd class.

Example 5.7. If $M = B\mu_r$, then $IM = \bigsqcup_{g\in\mu_r} B\mu_r$. If $M = \mathbb{P}(1,r)$ is a weighted $\mathbb{P}^1$, then $IM = M \sqcup \sum_{g\neq 1\in\mu_r} B\mu_r$.

Example 5.8. Let $M = BG$ for a finite abelian group G and V be a vector bundle on BG, which is the same thing as a representation of G. Then $\chi(M,V) = \dim V^G$. On the other hand, we have

$$\int_{IM} \widetilde{\mathrm{ch}}(V)\cdot\widetilde{\mathrm{td}}(T_M) = \int_{IM} \mathrm{tr}_g V = \sum_{g\in G} \frac{1}{|G|}\mathrm{tr}_g V = \dim V^G,$$

so we have verified Riemann-Roch in this setting.

5.5. **Axiomatic properties.**

Lemma 5.9. *We have the string equation*

$$\begin{aligned}\langle 1, x_1(L), \ldots, x_k(L), t, \ldots, t\rangle^{S_n}_{g,1+k+n,\beta} &= \left\langle x_1(L), \ldots, x_k(L), t^{\otimes n}\right\rangle^{S_n}_{g,k+n,\beta} \\ &\quad + \sum_i \left\langle \cdots, \frac{x_i(L)-x_i(1)}{L-1}, \ldots, t^{\otimes n}\right\rangle^{S_n}_{g,k+n,\beta}.\end{aligned}$$

Proof. The proof is the same as in GW theory and comes from computing π^*L_i along the map $\pi\colon \bar{M}_{g,n+1}(X,\beta) \to \bar{M}_{g,n}(X,\beta)$. □

Lemma 5.10. *We have the dilaton equation*

$$\begin{aligned}\left\langle L-1, x_1(L), \ldots, x_k(L), t^{\otimes n}\right\rangle^{S_n}_{0,1+k+n,\beta} &= (k-2)\left\langle x_1(L), \ldots, x_k(L), t^{\otimes n}\right\rangle^{S_n}_{g,k+n,\beta} \\ &\quad + \left\langle x_1(L), \ldots, x_k(L), t, t^{\otimes n-1}\right\rangle^{S_{n-1}}_{0,k+n,\beta}.\end{aligned}$$

Proof. We use the fact that

$$\pi_* L = k + \mathrm{Ind}^{S_n}_{S_{n-1}}(1) - 1,$$

where π forgets the first marked point and $L = \omega_\pi(p_1 + \cdots + p_n)$. This follows from the fact that there is a residue map

$$H^0(L) \xrightarrow{\mathrm{Res}} \bigoplus_{i=1}^{k+n} \mathbb{C}p_i \xrightarrow{\sum} \mathbb{C}.$$

The middle term is a direct sum of k trivial representations and one standard representation. □

Notation 5.11. Fix $\tau, t \in K(X)$. We will denote

$$\langle\langle x_1(L_1), \ldots, x_k(L_k)\rangle\rangle_{g,k} \coloneqq \sum_{m,n,\beta} \frac{Q^\beta}{m!} \langle x_1(L_1), \ldots, x_k(L_k), \tau^{\otimes m}, t^{\otimes n}\rangle^{S_n}_{g,k+m+n,\beta}.$$

We will let $\{\phi_\alpha\}$ be a basis of $K(X)$ and $\{\phi^\alpha\}$ be the dual basis. Then define the pairing

$$g_{\alpha\beta} = \chi(X, \phi_\alpha \otimes \phi_\beta).$$

This is not quite the right pairing, so we will consider

$$G_{\alpha\beta} = g_{\alpha\beta} + \langle\langle \alpha, \beta\rangle\rangle_{0,2}.$$

We will let $g^{\alpha\beta} = (g_{\alpha\beta})^{-1}$ and $G^{\alpha\beta} = (G_{\alpha\beta})^{-1}$.

Lemma 5.12. *Given* $A, B, C, D \in K(X)[q, q^{-1}]$, *we have the WDVV equation*

$$\sum_{\alpha,\beta} \langle\langle A(L), B(L), \phi_\alpha\rangle\rangle_{0,3} G^{\alpha\beta} \Big\langle\Big\langle C(L), D(L), \phi_\beta \Big\rangle\Big\rangle_{0,3}$$
$$= \sum_{\alpha,\beta} \langle\langle A(L), C(L), \phi_\alpha\rangle\rangle_{0,3} G^{\alpha\beta} \Big\langle\Big\langle B(L), D(L), \phi_\beta \Big\rangle\Big\rangle_{0,3}.$$

Proof. Pull back the equality of the boundary divisors on $\bar{M}_{0,4} = \mathbb{P}^1$. In K-theory, we need to apply the inclusion-exclusion principle when relating $\mathcal{O}_D$ to $\mathcal{O}_{C_i}$ when $D = \bigcup C_i$, which accounts for the appearance of $G^{\alpha\beta}$. □

5.6. **Loop space formalism.**

Definition 5.13. The *loop space* $(\mathcal{K}, \Omega)$ consists of

$$\mathcal{K} = K(X)(q)[\![Q]\!]$$

with the symplectic form

$$\Omega(f, g) = -\operatorname{Res}_{q=0,\infty} \Big\langle f(q^{-1}), g(q) \Big\rangle_X \frac{dq}{q}.$$

Remark 5.14. There is a polarization $\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-$, where

$$\mathcal{K}_+ = K(X)[q, q^{-1}][\![Q]\!]$$

and

$$\mathcal{K}_- = \left\{ \frac{f}{g} \mid f, g \in K(X)[\![Q]\!][q], g(0) \neq 0, \deg f < \deg g \right\}.$$

For example, $\frac{1}{1-q} \in \mathcal{K}_-$.

Definition 5.15. The *big J-function* $J \colon \mathcal{K}_+ \to \mathcal{K}$ is given by

$$\mathbf{t} = \mathbf{t}(q) \mapsto 1 - q + \mathbf{t}(q) + \sum_{\beta,\alpha} Q^\beta \phi^\alpha \left\langle \frac{\phi_\alpha}{1 - qL}, \mathbf{t}(L)^{\otimes n} \right\rangle^{S_n}_{0,1+n,\beta}.$$

Remark 5.16. Note that applying Riemann-Roch, we have

$$\chi\left([\bar{M}/S_n], \frac{1}{1 - qL}\right) = \int_{I[\bar{M}/S_n]} \operatorname{ch}\left(\frac{1}{1 - qL}\right) \operatorname{td}.$$

Each term becomes of the form $\mathrm{ch}\left(\mathrm{tr}_g \frac{1}{1-qL}\right)$, which gives us something of the form $\mathrm{ch}\left(\frac{1}{1-q\zeta L}\right)$ for some root of unity ζ. Then we can expand

$$\frac{1}{1-q\zeta L} = \frac{1}{(1-q\zeta)-q\zeta(L-1)} = \sum_j \frac{(q\zeta)^k}{(1-q\zeta)^{k-1}}(L-1)^k.$$

Definition 5.17. The *Givental cone* is defined to be $\mathcal{L} \coloneqq \operatorname{Im} J \subset \mathcal{K}$, where we are considering the germ near $1-q$ in the Q-adic topology.

Example 5.18. Let $X = \mathbb{P}^n$. Then we see that

$$J(0) = (1-q)\sum_{d\geqslant 0} \frac{Q^d}{\prod_{\ell=1}^d (1-Pq^\ell)^{n+1}},$$

where $P = \mathcal{O}(-1)$. As before, we can expand

$$\frac{1}{1-Pq^\ell} = \frac{1}{(1-q^\ell)-q^\ell(P-1)} = \sum_{k=0}^\infty \frac{q^{\ell k}}{(1-q^\ell)^{k+1}}(P-1)^k.$$

Proposition 5.19. *In fact, $\mathcal{L}$ is actually a cone in $\mathcal{K}$ and we have*

$$\mathcal{L} = \bigcup_{t\in K(X)[\![Q]\!]} (1-q)S_t^{-1}(q)\mathcal{K}_+.$$

We may also introduce a "huge" J-function

$$J(\mathbf{t},\tau) = 1-q+\mathbf{t}(q)+\tau+\sum_{Q^\beta} k!\phi^\alpha\left\langle \frac{\phi_\alpha}{1-qL}, \mathbf{t}(L)^{\otimes n}, \tau^{\otimes k}\right\rangle_{0,1+n+k,\beta}^{S_n}.$$

If we fix $\mathbf{t}(q)$, then $J(\tau)$ comes from a fundamental solution of a *differential equation* on the Frobenius manifold in τ. In addition, $\mathcal{L}^{\mathbf{t}} = \operatorname{Im} J(\tau)$ is actually a Lagrangian cone.

Example 5.20. Consider $f\colon \mathbb{C}[\Lambda_1^\pm,\Lambda_2^\pm] \to \mathbb{C}[\Lambda_1^\pm,\Lambda_2^\pm]$ which is given by the Adams operation

$$t \mapsto \mathrm{tr}_{(1\cdots n)}\, t^{\otimes n} = \Psi^n(t).$$

Therefore, this sends $\Lambda_1 \mapsto \Lambda_1^n$ and $\Lambda_2 \mapsto \Lambda_2^n$. However, it is not clear how to take derivatives in t because we would need $n\nu t^{n-1} = \Psi^n(\nu)$, so the J-function does not satisfy a quantum differential equation in $\mathbf{t}$.

5.7. **Equivariant K-theory and localization.** Let a torus $T = (\mathbb{C}^\times)^r$ act on a smooth projective variety X.

Definition 5.21. The *equivariant K-theory* $K_T(X)$ is the Grothendieck group of T-equivariant vector bundles on X.

Example 5.22. For example, we have

$$K_T(\mathrm{pt}) = \mathrm{Rep}(T) = \mathbb{C}[\Lambda_1^\pm,\ldots,\Lambda_r^\pm],$$

where the Λ_i are the 1-dimensional representations with weight $(0,\ldots,1,\ldots,0)$.

Remark 5.23. For any X, $K_T(X)$ is a $K_T(\mathrm{pt})$-algebra.

Example 5.24. Consider the fully equivariant K-theory of $\mathbb{P}^n$. Then we have

$$K_T(\mathbb{P}^n) = \mathbb{C}[P^{\pm}, \Lambda_0^{\pm}, \dots, \Lambda_n^{\pm}] / \prod_{i=1}^{n} (P - \Lambda_i).$$

When we specialize $\Lambda_i = 1$, we recover the non-equivariant K-theory.

Definition 5.25. We can define the equivariant Euler characteristic as the pushforward

$$\chi_T = \pi_* \colon K_T(X) \to K_T(\mathrm{pt}).$$

Example 5.26. For example, we have $\chi_T(\mathcal{O}_{\mathbb{P}^1}(1)) = \Lambda_0^{-1} + \Lambda_1^{-1}$. Similarly, we have

$$\chi_T(\mathcal{O}(2)) = \Lambda_0^{-2} + \Lambda_0^{-1}\Lambda_1^{-1} + \Lambda_1^{-2}.$$

Notation 5.27. Note that $\mathrm{Frac}(T) = \mathrm{Frac}(\mathrm{Rep}(T))$. Then we will set

$$K_T^{\mathrm{loc}}(X) \coloneqq \mathrm{Frac}(T) \otimes_{\mathrm{Rep}(T)} K_T(X).$$

We will need the following fact. Let F be the fixed point set of T on X and consider $\iota\colon F \hookrightarrow X$. Then there is an isomorphism of $\mathrm{Frac}(T)$-algebras

$$K_T^{\mathrm{loc}}(X) \xrightarrow{\iota^*} K_T^{\mathrm{loc}}(F).$$

Remark 5.28. The inverse is given by

$$(\iota^*)^{-1} = \bigoplus_{i=1}^{r} \frac{\iota_{i*}}{\mathrm{Eu}(N_{F_i/X})},$$

where $\mathrm{Eu}(V) \coloneqq \sum_{j=0}^{\mathrm{rk}\, V} (-1)^j \Lambda^j V^*$ is the K-theoretic Euler class.

Example 5.29. On $\mathbb{P}^n$, we have $n+1$ isolated fixed point F_i, and then we obtain

$$\frac{\iota_{i*}\mathcal{O}_{F_i}}{\mathrm{Eu}(N_{F_i/\mathbb{P}^n})} = \prod_{j \neq i} \frac{P - \Lambda_j}{\Lambda_i - \Lambda_j}.$$

These are idempotents of $K_T^{\mathrm{loc}}(\mathbb{P}^n)$.

Using equivariant localization, we obtain

$$\chi_T(X; V) = \sum_i \chi\left(F_i; \frac{\iota_i^* V}{\mathrm{Eu}(N_{F_i/X})}\right).$$

Remark 5.30. In the $\Lambda = 1$ limit, we can recover the non-equivariant Euler characteristic.

Example 5.31. Let $X = \mathbb{P}^1$. Then we compute

$$\begin{aligned} \chi_T(\mathbb{P}^1, \mathcal{O}(1)) &= \chi\left(F_0, \frac{\mathcal{O}(1)|_{F_0}}{\mathrm{Eu}(T_{F_0/\mathbb{P}^1})}\right) + \chi\left(F_1, \frac{\mathcal{O}(1)|_{F_1}}{\mathrm{Eu}(T_{F_1/\mathbb{P}^1})}\right) \\ &= \frac{\Lambda_0^{-1}}{1 - \frac{\Lambda_0}{\Lambda_1}} + \frac{\Lambda_1^{-1}}{1 - \frac{\Lambda_1}{\Lambda_0}} \\ &= \Lambda_0^{-1} + \Lambda_1^{-1}. \end{aligned}$$

From now on, we will assume that X is a GKM manifold. Our goal now is to use localization on $\bar{M}_{g,n}(X,\beta)$ to study the big J-function of X. Any T-fixed stable map $f\colon (\Sigma, \underline{p}) \to X$ has two types of irreducible components. There are those components which map entirely to fixed points and those components who map via ramified covers of 1-dimensional orbits. Unfortunately, we cannot compute the big J-function via virtual localization, but we can still study it.

Let $f = f(q) \in \mathcal{K} = K_T^{\mathrm{loc}}(X)(q)[\![Q]\!]$. Denote the fixed points by F_α for $\alpha = 1, \dots, r$ with inclusions $\iota_\alpha \colon F_\alpha \hookrightarrow X$. Then denote

$$f_\alpha(q) = \iota_\alpha^* f(q) \in K_T^{\mathrm{loc}}(\mathrm{pt})(q)[\![Q]\!].$$

Then $f \in \mathcal{L}$ if and only if for all α, the following conditions hold:

(1) Given $\lambda \in T_{F_\alpha}X$ and $m \in \mathbb{N}_+$, $f_\alpha(q)$ has at most simple poles at $q = \lambda^{1/m}$, and

$$\operatorname{Res}_{q=\lambda^{1/m}} f_\alpha(q)\frac{dq}{q} = -\frac{Q^m}{m} \cdot \frac{\mathrm{Eu}(T_{F_\alpha/X})}{\mathrm{Eu}(T^{\mathrm{vir}}_{\phi_{\alpha,\alpha',m}} \bar{M}_{0,2}(X, m\cdot\beta(\alpha,\alpha')))} \cdot f_{\alpha'}(\lambda^{1/m}).$$

Here, $\lambda \in T_{F_\alpha}X$ means that λ is a 1-dimensinoal subrepresentation of $T_{F_\alpha}X$. Moreover, $\beta(\alpha,\alpha')$ is the degree of the 1-dimensional orbit of T determined by the weight λ and $\phi_{\alpha,\alpha',m}$ parameterizes $m:1$ covers of this 1-dimensional orbit.

(2) $f_\alpha(q)$ has its other poles at $0, \infty$, and roots of unity, and is regular elsewhere. In addition, there are various minor conditions.

Example 5.32. Let $X = \mathbb{P}^n$. Then

$$J_T^X(0) = (1-q)\sum_{d\geqslant 0} \frac{Q^d}{\prod_{i=0}^n \prod_{\ell=1}^d}\left(1 - \frac{P}{\Lambda_i}q^\ell\right) \in K_T(X)(q)[\![Q]\!].$$

This then implies that

$$J_T^X(0)|_{F_j} = (1-q)\sum_{d\geqslant 0} \frac{Q^d}{\prod_{i=0}^n \prod_{\ell=1}^d \left(1 - \frac{\Lambda_j}{\Lambda_i}q^\ell\right)},$$

so the poles are at $q = \left(\frac{\Lambda_i}{\Lambda_j}\right)^{1/\ell}$. Note also that

$$T_{F_j}X = \frac{\Lambda_0}{\Lambda_j} + \cdots + \frac{\Lambda_n}{\Lambda_j} - 1.$$

5.8. **Quantum K/GV correspondence.** Our goal here will be to compute $J^K(0)$ on a Calabi-Yau threefold. The following result was conjectured by Jockers-Mayr and by Garoufalidis-Scheidegger.

Theorem 5.33. *We have*

$$\begin{aligned} J^K(0) &:= (1-q) + \sum_{\beta,\alpha} Q^\beta \Phi^\alpha \left\langle \frac{\Phi_\alpha}{1-qL} \right\rangle_{0,1,\beta}^{X,K} \\ &= (1-q) + \sum_\beta \sum_{r=1}^\infty \mathrm{GV}_\beta Q^{r\beta}\left[\sum_{j=1}^{n_1} \Phi^{1j} a(r,q^r) + \Phi^{01} b(r,q^r)\right], \end{aligned}$$

where $\{\Phi_\alpha\}_{\alpha=1}^N$ *is a basis of* $K^0(X)$ *decomposed such that*

$$\{\Phi_\alpha\}_{\alpha=1}^N = \bigsqcup_{i=0}^{3} \{\Phi_{ij}\}_{j=1}^{n_i}$$

such that the collection $\mathrm{ch}(\Phi_{ij})$ *for fixed* i *form a basis of* $H^{2i}(X)$. *We then select a dual basis* $\{\Phi^\alpha\}_\alpha = \{\Phi^{ij}\}_{i,j}$. *Moveover, we set the multiple cover formulae*

$$a(r, q^r) = \frac{r-1}{1-q^r} + \frac{1}{(1-q^r)^2},$$
$$b(r, q^r) = \frac{r^2-1}{1-q^r} + \frac{3}{(1-q^r)^2} - \frac{2}{(1-q^r)^3}.$$

Finally, GV denotes Gopakumar-Vafa invariants, which are defined in genus zero by

$$\mathrm{GW}_{0,\beta} = \sum_{r|\beta} \frac{1}{r^3}\mathrm{GV}_{\beta/r}.$$

By Möbius inversion, we see that

$$\mathrm{GV}_\beta = \sum_{r|\beta} \frac{\mu(r)}{r^3}\mathrm{GW}_{0,\beta/r}.$$

Corollary 5.34. *We have*

$$\langle 1\rangle^K_{0,1,\beta} = \sum_{r|\beta} r^2 \mathrm{GV}_{\beta/r}.$$

Applying Möbius inversion, we obtain[9]

$$\mathrm{GV}_\beta = \sum_{r|\beta} r^2\mu(r)\langle 1\rangle^K_{0,1,\beta/r} \in \mathbb{Z}.$$

In addition, we can recover the usual GW invariants by[10]

$$\lim_{q\to 1}(1-q)^3\left\langle \frac{1}{1-qL}\right\rangle^{X,K}_{0,1,\beta} = \sum_{r|\beta}\frac{-2}{r^3}\mathrm{GV}_{\beta/r} = -2\mathrm{GW}_\beta = \langle\psi\rangle^{GW}_{0,1,\beta}.$$

Naively, we can think of $J^K(0) \in \mathbb{Z}[\![q]\!][\![Q]\!]$, but to be a bit more precise, we can consider

$$J^K(0) \in 1 - q + \mathcal{K}_-.$$

Here, we expand $\frac{1}{1-qL} = \sum q^i L^i$. We also need to consider the fact that if M is a DM stack and L is a line bundle on L, then the function

$$f^\Phi(q) \coloneqq \chi\left(M, \frac{\Phi}{1-qL}\right) \in \mathbb{Z}[\![q]\!]$$

converges to a rational function for $|q| < \varepsilon$. Note that there are poles only at roots of unity.

[9] Note that this result was obtained in all genus by Ionel-Parker.

[10] This is like if we apply Hirzebruch-Riemann-Roch to compute the K-theoretic pairing, the top degree term only sees the Chern characters of the two inputs.

This fact can be proved using the orbifold Riemann-Roch theorem. Decompose the inertia as

$$\begin{aligned} IM &= \bigsqcup_{i \in I} M_i \\ &= \bigsqcup_{\zeta} M_\zeta, \end{aligned}$$

where M_ζ is the union of twisted sectors such that g acts on L with eigenvalue ζ. Using this, we can compute

$$\begin{aligned} \chi\left(M, \frac{1}{1-qL}\right) &= \sum_{i \in I} \int_{M_i} \mathrm{td}(M_i)\, \mathrm{ch}\left(\frac{\mathrm{Tr}_g\left(\frac{1}{1-qL}\right)}{\lambda_{-1}(N^*_{M_i/M})} \right) \\ &= \sum_{\zeta} \int_{M_\zeta} \mathrm{td}(M_\zeta) \sum_{k \geqslant 0} \frac{(\zeta q)^k}{(1-\zeta q)^{k+1}} \frac{(\mathrm{ch}(L)-1)^k}{\mathrm{ch}\, \lambda_{-1}(N^*)}, \end{aligned}$$

and therefore all poles must be at roots of unity. Here, we use the expansion

$$\frac{1}{1-\zeta q L} = \sum_{k \geqslant 0} \frac{(\zeta q)^k}{(1-\zeta q)^{k+1}} (L-1)^k$$

and take the Chern character.

Example 5.35. Let $L = \mathcal{O}(1)$ on $\mathbb{P}^1$. Then we have

$$\begin{aligned} \chi\left(\mathbb{P}^1, \frac{1}{1-qL}\right) &= \chi\left(\mathbb{P}^1, \frac{1}{1-q} + \frac{q}{(1-q)^2}(L-1)\right) \\ &=\sim_{i \geqslant 0} q^i (i+1) \\ &= \frac{1}{(1-q)^2}. \end{aligned}$$

Example 5.36. We will now consider weighted projective space. Then we have

$$\chi\left(\mathbb{P}(w_1, \ldots, w_n), \frac{1}{1-qL}\right) = \prod_{i=0}^{n} \frac{1}{1-q^{w_i}}.$$

5.9. **Adelic characterization.** Recall that there is a relationship between GW and GV invariants using an explicit formula in all genus. On the other hand, we can recover GW invariants from KGW invariants via the $q \to 1$ limit. In the other direction, we need the Givental-Tonita *adelic characterization*. The idea is that if we have a rational function f, then we can recover it from the Laurent expansions

$$f_{q=\zeta} = \sum f_{\zeta,k} (1 - \zeta^{-1} q)^k$$

at roots of unity ζ.

We will study the inertia stack $I\bar{M}_{0,1}(X, \beta)$. One can keep track of the components via Figure 3 or via Figure 4.

We will now explain Figure 4. Let L_1 be the cotangent line at the first marked point and let g be the automorphism of a point of $I\bar{M}_{0,1}(X, \beta)$. Then ζ is defined by the condition that g acts on L_1 with eigenvalue ζ. In both pictures, the bullet denotes the first marked point and its image in X, while a thick component is one which is a multiple cover of its image.

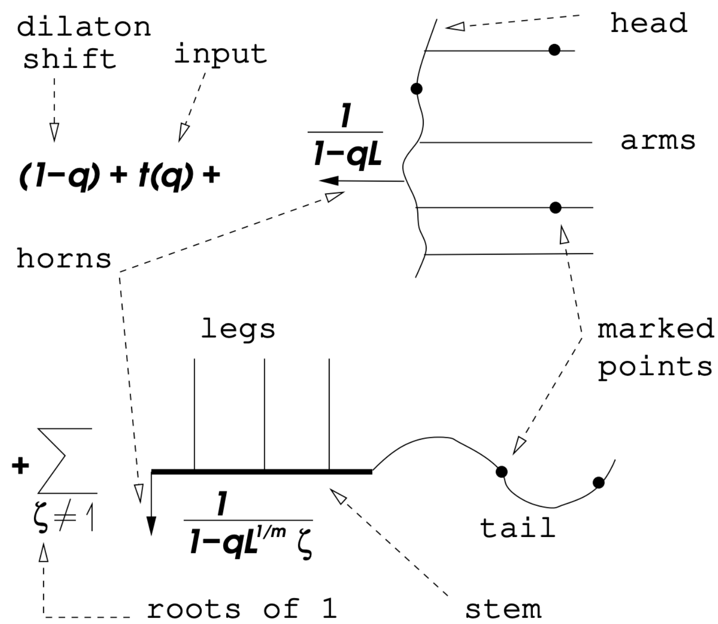
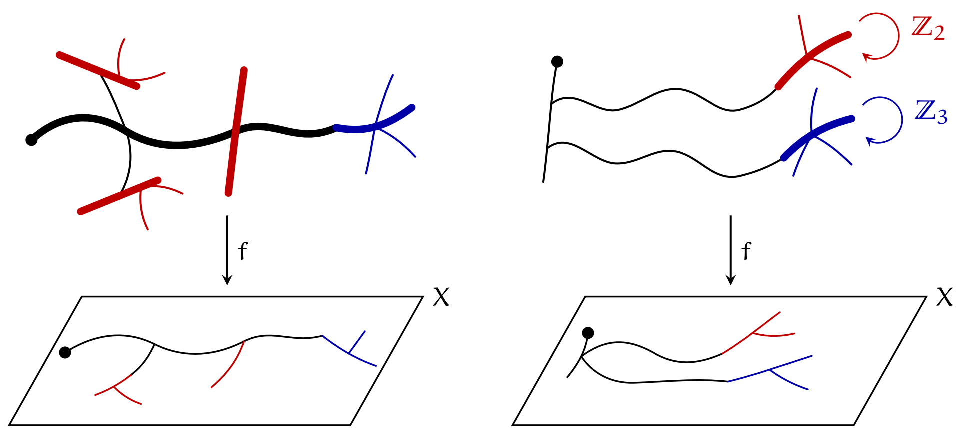


FIGURE 3. A way to keep track of the inertia components due to Givental-Tonita.



(a) $\zeta^r = 1$, $\zeta \neq 1$ (b) $\zeta = 1$

FIGURE 4. The components of $I\bar{M}_{0,1}(X, \beta)$, due to You-Cheng.

First, suppose that $\zeta = 1$. Then the black component is the maximal connected component containing the first marked point on which g acts as the identity. The red and blue components are precisely those on which g acts nontrivially, and we refer to them as *arms*.

Now suppose that ζ is an r-th root of unity for some $r \geqslant 2$. In this case, the nodes of the black component are required to be balanced with respect to the

g-action, so we cut off the nonbalanced part, which is colored blue, and refer to it as the *tail*. On what remains, the black component is the maximal connected component containing the first marked point on which g^r acts as the identity. Finally, after taking out the black component and the tail, we refer to the remaining components as *legs*.

The following definition is due to Givental. We write

$$\left\langle \frac{\Phi}{1-qL} \right\rangle_{0,1,\beta}^{X,K} := \sum_{\zeta} \left\langle \frac{\Phi}{1-qL} \right\rangle_{0,1,\beta}^{X_\zeta},$$

where the X_ζ term is simply the contribution of M_ζ under the Riemann-Roch theorem. Now, we will define arms, legs, and tails. Here, we define the *arms*

$$\operatorname{arm}(q) := \sum_{\beta,\alpha,\zeta\neq 1} Q^\beta \Phi^\alpha \left\langle \frac{\Phi}{1-qL} \right\rangle_{0,1,\beta}^{X_\zeta},$$

the *legs*

$$\operatorname{leg}_\zeta(q) := \operatorname{leg}_r(q) = \Psi^r(\operatorname{arm}(q)),$$

and the *tails*

$$\operatorname{tail}_\zeta(q) := \sum_{\beta,\alpha,\zeta'\neq\zeta} Q^\beta \Phi^\alpha \left\langle \frac{\Phi^\alpha}{1-qL} \right\rangle_{0,1,\beta}^{X_{\zeta'}}.$$

Proposition 5.37. *The Laurent expansion of the J-function at $q = 1$ is given by*

$$\begin{aligned} J^K(0)_{q=1} &= J^{\text{fake}}(\operatorname{arm}(q)) \\ &:= 1 - q + \operatorname{arm}(q) + \sum_{n,\beta,\alpha} \left\langle \frac{1}{1-qL}, \operatorname{arm}(L), \ldots, \operatorname{arm}(L) \right\rangle_{0,n+1,\beta}^{\text{fake}}, \end{aligned}$$

where the fake invariants are computed by applying Riemann-Roch and only considering the contribution of the untwisted sector.

Proposition 5.38. *At any other root of unity $\zeta \neq 1$, we have*

$$\begin{aligned} &J^K(0)_{q=\zeta} = (1-q) + \operatorname{tail}_\zeta(q) \\ &+ \sum_{n,\beta,\alpha} \frac{Q^{r\beta}\Phi^\alpha}{n!} \left[\frac{\Phi_\alpha}{1-\zeta q L^{1/r}}, \operatorname{leg}_r(L), \ldots, \operatorname{leg}_r(L), 1 - \zeta^{-1}L^{1/r} + \operatorname{tail}_\zeta(\zeta^{-1}L^{1/r}) \right]^{\text{stem}}, \end{aligned}$$

where the stem is some fake invariant of $X \times B\mu_r$ defined on the moduli space

$$\bar{M}^X_{-,n+2,\beta}(\beta) := \bar{M}_{0,2}(X \times B\mu_r, \beta, (g, 1, \ldots, 1, g^{-1})).$$

It is defined by the formula

$$[\cdots]^{\text{stem}} := \int_{[\bar{M}^X_{0,n+2,\beta}(\zeta)]^{\text{vir}}} \operatorname{td}(T_{\bar{M}}) \operatorname{ch}\left(\frac{\operatorname{Tr}(\cdots)}{\operatorname{Tr}(\lambda_{-1}(N^*))} \right),$$

where N denotes the normal bundle of $\bar{M}^X_{0,n+2,\beta}(\zeta) \subset \bar{M}_{0,nr+2}(X, r\beta)$.

Remark 5.39. If we can calculate the fake and stem theories, then we can compute genus zero quantum K invariants.

5.10. **Quantum K-theory of a point.** One natural question is to study higher genus quantum K-theory. We will begin with the simplest target, which is a point. One motivation for studying higher-genus would be to try to prove integrality of the Klemm-Pandharipande definition of GV invariants for Calabi-Yau fourfolds or the Pandharipande-Zinger definition of GV invariants for Calabi-Yau fivefolds (in this situation, we only need to consider genus 1). One reason we need to consider the quantum K-theory of a point is because in $I\bar{M}_{1,1}(X,\beta)$, there are some components where an entire curve of genus 1 is contracted to a point.

Question 5.40. *What is the K-theoretic Witten-Kontsevich theorem?*[11]

We will first recall the cohomological Witten's conjecture. We will begin with the operator

$$L = \partial_x^2 + u(x)$$

and deform it to

$$\tilde{L} = \partial_x^2 + u(x, T_3, T_5, \ldots),$$

where we should think of $x = T_1$ and $u(x) = u(x, 0, 0, \ldots)$. Formally, write

$$L^{1/2} = \partial_x + \sum_{k>0} a_{-k}(x)\, \partial_x^{-k}$$

using the property

$$\partial^n \circ f = \sum_{m=0}^{\infty} \binom{n}{m} f^{(m)}\, \partial^{n-m}$$

for all integers $n \in \mathbb{Z}$. In particular, we construct the (2-)KdV hierarchy by the equation

$$\frac{\partial \tilde{L}}{\partial T_{2k+1}} = \left[(\tilde{L}^{\frac{2k+1}{2}})_{\geqslant 0}, \tilde{L} \right]$$

for all $k \geqslant 0$. We define a τ-*function* to be a function τ such that

$$2 \frac{\partial^2}{\partial x^2} \log \tau(x, T_3, T_5, \ldots) = u(x, T_3, T_5, \ldots).$$

Now perform the above process using $u = 2x$. Define the function

$$F(t_0, t_1, \ldots) := \sum_{n \geqslant 0} \sum_{k_0, \ldots, k_n} \left\langle \tau_0^{k_0} \cdots \tau_n^{k_n} \right\rangle \frac{t_0^{k_0} \cdots t_n^{k_n}}{k_0! \cdots k_n!}.$$

Note that because the genus is determined entirely by the number of insertions and the degree of the insertions, the notation

$$\left\langle \tau_0^{k_0} \cdots \tau_n^{k_n} \right\rangle$$

uniquely determines a descendant GW invariant of a point.

Theorem 5.41 (Kontsevich). *F is a τ-function of the KdV hierarchy constructed starting with $L = \partial_x^2 + 2x$ under the change of variables $t_i = (2i+1)!!T_{2i+1}$.*

[11] According to You-Cheng, the difficulty lies everywhere.

One application of this theorem is to consider virtual localization. For example, suppose we consider $\bar{M}_{g,0}(X, d[C])$ where C is a unique rational curve in a Calabi-Yau threefold with normal bundle $N = \mathcal{O}(-1) \oplus \mathcal{O}(-1)$. Then in this setting we can reduce to the local model and compute

$$\int_{[\bar{M}_{g,0}(\mathbb{P}^1,d)]^{\mathrm{vir}}} c_{\mathrm{top}}(R^1\pi_* N) = \sum_{m \vdash d} \frac{(-1)^{d-L(m)}}{|\mathrm{Aut}(m)|} \prod_{i=1}^{L(m)} \int_{\bar{M}_{g,L(m)}} \frac{1 + \cdots + c_g(\mathcal{H})}{\prod_i (1 - m_i \psi_i)}$$

via virtual localization. If we would like a similar result in K-theory, then we will need to consider terms of the form

$$\chi\left(\bar{M}_{g,L(m)} / \mathrm{Aut}(m), \frac{\lambda_{-1}(\mathcal{H})}{1 - m_i L_i}\right),$$

so we will need to consider the permutation-equivariant quantum K-theory of a point.

We now proceed with a literature review. First, Lee computed the formula

$$\left\langle \frac{1}{1-q_1 L}, \ldots, \frac{1}{1-q_n L} \right\rangle_{0,n}^{\mathrm{pt},K} = \left(\prod_{i=1}^n \frac{1}{1-q_i}\right)\left(1 + \sum_{i=1}^n \frac{q_i}{1-q_i}\right)^{n-3}.$$

Later, Lee-Qu gave a way to compute

$$\left\langle \frac{1}{1-q_1 L}, \ldots, \frac{1}{1-q_n L} \right\rangle_{1,n}^{\mathrm{pt},K}$$

in principle. Givental computed the formula

$$\sum_{n \geqslant 2} \left\langle \frac{1}{1-qL}, 1, \ldots, 1 \right\rangle_{0,1+n}^{S_n} = (1-q) \exp\left(\sum_{k>0} \frac{\Psi^k}{k}\left(1 + \frac{q}{1-q}\right)\right).$$

Finally, in 2024, Milanov computed

$$\begin{aligned}\sum_{n \geqslant 0} &\left\langle \frac{1}{1-q_1 L}, \ldots, \frac{1}{1-q_r L}, 1, \ldots, 1 \right\rangle_{0,r+n}^{S_n} \\ &= \left(\prod_{i=1}^r \frac{1}{1-q_i}\right)\left(1 + \sum_{i=1}^r \frac{q_i}{1-q_i}\right) \exp\left(\frac{\Psi^k}{k}\left(1 + \sum_{i=1}^r \frac{q_i}{1-q_i}\right)\right).\end{aligned}$$

Let $s \in \mathbb{Z}$ and define the pluri-Hodge sheaf $\mathcal{H}_s \coloneqq R^0\pi_*(\omega_\pi^s)$. Considering the map $\bar{M}_{g,n+1} \xrightarrow{\pi} \bar{M}_{g,n}$, we will denote the upstairs cotangent lines by L_i and the downstairs ones by ℓ_i.

Lemma 5.42. *For all integers s and d_i, we have*

$$\pi_*\left(\omega_\pi^s\left(\sum_{i=1}^r d_i D_i\right)\right) = \mathcal{H}_s \mathcal{H}_{1-s}^* + \sum_{i=1}^n \sum_{k=1}^{d_i} \ell_i^{-k+s}.$$

Remark 5.43. Note that if $f(q) \in \mathbb{Z}[q]$, then $\exp \sum_k \frac{\Psi^k}{k}(f(q)) \in \mathbb{Z}[\![q]\!]$. The reason is that this operation sends $f = \sum_{i=1}^n a_i q^i$ to $\prod_{i=1}^n \frac{1}{(1-q_i)^{a_i}}$.

Proof. When all d_i are zero, we see that

$$R^0\pi_*\omega_\pi^s - R^1\pi_*\omega_\pi^s = \mathcal{H}_s - \mathcal{H}_{1-s}^*$$

by Serre duality. The next step is to show that

$$\pi_*\omega_\pi^s((r+1)D_i) = \pi_*\omega_\pi^s(rD_i) + \ell_i^{s-r-1}.$$

Here, we first consider the exact sequence

$$0 \to \mathcal{O}(-D_i) \to \mathcal{O} \to \mathcal{O}_{D_i} \to 0$$

and twist by ω_π^s to obtain

$$0 \to \omega_\pi^s(rD_i) \to \omega_\pi^s((r+1)D_i) \to \omega_\pi^s((r+1)D_i) \otimes \mathcal{O}_{D_i} \to 0$$

and then compute the pushforward. $\square$

Lemma 5.44. *Let f_i be rational functions, for example $\frac{1}{1-qL}$. Then*

$$\left\langle f_1(L), \ldots, f_n(L), L_{n+1}^r \right\rangle_{g,n+1} = \sum_{i=1}^n \langle f_1(L), \ldots, D(f_i f_{n+1}), \ldots, f_n(L_n) \rangle_{g,n} + \left\langle f_1(L), \ldots, f_n(L) \mid \mathcal{H}_r - \mathcal{H}_{1-r}^* \right\rangle_{g,n},$$

where $f_{n+1} = L_{n+1^r}$ *and* $(Df)(x) := \frac{f(x)-f(1)}{x-1}$.

To prove this, we need to consider the pushforward

$$\pi_*(L_1^{d_1} \cdots L_{n+1}^{d_{n+1}}) = \pi_*\Big(\pi^*(\ell_1^{d_1} \otimes \cdots \otimes \ell_n^{d_n}) \otimes \omega_\pi^{d_{n+1}}(d_i + d_{n+1})D_i\Big).$$

Here, we use the fact that $L_i = (\pi^*\ell_i)(D_i)$ and $L_{n+1} = \omega_\pi(D_1 + \cdots + D_n)$. Applying the previous lemma, we see that the pushforward becomes

$$(\ell_1^{d_1} \cdots \ell_n^{d_n}) \left(\mathcal{H}_{d_{n+1}} - \mathcal{H}_{1-d_{n+1}}^* + \sum_{i=1}^n \sum_{k=1}^{d_i+d_{n+1}} \ell_i^{-k+d_{n+1}} \right).$$

Now, Lee's formula computes the correlators

$$\Big\langle (L-1)^{k_1}, \ldots, (L-1)^{k_n} \Big\rangle_{0,n} = \frac{(n-3)!}{\prod_{i=1}^n k_i (n-3-\sum_i k_i)!}$$

via the expansion

$$\frac{1}{1-qL} = \frac{1}{1-q} \sum_{k \geqslant 0} \frac{q^k (L-1)^k}{(1-q)^k}.$$

Note here that there is no Hodge bundle in genus 0, so in fact the lemma can be used to obtain Lee's formula.

Lemma 5.45. *We have*

$$\mathcal{H}_r - \mathcal{H}_{1-r}^* = \pi^*(\mathcal{H}_r - \mathcal{H}_{1-r}^*) - \sum_{i=1}^n (D^2(\pi^*\ell_i)^r)\mathcal{O}_{D_i}.$$

The lemmas collectively imply that we can reduce to the case of a single marking at the cost of making the insertions very complicated and introducing pluri-Hodge terms everywhere. In genus 1, life becomes easier because on $\bar{M}_{1,1}$, we have

$$\mathcal{H}_r = \mathcal{O}(r) = L_1^r.$$

In this case, recall the classical formula

$$\chi\left(\bar{M}_{1,1}, \frac{1}{1-qL}\right) = \frac{1}{(1-q^4)(1-q^6)}.$$

Unfortunately, computing

$$\chi\left(\bar{M}_{1,n}; \frac{1}{1-q_1 L} \cdots \frac{1}{1-q_n L}\right)$$

is too hard, so we can try to compute

$$\chi\left(\bar{M}_{1,1}; \frac{1}{1-q_1 L_1} \cdots \frac{1}{1-q_n L_1}\right)$$

which is known, but it is unclear how to write a good formula for it. The reason we would consider this is the operation

$$\mathbb{Z}[\![q]\!] \xrightarrow{[]_r} \mathbb{Z}[\![q_1, \ldots, q_r]\!] \qquad q^k \to h_k(q_1, \ldots, q_r) \coloneqq \sum_{\ell_1 + \cdots + \ell_r = k} q_1^{\ell_1} \cdots q_r^{\ell_r}.$$

Unfortunately, You-Cheng does not know how to write

$$\chi\left(\bar{M}_{1,1}; \frac{1}{1-q_1 L_1} \cdots \frac{1}{1-q_n L_1}\right) = \left[\frac{1}{(1-q^4)(1-q^6)}\right]_n$$

in a nice form. In principle we can write formulae like

$$[f(q)]_2 \coloneqq \frac{q_1 f(q_1) - q_2 f(q_2)}{q_1 - q_2},$$

but this does not seem to help much.

5.11. **Permutation-equivariant theory of a point.** We can compute $\chi\left(\bar{M}_{1,1}, \frac{1}{1-qL}\right)$ in another way. Consider the involution on an elliptic curve, which gives a map

$$\bar{M}_{1,1} \to \bar{M}_{0,1+3}/S_3.$$

Then, Givental computes

$$\chi\left([\bar{M}_{0,1+n-1}/S_{n-1}], \frac{1}{1-qL}\right) = \prod_{i=2}^{n-1} \frac{1}{1-q^i}.$$

In particular, we obtain

$$\frac{1}{1-q^2} \cdots \frac{1}{1-q^3},$$

and then because the map is a μ_2-gerbe (or a 2 : 1-cover on objects), we obtain the desired formula.

Consider $\mathcal{H}_{g,1} \subset \bar{M}_{g,1}$ to be the hyperelliptic locus. Note that the marking lies on a Weierstrass point. Note that the fixed points of the hyperelliptic involution go to $2g+1$ points on $\mathbb{P}^1$, and therefore one would want to use the calculation

$$\chi\left(\bar{M}_{0,1+2g+1}/S_{2g+1}, \frac{1}{1-qL}\right) = \prod_{i=2}^{2g+1} \frac{1}{1-q^i}$$

to compute the quantity $\chi\left(\mathcal{H}_{g,1}, \frac{1}{1-qL}\right)$. However, because of the way that the curves upstairs degenerate, we will need to consider the Hassett moduli space where the last $2g+1$ markings have weight $\frac{1}{2}$.

6. Quasimap wall-crossing (Yang Zhou)

Let X be the quintic threefold. Our goal is to count genus zero curves on X, which is a problem that was solved by Givental and Lian-Liu-Yau. Via the closed immersion $X \subset \mathbb{P}^4$, we can write

$$\bar{M}_0(X, d) \subset \bar{M}_0(\mathbb{P}^4, d),$$

which is smooth and has a torus action by $(\mathbb{C}^\times)^5$. If a map is fixed by this action, the domain must have some components which map to the fixed points and some components which form an $n:1$ cover of the 1-dimensional orbits. In particular, the fixed locus is indexed by a set of graphs.

In this setting, we have the quantum Lefschetz principle, which states that

$$[\bar{M}_0(X, d)]^{\mathrm{vir}} = e(E) \cap [\bar{M}_0(\mathbb{P}^4, d)].$$

Here, the vector bundle E is defined by

$$E = \pi_* f^* \mathcal{O}(5),$$

where π is the universal curve and f is the universal stable map. We will give a different perspective on the result that if

$$\begin{aligned} I(q,z) &= 1 + \sum_{d>0} \frac{\prod_{k=1}^{5}(5H+kz)}{\prod_{k=1}^{d}(H+kz)^5} q^d \in H^*(X)[z^{-1}][\![q]\!] \\ &= I_0(q)\cdot 1 + I_1(q)\frac{H}{z} + \frac{I_2}{(}q)\frac{H^2}{z^2} + I_3(q)\frac{H^3}{z^3}, \end{aligned}$$

then

$$J(Q,z) = 1 + \sum_{d>0} \phi^\alpha \left\langle \frac{\phi_\alpha}{z(z-\psi)} \right\rangle_{0,1}^{X} Q^d = \frac{I}{I_0}$$

up to the mirror map $Q = q\exp\left(\frac{I_1}{I_0}\right)$.

6.1. **Naive counting.** We will consider "maps" $\mathbb{P}^1 \to X \subset \mathbb{P}^4$ from a fixed $\mathbb{P}^1$. For simplicity, we will assume $X = \left(\sum x_i^5 = 0\right)$ is the Fermat quintic. This is given by a collection $[f_0, \ldots, f_4]$ where $f_i \in \mathbb{C}[u,v]$ such that $\deg f_i = d$ and $\sum f_i^5 = 0$. To be an actual morphism, we need the f_i to have no common zeroes, but this moduli is not proper, so we relax this condition. Then we consider the moduli space

$$QG(X,d) = \left\{ [f_0, \ldots, f_4] \,\middle|\, \begin{matrix} f_i \in \mathbb{C}[u,v] \\ \deg(f_i = d)' \end{matrix} \sum f_i^5 = 0, (f_0, \ldots, f_4) \neq (0, \ldots, 0) \right\} \Big/ \mathbb{C}^\times.$$

This lives inside the corresponding moduli space

$$\begin{aligned} QG(\mathbb{P}^4, d) &:= \{[f_0, \ldots, f_4] \mid f_i \in \mathbb{C}[u, v], \deg f_i = d, (f_0, \ldots, f_4) \neq (0, \ldots, 0)\} \\ &\cong \mathbb{P}^{5d+5-1} \end{aligned}$$

of rational maps to projective space. There are two problems. First, the domain is parameterized, so we need to account for the automorphisms of $\mathbb{P}^1$. Second, to obtain stable maps, we need to blow up the domain.

First, let $\mathbb{C}^\times$ act on $\mathbb{P}^1$ by scaling u. Consider the special fixed locus $F_\star \subset QG(X, d)$ defined by

$$F_{\star,d} = \{[f_0, \ldots, f_4] \text{ fixed and } (v = 0) \text{ is not a common zero}\}.$$

Therefore, $F_{\star,d} = \left\{[a_0 u^d, \ldots, a_4 u^d] \mid [a_0, \ldots, a_4] \in X\right\} \cong X$. From this perspective, we can rewrite the I-function as

$$\begin{aligned} I(q, z) &= \sum_{d > 0} \left(\frac{[F_{\star,d}]^{\mathrm{vir}}}{e_{\mathbb{C}^\times}(N^{\mathrm{vir}})} \right)^{\mathrm{PD}} q^d \\ &= 1 + \sum_{d > 0} \left(\frac{[X]}{e_{\mathbb{C}^\times}(N^{\mathrm{vir}})} \right)^{\mathrm{PD}} q^d \end{aligned}$$

using standard arguments in deformation theory. This lives in $H^*_{\mathbb{C}^\times}(X)[\![q]\!][z^{-1}]$.

6.2. **Wall-crossing without rigor.** Second, consider a nodal curve C with a marking plus a tail with an action of $\mathbb{C}^\times$ and suppose we have a map $f\colon C \to X$. If we take the limit

$$\lim_{t \to 0} [f_0(t^{-1}u, v), \ldots, f_4(t^{-1}u, v)] = [a_0 u^{d_0}, \ldots, a_4 u^{d_0}],$$

we see that the limit of the bubble lands in $F_{\star, d_0}$, where the common zero is away from the node. Then we can replace this tail by a new marking which carries some terms of the I-function. For the quintic, this is simply $I_1 H - I_0 \psi = [zI - z]_{+, z = -\psi}$. At the end of this process, we are left with $C = \mathbb{P}^1$ with various contributions of the I-function.

Note that if we consider $\mathbb{C}^\times$ acting on $\mathbb{P}^4$, then $f\colon C \to \mathbb{P}^4$ is fixed if for all $t \in \mathbb{C}^\times$, there exists φ_t such that

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "\varphi_t"'] & \mathbb{P}^4 \\ C \arrow[ur, "t \cdot f"'] & \end{tikzcd}$$

commutes. Therefore, we obtain an action of $\mathbb{C}^\times$ on the domain.

We will define the moduli space $Q^\varepsilon_{g,n}(X, d)$ of ε-stable quasimaps to X, which is analogous to the moduli of stable maps. We want to think of these moduli spaces as being virtually birational to each other, and each of these is a proper DM stack with perfect obstruction theory, and therefore we can define quasimap invariants

$$\left\langle \gamma_1 \psi_1^{a_1} \cdots \gamma_n \psi_n^{a_n} \right\rangle^\varepsilon_{g,n} := \int_{[Q^\varepsilon_{g,n}(X,d)]^{\mathrm{vir}}} \prod_{i=1}^n \mathrm{ev}_i^*(\gamma_i) \psi_i^{a_i}.$$

Here, $\varepsilon \in \mathbb{Q}_{>0}$ is a positive rational number. The space of possible ε has a wall and chamber structure, where the walls occur at $\frac{1}{n}$ for positive integers n. When

$\varepsilon > 1$ we obtain Gromov-Witten theory (namely stable maps), and as we move to the left, we allow fewer rational tails but more basepoints. Eventually, in the $\varepsilon \to 0^+$ limit, no rational tails are allowed. In the end, we obtain the following wall-crossing formula around the wall at $\frac{1}{d_0}$:

$$\langle - \rangle^{\varepsilon_+}_{g,n,d} - \langle - \rangle^{\varepsilon_-}_{g,n,d} = \sum_k \frac{1}{k!} \langle -, \mu, \dots, \mu \rangle^{\varepsilon_+}_{h,n+k,d-kd_0}.$$

Here, $\mu = [zI - z]_{+,q^{d_0},z=-\psi}$. In the special case, where $g = 0$, $n = 1$, and $\varepsilon = 0^+$, we obtain

$$\sum \varphi^\alpha \left\langle \frac{\varphi_\alpha}{z(z-\psi)} \right\rangle^{\varepsilon}_{0,1} = I_{z<0}.$$

Crossing all walls from the GW chamber, we obtain

$$I = 1 + \sum_{\substack{d \geqslant 1 \\ k \geqslant 0}} \frac{1}{k!} \sum_\alpha \varphi^\alpha \left\langle \frac{\varphi_\alpha}{z(z-\psi)}, I_1 H - I_0 \psi, \dots, I_1 H - I_0 \psi \right\rangle^{\infty}_{0,1+k,d} q^d,$$

which produces the mirror map via the dilaton and divisor equations.

6.3. **Quasimaps.**

Definition 6.1. A *prestable quasimap* to the quintic threefold X consists of

$$(C, x_1, \dots, x_n, \mathcal{L}, \varphi_0, \dots, \varphi_4),$$

where $(C, x_1, \dots, x_n)$ is a prestable curve, $\mathcal{L} \in \operatorname{Pic} C$, and $\varphi_0, \dots, \varphi_4 \in H^0(C, \mathcal{L})$, such that

$$\varphi^{-1}(0) := \{x \in C \mid \varphi_0(x) = \dots = \varphi_4(x) = 0\}$$

is finite and away from the special points. In addition, we require that $\sum \varphi_i^5 = 0$.

Importantly, we will refer to $\varphi^{-1}(0)$ as the set of *base points*. Away from $\varphi^{-1}(0)$, we have an actual morphism $C \setminus \varphi^{-1}(0) \to X$. Geometrically, suppose we have a family over $\mathbb{A}^1 \setminus 0$ where in the limit as $t \to 0$, the zero loci $\varphi_i^{-1}(0)$ will end up colliding. We can either preserve this quasimap in the central fiber, or we can blow up the central fiber some number of times to obtain an actual morphism at the cost of bubbling out a rational tail. Therefore, any moduli space of prestable quasimaps cannot be separated, so we must consider some kind of stability condition.

Definition 6.2. A prestable quasimap $(C, \mathbf{x}, \mathcal{L}, \varphi)$ is *ε-stable* if

(1) For all $x \in \varphi^{-1}(0)$, we have $\operatorname{len}(x) \leqslant \frac{1}{\varepsilon}$, where $\operatorname{len}(x) = \min\{\operatorname{ord}_x(\varphi_i)\}$;
(2) The $\mathbb{Q}$-line bundle $\mathcal{L}^\varepsilon \otimes \omega_C^{\log}$ has positive degree on every irreducible component of C.

Note that for all irreducible components $C' \subseteq C$, we have $\deg \mathcal{L}|_{C'} \geqslant 0$. Therefore, the things which change are curves of genus 0 with $\leqslant 2$ special points or curves of genus 1 with no marked points. Note that in the $(0,0)$ and $(1,0)$ cases, we have $C = C'$, and in the $(0,2)$ case, stability is equivalent to $\deg \mathcal{L} > 0$, so in the $(0,1)$ case, the second condition is equivalent to

$$\deg \mathcal{L}|_{C'} > \frac{1}{\varepsilon}.$$

Example 6.3. In the $\varepsilon > 1$ chamber, note that the first condition implies that there are no basepoints, and the second condition implies that all unstable components have positive degree, and therefore we have recovered stable maps.

If we move into the chamber where $\frac{1}{2} < \varepsilon < 1$, then we must contract rational tails of degree 1. For example if we have a chain of two degree 1 rational curves ending in a tail, then we contract the outermost tail and are left with a single degree 2 rational tail with a basepoint of length 1.

If we move into the $\frac{1}{3} < \varepsilon < \frac{1}{2}$ chamber, then we must contract all rational tails of degree 2 and replace them with basepoints of length 2.

Remark 6.4. In the case of projective space, there is an actual morphism

$$Q_g^{\varepsilon+}(\mathbb{P}^N, d) \to Q_g^{\varepsilon-}(\mathbb{P}^N, d) \qquad (C_g \cup C_0, \mathcal{L}, \varphi) \mapsto (C_g, \mathcal{L}|_{C_g}(p), \varphi(p)).$$

For a general target, this is not possible, but this idea gives intuition for the phenomenon of quasimap wall-crossing.

We will now discuss how to generalize this to more general target spaces. Let

$$X = V /\!\!/_\theta G,$$

where V is affine, G is reductive, and θ is a character of G. We need to require that

(1) $V^{ss}(\theta) = V^s(\theta) \neq \emptyset$;
(2) V has at worst lci singularities and $V^s(\theta)$ is smooth.

Note that we require the stable locus to have finite stabilizers. For convenience, we will also require

(1) $V /\!\!/ G$ is projective;
(2) G acts freely on the stable locus.

Example 6.5. In the case of the quintic, V is the affine cone over the quintic. The unstable locus consists of the origin, which is the unique singular point.

Example 6.6. Another example is that we can consider $\mathbb{C}^N /\!\!/ (\mathbb{C}^\times)^r$ with generic stability condition and obtain a toric variety.

Example 6.7. We can consider the Grassmannian

$$G(2,4) = M_{2\times 4} /\!\!/ GL_2,$$

where the unstable locus consists of matrices with rank strictly less than 2.

Remark 6.8. The condition on the stable locus is always satisfied when G is abelian for generic stability conditions, but for non-abelian G this is not always possible.

Remark 6.9. For a general smooth projective variety, the origin is not necessarily lci, so it is unclear how to define quasimaps in a way where we can still define a virtual cycle.

Definition 6.10. A *prestable quasimap* to $V /\!\!/ G$ is a map $u\colon C \to [V/G]$ such that the base locus

$$u^{-1}([V^{un}(\theta)/G])$$

is finite and away from the special points. By definition, u consists of a principal G-bundle $P \to C$ together with an equivariant map $P \to V$, which is equivalent to a section of the associated bundle $P \times_G V$.

Example 6.11. For the quintic, this exactly recovers a line bundle $\mathcal{L} \in \operatorname{Pic} C$ (which is equivalent to a $\mathbb{C}^\times$-torsor) combined with a section of $\operatorname{Tot}(\mathcal{L}^{\oplus 5})$ which satisfies the defining equation.

In order to define degrees, we can consider the line bundle $\mathcal{L}_\theta$ on $[V/G]$. Then we can define the degree of $C' \subset C$ under u to be

$$\deg(u^*\mathcal{L}_\theta|_{C'}).$$

Example 6.12. In the $G(2,4)$ example, note that a principal GL_2-bundle is equivalent to rank 2 vector bundle $\mathcal{E}$ on C. Therefore, $P \times_G V$ is equivalent to the data of $\mathcal{E}^{\oplus 4}$, and therefore quasimaps to the Grassmannian are equivalent to stable quotients. In this case, we are looking for generically surjective

$$\mathcal{O}^{\oplus 4} \to \mathcal{E},$$

and because $\theta = \det$, we see that $u^*\mathcal{L}_\theta = \det \mathcal{E}$.

Example 6.13. Consider $V = \mathbb{C}^{N+1}$ and note that $\mathbb{P}^N = V /\!\!/ \mathbb{C}^\times$. If we consider $\theta_1 : z \mapsto z$, then we have

$$V^{\mathrm{un}} = V((x_i)).$$

However, if we consider $\theta \colon z \mapsto z^2$, then we have

$$V^{\mathrm{un}} = V((x_i x_j)).$$

Here, we see that the scheme structure is sensitive to the choice of character, so we cannot simply naively define the length of a basepoint. Note that

$$V^{\mathrm{un}} = V\left(\left\langle \bigoplus_{k \geqslant 1} R_k \right\rangle\right), \qquad R_k = H^0([V/G], L_\theta^{\otimes k}).$$

In particular, we must define

$$\operatorname{len}(x) \coloneqq \min_{\substack{f \in R_k \\ k \geqslant 1}} \frac{\operatorname{ord}_x u^*(f)}{k}.$$

Example 6.14. Let $X = \mathrm{pt} = \mathbb{C} /\!\!/_\theta \mathbb{C}^\times$. Then a prestable quasimap is simply

$$(C, \mathcal{L}, \varphi \in H^0(\mathcal{L})),$$

where $\varphi^{-1}(0)$ is finite and away from the special points. In addition, if $\deg \mathcal{L} = 1$, we have precisely recovered the data of a new marked point in C^{sm}.

Example 6.15. Now consider $X = \mathrm{pt} = \mathbb{C}^n /\!\!/ (\mathbb{C}^\times)^n$. Also suppose that the degree is $(1, \ldots, 1)$. Now we will set

$$\theta = (a_1, \ldots, a_n).$$

Therefore, note that

$$\langle R_1 \rangle = \left\langle (x_1^{a_1} \cdots x_n^{a_n}) \right\rangle.$$

Because these all live in degree 1, we see that for a smooth point $x \in C^{\mathrm{sm}}$, we have

$$\operatorname{len}_x = \sum \operatorname{ord}_x(\varphi_i^{a_i}) = \sum a_i \operatorname{ord}_x \varphi_i.$$

If we set $\{p_i\} = \varphi_i^{-1}(0)$, then the condition that $\varepsilon \cdot \mathrm{len}(x) \leqslant 1$ is equivalent to requiring that

$$\sum_{x=p_i} \varepsilon a_i \leqslant 1.$$

The degree condition now states that

$$\varepsilon \sum_i a_i + \deg \omega_C > 0.$$

Equivalently, this states that

$$\omega_C(\varepsilon a_1 p_1 + \cdots + \varepsilon a_n p_n) > 0,$$

so we recover a weighted pointed curve with weights $(\varepsilon a_1, \ldots, \varepsilon a_n)$.

6.4. **Master space.** Our method will be to perform virtual localization on a master space which interpolates the two moduli. Consider the $\mathbb{C}^\times$-action on $\mathbb{P}^n$ given by

$$t \cdot [x_0 : \cdots : x_n] = [tx_0, x_1, \ldots, x_n].$$

The fixed loci consists of a $\mathbb{P}^{n-1}$ where $x_0 = 0$ and a point $[1, 0, \ldots, 0]$. Then we can compute

$$\int_{\mathbb{P}^{n-1}} H^{n-1} = 1$$

by transfering the computation to the other fixed locus. In particular, we lift to $H \in H^*_{\mathbb{C}^\times}(\mathbb{P}^n) = \mathbb{Q}[H, z]/(H - z)H^n$ and compute

$$\int_{\mathbb{P}^n} H^{n-1} = \int_{\mathbb{P}^{n-1}} \frac{H^{n-1}}{-z + H} + \int_{\mathrm{pt}} \frac{z^{n-1}}{z^n}.$$

Taking residues, we see that

$$\begin{aligned} = 0 \operatorname{Res}_{z=0} \int_{\mathbb{P}^{n-1}} \frac{H^{n-1}}{-z + H} + 1 \\ = 1 - \int_{\mathbb{P}^{n-1}} H^{n-1}, \end{aligned}$$

which recovers the desired integral. The key fact we used here is that projective space is proper.

Example 6.16 (Second toy example)**.** We will consider a second toy example. Recall Hassett's moduli of weighted pointed curves

$$(C, x_1, \ldots, x_n)$$

where C is nodal and the $x_i \in C^{\mathrm{sm}}$ can collide. The collisions are regulated by the fact that each marked point is given a weight $\delta_i \in Q \cap (0, 1]$ and are required to satisfy the following conditions:

(1) For all $x \in C^{\mathrm{sm}}$, then
$$\sum_{x=x_i} \delta_i \leqslant 1;$$
(2) We have the stability condition $\omega_C(\delta_1 x_1 + \cdots + \delta_n x_n) > 0$. For example, if we have a rational tail, then $\sum_{x_i \in C'} \delta_i > 1$.

Example 6.17 (Another toy example)**.** Our goal will be to once again compute

$$\int_{\bar{M}_{0,4}} \psi_1 = 1.$$

Our first goal will be to change the stability condition. If we set

$$\delta_+ = \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right),$$

we see that the stability condition does not change. However, if we set

$$\delta_- = \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right),$$

the moduli space does not change, but the universal curve in fact becomes $\mathbb{P}^1 \times \mathbb{P}^1$ with the section x_4 being the diagonal. This tells us that

$$\int_{\bar{M}_{0,4}^{\delta_-}} \psi_1 = 0,$$

and we will use this to compute in the δ_+ chamber. We can try to form a master space by both allowing rational tails with x_i, x_4 and allowing $x_1 = x_4$, but if we do this, we end up with a semistable point with a rational tail where $x_1 = x_4$. We want to end up with this by taking a quotient from a master space, so we will add the data of a tangent vector to the moduli. Therefore, define the $\delta_\pm$ master space to consist of objects of the form

$$\xi = (C, x_1, \dots, x_4, \nu \in T_{x_4} C \cup \infty)$$

where C is nodal, x_i are smooth points, x_1, x_2, x_3 are disjoint,

$$\sum_{x_i \in C'} \frac{2}{3}\delta_i > 1$$

for all rational tails C', if $\nu = 0$, then x_4 is disjoint from the other x_i, and whe $\nu = \infty$, there are no rational tails where $x_4 \neq x_i$ for $i = 1, 2, 3$.

We now give this master space an action of $\mathbb{C}^\times$ by scaling ν. There are various fixed loci:

(1) When $\nu = 0$, we obtain $\bar{M}_{0,4}^{\delta_+}$;
(2) When $\nu = \infty$, we obtain $\bar{M}_{0,4}^{\delta_-}$;
(3) There are fixed loci where $\nu \neq 0, \infty$ and we have a rational tail with $x_i = x_4$.

If we consider a general point (with all four marked points disjoint), it will travel directly to itself along the $\mathbb{C}^\times$-action. However, if we start with a rational tail containing x_1 and x_4, then we first travel to a rational tail with general ν but $x_1 = x_4$, and if we smooth out the node, we obtain a single curve with $x_1 = x_4$, but we allow $\nu \to \infty$, which is another fixed point. Now we can integrate

$$\int_{\mathsf{Master}} \psi_1 = \int_{\bar{M}_{0,4}^{\delta_+}} \frac{\psi_1}{-z-?} + \int_{\bar{M}_{0,4}^{\delta_-}} \frac{\psi_1}{z+?} + \mathrm{Cont}(12|34) + \mathrm{Cont}(13|24) + \mathrm{Cont}(14|23).$$

Taking residues at $z = 0$, we obtain

$$0 = -\int_{\bar{M}_{0,4}^{\delta_+}} \psi_1 + 0 + 0 + 0 + \int_{\mathrm{pt}} \frac{(-z)}{(\pm z)(\pm z)},$$

where in the final term the $\pm z$ in the denominator are node smoothing parameters, and therefore we recover the desired integral.

With all of our toy examples, we are finally ready to discuss the quasimap master space. Our first step at the $\frac{1}{d_0}$ wall is to allow both degree d_0 rational tails and length d_0 base points. However, we can consider curves with a rational tail and a quasimap of the form

$$[x_0 u^{d_0}, \ldots, a_n u^{d_0}],$$

which have infinitely many automorphisms. However, we cannot attach tangent vectors to the base points (because deformations will remove them), and there are also situations where there exists more than one degree d_0 with a base point of length d_0 (and then the automorphism group contains $\mathbb{C}^\times$).

Remark 6.18. There is a key observation. Suppsose that we have two rational tails of degree d_0 and call the two nodes p_1 and p_2. Now define the space of smoothing parameters

$$\mathbb{C} \cong \Theta_i = T_{p_i} E_i \otimes T_{p_i} C_g,$$

where E_i is the rational tail and C_g is the rest of the curve. We will consider the moduli

$$\mathfrak{M}^{\mathrm{wt}}_{g,n,d} \xrightarrow{\text{ét}} \mathfrak{M}_{g,n}$$

of weighted semistable curves labelled by degrees. Note that this morphism is highly non-separated. We will consider the open substack $\mathfrak{M}^{\mathrm{wt},\varepsilon_0}_{g,n,d}$ where the curves have no tail of degree strictly less than d_0. Then we can consider the Cartier divisor $\mathcal{Z}$ where there exists at least one tail of degree d_0. In particular, we see that

$$\Theta_i = N_{\mathcal{Z}_i}|_\xi$$

for the two branches $\mathcal{Z}_1$ and $\mathcal{Z}_2$ at the point of interest. We also see that $\Theta_1 \otimes \Theta_2 = \mathcal{O}(\mathcal{Z})|_\xi$, so we can add the data of

$$v \in \left(\bigotimes_i \Theta_i \right)^\vee \cup \{\infty\}$$

and the data of

$$w \in \mathbb{P}(\Theta_1 \oplus \Theta_2).$$

The actual construction proceeds as follows. Consider $\mathcal{Z} \subset \mathfrak{M} = \mathfrak{M}^{\mathrm{wt},\varepsilon_0}_{g,d}$. Define $\tilde{\mathfrak{M}} \to \cdots \to \mathfrak{M}$ to be an iterated blowup along strata of $\mathcal{Z}$. Then an object of the master space (a quasimap with calibrated tails) over a scheme S is given by the data of a quasimap

$$\begin{tikzcd} C \arrow[r, dashed, "f"] \arrow[d] & X \\ S & \end{tikzcd}$$

inducing a morphism $p\colon S \to \mathfrak{M}^{\mathrm{wt},\varepsilon_0}_{g,d}$, a choice of a lift

$$\begin{tikzcd} & \tilde{\mathfrak{M}} \arrow[d, "\tau"] \\ S \arrow[ur, dashed, "w"] \arrow[r, "p"] & \mathfrak{M}^{\mathrm{wt},\varepsilon_0}_{g,d}, \end{tikzcd}$$

a line bundle $\mathcal{N} \in \operatorname{Pic} S$, and $\nu = (\nu_1, \nu_2)$ where $\nu_1 \in H^0(\rho^*\mathcal{O}(-\mathcal{Z}))$ and $\nu_2 \in H^0(\mathcal{N})$, where (ν_1, ν_2) is nonvanishing. We also disallow basepoints of length strictly larger than d_0 and require stability such that

(1) When $\nu = 0$, we obtain ε_+-stability;
(2) When $\nu = \infty$, we obtain ε_--stability;
(3) Any constant tail (degree d_0 with base point of length d_0) is an entangled tail. Here, we consider the last blowup such that w maps to an exceptional locus $\bigcap_{i \in I} \tilde{\mathcal{Z}}_i$, and then the E_i are entangled tails for $i \in I$;
(4) If there exists a degree d_0 tail, then all length d_0 base points are on degree d_0 tails.

7. Log GLSM (Qile Chen)

Let X be a smooth projective variety (or even a DM stack). Let E be a vector bundle on X with a section $s \in H^0(E)$ which defines a smooth zero section Z of the expected codimension. We will refer to X and E as ambient and s and Z as the complete intersection.

Question 7.1. *What is the relationship between the GW theory of Z and the GW theory of (X, E)?*

When $g = 0$, Givental and Lian-Liu-Yau used this relationship to compute the GW invariants of complete intersections, and when $g = 1$, this was done by Zinger. In general, there are two approaches: (N)MSP and log GLSM. In this series of lectures, we will discuss log GLSM.[12]

7.1. **R-maps.** We will consider the *R-torus* $\mathbb{C}^\times_\omega \coloneqq \mathbb{C}^\times$. This has a weight 1 representation, which we will call $\mathbb{C}_\omega$. It descends to a line bundle $\mathcal{L}_\omega \coloneqq [\mathbb{C}_\omega/\mathbb{C}^\times_\omega]$ on $B\mathbb{C}^\times_\omega$. The targets of R-maps will be stacks

$$\mathbb{P} \to B\mathbb{C}^\times_\omega$$

which are DM-type, separated, and (log) smooth. Now, an R-map consists of a (2-)commutative diagram

$$\begin{tikzcd} & \mathfrak{P} \arrow[d] \\ C \arrow[ur, "f"] \arrow[r, "\omega^{\log}"'] \arrow[d] & B\mathbb{C}^\times_\omega \\ S. & \end{tikzcd}$$

Here, the curve could be prestable, twisted, log, or even punctured.

[12]Qile referred the audience to the author of these notes and Yang Zhou if they have questions about MSP.

Example 7.2. Our first example is that of stable maps. Let $\mathfrak{P} = B\mathbb{C}^\times_\omega$. Because in the diagram

$$\begin{tikzcd} & X \times B\mathbb{C}^\times_\omega \arrow[d] \\ C \arrow[ur] \arrow[r] & B\mathbb{C}^\times_\omega \end{tikzcd}$$

the vertical arrow is the second projection, we see that this reduces to a map to X.

Example 7.3. Consider maps

$$\begin{tikzcd} & \mathcal{L}_\omega \arrow[d] \\ C \arrow[ur, "f"] \arrow[r] & B\mathbb{C}^\times_\omega. \end{tikzcd}$$

If we assume that C is unmarked, we see that the data of this map is simply a section of ω_C, and therefore we recover holomorphic differentials.

Example 7.4. Consider a morphism $\mathbb{C}^\times_R \to \mathbb{C}^\times_\omega$ given by $z \mapsto z^{r_{\text{in}}}$. Then we see that

$$\mathcal{L}_R^{\otimes r_{\text{in}}} \simeq \mathcal{L}_\omega|_{B\mathbb{C}^\times_R}.$$

Now a diagram

$$\begin{tikzcd} & B\mathbb{C}^\times_R \arrow[d] \\ C \arrow[ur, "f"] \arrow[r] & B\mathbb{C}^\times_\omega \end{tikzcd}$$

with C unmarked is equivalent to the data of a line bundle $\mathcal{L}_R|_C$ and an isomorphism $\mathcal{L}_R^{\otimes r_{\text{in}}} \simeq \omega_C$. Alternatively, we can consider R-maps of the form

$$\begin{tikzcd} & \mathcal{L}_R \arrow[d] \\ & B\mathbb{C}^\omega_R \arrow[d] \\ C \arrow[uur, "f"] \arrow[r] & B\mathbb{C}^\times_\omega, \end{tikzcd}$$

which parameterize curves with an R-spin line bundle together with a section.

Example 7.5. Let X be a smooth projective variety. Then consider R-maps of the form

$$\begin{tikzcd} & E^\vee \boxtimes \mathcal{L}_\omega \arrow[d] \\ & X \times B\mathbb{C}^\times_\omega \arrow[d] \\ C \arrow[uur, "f"] \arrow[r] & B\mathbb{C}^\times_\omega. \end{tikzcd}$$

This equivalent to a map $C \to X$ together with a section $\rho \in H^0(E^\vee|_C \otimes \omega_C)$.

We will consider targets in the GW case. Consider the data of an ambient (X, E). Then we will set $\mathbb{P}^\circ := E^\vee \boxtimes \mathcal{L}_\omega$. The morphism $\mathfrak{P}^\circ \to B\mathbb{C}^\times_\omega$ is very nice, but it is not proper. To fix this, we will consider the log stack

$$\mathfrak{P} = (\mathbb{P}(E^\vee \boxtimes \mathcal{L}_\omega \oplus \mathcal{O}), \infty).$$

Because we have a smooth divisor in a smooth stack, the map $\mathfrak{P} \to B\mathbb{C}^\times_\omega$ is proper and log smooth. Note that we have the zero section

$$0_{\mathfrak{P}} \subset \mathfrak{P}^\circ \subset \mathfrak{P} \supset \infty.$$

We will now consider targets in the r-spin case. Consider the log stack

$$(\mathbb{P}(\mathcal{L}_R^{\oplus n} \oplus \mathcal{O}), \infty) = \mathfrak{P} \to B\mathbb{C}^\times_R \to B\mathbb{C}^\times_\omega.$$

This morphism is proper and log smooth.

7.2. **Log and punctured R-maps.** We will consider diagrams

$$\begin{tikzcd} & \mathfrak{P} \arrow[d] \\ C \arrow[ur, "f"] \arrow[r, "\omega^{\log}"'] \arrow[d] & B\mathbb{C}^\times_\omega \\ S & \end{tikzcd}$$

where $C \to S$ is a log or a punctured (possibly twisted) curve and S is an fs log scheme. The stability condition in the GW case is that f is representable and

$$(\omega_{\log})^{1+\delta} + H^{\otimes k} \otimes f^*\mathcal{O}(\infty) > 0$$

for all $0 < \delta \ll 1$ and $k \gg 1$, where H is an ample line bundle on the coarse moduli space of X. For punctured R-maps, we further require pre-stability.

Example 7.6. Consider the case of holomorphic differentials. Then consider $f \in H^0(\omega_C)$ such that all zeroes are concentrated in a single Weierstrass point, where C has genus 2. If we consider $\lim_{\lambda \to \infty} \lambda \cdot f$, then the limit consists of a genus 2 component mapping entirely to ∞ and a rational component which remembers the zero.

We will now discuss the tropicalization of log or punctured R-maps. Recall that a graph

$$\underline{G} = (\mathbb{V}, \mathbb{H}, \iota \colon \mathbb{H} \to \mathbb{H}, \nu \colon \mathbb{H} \to \nu)$$

consists of a set $\mathbb{V}$ of vertices, a set $\mathbb{H}$ of half edges, an involution ι on the half edges, and a root map from half edges to vertices. Then we will let the set $\mathbb{L}$ of legs be

$$\mathbb{L} = \{h \in \mathbb{H} \mid \iota(h) = h\}.$$

The edges are pairs of distinct half-edges which are sent to each other by the involution:

$$\mathbb{E} = \left\{\{h, \hat{h}\} \mid \iota(h) = \hat{h} \neq h\right\}.$$

Now a decorated graph is a graph combined with a genus function

$$g \colon \mathbb{V} \to \mathbb{N},$$

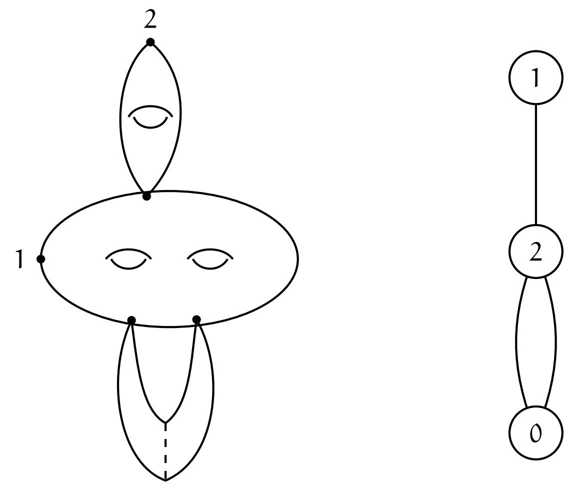
an assignment

$$r \colon \mathbb{H} \to \mathbb{N},$$

and an ordering

$$\mathfrak{m}\colon \mathbb{L} \to \{0, \ldots, n\}.$$

Example 7.7.



For a log or punctured R-map

$$\begin{tikzcd} & \mathfrak{P} \arrow[d] \\ C \arrow[ur, "f"] \arrow[r] \arrow[d] & B\mathbb{C}_\omega^\times \\ S, & \end{tikzcd}$$

we will tropicalize it by taking the associated generalized cone complex of the entire diagram. Because $B\mathbb{C}_\omega^\times$ has no log structure, what we see is

$$\begin{tikzcd} \Sigma(C) \arrow[r, "\Sigma(f)"] \arrow[d] & \Sigma(\mathfrak{P}) = \mathbb{R}_{\geqslant 0} \\ \Sigma(S). & \end{tikzcd}$$

Now suppose that $S = \operatorname{Spec}(Q \to \mathbb{C})$ is a log point with Q an fs sharp monoid. In particular, this means that $\Sigma(S) = Q_{\mathbb{R}}^\vee$. Because everything is (piecewise) linear, it suffices to consider an interior point $t \in \Sigma(S)$ and consider $\Sigma(C)_t$.

From each half edge, we treat it as pointing outward from the attached vertex, and in particular, this means that a linear map to $\mathbb{R}_{\geqslant 0}$ corresponds to a slope. We will call these *contact orders*. In particular, we have $c(h) = -c(\hat{h})$ if $h \neq \hat{h}$ are related by ι. For all $v \in \mathbb{V}$, we also define the *degeneracy*

$$e_v\colon \mathbb{Q}_{\mathbb{R}}^\vee \to \mathbb{R}_{\geqslant 0} \qquad t \mapsto \Sigma(f)_t(v).$$

In particular, $e_v \in Q$ is an integral point.

The next thing we need to introduce are punctured legs. They will be denoted by

$$\mathbb{L}^\circ = \left\{ h \in \mathbb{H} \mid \Sigma(f)_t(h) \text{ is bounded and nontrivial for all } t \in Q^\vee_{\mathbb{R}} \right\}.$$

We will define prestability at $h \in \mathbb{L}^\circ$ such that $0 \in \overline{\Sigma(f)_t(h)}$, which is a technically necessary assumption. Under the assumption that all punctured legs are prestable, we can define the *edge length* function

$$\ell \colon \mathbb{E} \cup \mathbb{L}^\circ \to \mathsf{Map}(Q^\vee_{\mathbb{R}}, \mathbb{R}_{\geqslant 0}) \qquad x \mapsto \ell(x) \colon t \mapsto \ell(x)(t).$$

Note that $\ell(E) \in Q$ is also an integral point for edges E. For punctured legs, this is true up to subdivision.

7.3. **Log GLSM invariants.**

Definition 7.8. We will define the *decorated tropical type* of a log map f as

$$\boldsymbol{\tau} = (G, c \colon \mathbb{H} \to \mathbb{Z}, \beta \colon \mathbb{V} \to H_2(X)).$$

Definition 7.9. A log or punctured map f is of *compact type* if

(1) $\operatorname{Im} \Sigma(f)_t$ is bounded for all t;
(2) $f(p_h) \in 0_{\mathfrak{P}}$ for all half-edges h with contact order 0.

We are now able to define the category $\mathcal{R}(\mathfrak{P}, \tau)$ of stable punctured R-maps fibered over the category LogSch of fs log schemes and marked by τ.

Theorem 7.10. *The stack $\mathcal{R}(\mathfrak{P}, \boldsymbol{\tau})$ is represented by a proper DM stack with basic/minimal log structure.*

Remark 7.11. If $\mathbb{V} = \{v\}$ is a singleton, then we will write

$$\mathcal{R}_{g,c}(\mathfrak{P}, \beta) \coloneqq \mathcal{R}(\mathfrak{P}, \boldsymbol{\tau}).$$

Consider the diagram

$$\begin{tikzcd} & \mathfrak{P} \arrow[d] \\ \mathcal{C} \arrow[ur, "f"] \arrow[r] \arrow[d, "\pi"] & B\mathbb{C}^\times_\omega \\ \mathcal{R}, & \end{tikzcd}$$

where we set $\mathcal{R} \coloneqq \mathcal{R}(\mathfrak{P}, \boldsymbol{\tau})$. Then the perfect obstruction theory is defined by

$$T_{\mathcal{R}/\mathsf{Log}_{\mathfrak{M}_G}} \to \mathbb{E}_\square = R\pi_* f^* T_{\mathfrak{P}/B\mathbb{C}^\times_\omega}.$$

Here, both tangent bundles refer to the log tangent complex. From the perfect obstruction theory, this defines the canonical virtual cycle

$$[\mathcal{R}(\mathfrak{P}, \boldsymbol{\tau})]^{\mathrm{vir}}.$$

The main drawback of this virtual cycle is that it does not reproduce Gromov-Witten theory. However, we can consider how it differs from GW theory, but for this we need a birational modification. Let Q be a sharp monoid. It has a partial ordering by $a < b$ if there exists a nonzero $c \in Q$ such that $a + c = b$.

Definition 7.12. We say that either f or $\Sigma(f)$ has *uniform maximal degeneracy* if there exists $e_{\max} \in \{e_v \mid v \in \mathbb{V}\}$ such that $e_{\max} \geqslant e_v$ for all $v \in \mathbb{V}$.

Remark 7.13. If f is compact type and has uniform maximal degeneracy, then $\Sigma(f)$ is uniformly bounded by e_{max}.

Definition 7.14. Define $\mathcal{U}(\mathfrak{P}, \boldsymbol{\tau})$ to be the stack of stable punctured or log R-maps marked by $\boldsymbol{\tau}$ with uniform maximal degeneracy.

Remark 7.15. If $\mathbb{V} = \{v\}$ is a singleton, then we will write

$$\mathcal{U}_{g,c}(\mathfrak{P}, \beta) \coloneqq \mathcal{U}(\mathfrak{P}, \boldsymbol{\tau}).$$

If one of the contact orders is negative, then we will write

$$\mathcal{U}_{g,c}(\infty, \beta) \coloneqq \mathcal{U}(\mathfrak{P}, \boldsymbol{\tau}).$$

Note that there are two types of vertices: if a vertex is mapped to 0, then either there is at least one half-edge which points upwards or all contact orders are 0, in which case all are contracted. These are called *log vertices*. The other type of vertex is one that does not go to 0. Either all half-edges point downwards, or there are some half-edges which point upwards. These are called ∞*-vertices*. Moreover, the vertices which are compact types are the ones without upwards-pointing half-edges.

If we forget the condition of having uniform maximal degeneracy, then there is a map

$$\mathcal{U}(\mathfrak{P}, \boldsymbol{\tau}) \xrightarrow{F} \mathcal{R}(\mathfrak{P}, \boldsymbol{\tau})$$

which is a subcategory over LogSch. However, as a morphism of algebraic stacks, it is log étale, proper, and surjective. In addition, $\mathcal{U}$ has a canonical perfect obstruction theory which is pulled back from $\mathcal{R}$. In the case when the set of vertices is a singleton, we have

$$F_*[\mathcal{U}_{g,c}(\mathfrak{P}, \beta)]^{\mathrm{vir}} = [\mathcal{R}_{g,c}(\mathfrak{P}, \beta)]^{\mathrm{vir}};$$
$$F_*[\mathcal{U}_{g,c}(\infty, \beta)]^{\mathrm{vir}} = [\mathcal{R}_{g,c}(\infty, \beta)]^{\mathrm{vir}}.$$

In the case of a log vertex, we in fact have inclusions from the moduli of stable maps with p-fields as

$$\begin{tikzcd} \mathcal{U} \arrow[rr, "F"] & & \mathcal{R} \\ & \mathcal{R}_{g,c}(\mathfrak{P}^\circ, \beta). \arrow[ul, hook] \arrow[ur, hook] & \end{tikzcd}$$

If we denote the moduli of p-fields by $\mathcal{R}^\circ$, then the boundary

$$\Delta \coloneqq \mathcal{U} \setminus \mathcal{R}^\circ$$

is a Cartier divisor. By the condition of having uniform maximal degeneracy, we have a section

$$e_{\mathrm{max}} \in H^0(\bar{M}_{\mathcal{U}}).$$

This is equivalent to the data of a morphism

$$\mathcal{U} \to ([\mathbb{A}^1/\mathbb{G}_m], 0)$$

which pulls 0 back to Δ.

We will define a *superpotential* to be a function

$$\begin{tikzcd} \mathfrak{P}^\circ \arrow[rr, "W"] \arrow[dr] & & \mathcal{L}_\omega \arrow[dl] \\ & \mathbb{BC}^\omega_\times. & \end{tikzcd}$$

Its *critical locus* is denoted by

$$\mathrm{Crit}(W) = (\mathrm{d}W = 0).$$

We say that W has *proper critical locus* if $\mathrm{Crit}(W) \to \mathbb{BC}^\times_\omega$ is proper.

For example, suppose we are interested in GW theory. Let X be a smooth projective variety, V be a vector bundle on X, and $s \in H^0(V)$. Set $Z = (s = 0)$. Then we obtain

$$W\colon \mathfrak{P}^\circ = V^\vee \boxtimes \mathcal{L}_\omega \xrightarrow{s \otimes 1_\omega} \mathcal{L}_\omega.$$

Lemma 7.16. *W has proper critical locus if and only if Z is smooth of expected codimension.*

Example 7.17. Let X be a point, $V = \mathbb{C}$, and $s \neq 0$. Then $\mathrm{d}W = s \neq 0$, so the critical locus is empty.

Using the Kiem-Li method of cosection localization, we can obtain a virtual cycle

$$[\mathcal{R}_{G,c=n}(\mathfrak{P}^\circ, \beta)]^{\mathrm{vir}}_W,$$

which is supported on the critical locus. In GW theory, this equals the GW virtual cycle,

$$\pm[\bar{M}_{g,n}(\mathrm{Crit}(W), \beta)]^{\mathrm{vir}},$$

where the sign can be explicitly calculated.

Theorem 7.18. *Let $\boldsymbol{\tau}$ be a of compact type. In the compact type setting,*

(1) $\mathcal{U}(\mathfrak{P}, \boldsymbol{\tau})$ has a reduced perfect obstruction theory, which yields a reduced virtual cycle $[\mathcal{U}(\mathfrak{P}, \boldsymbol{\tau})]^{\mathrm{vir}}$;

(2) Δ has a reduced perfect obstruction theory, which yields a reduced virtual cycle $[\Delta]^{\mathrm{vir}}$;

(3) We have $[\mathcal{U}_{g,c=n}(\mathfrak{P}, \beta)]^{\mathrm{red}} = \pm[\bar{M}_{g,n}(\mathrm{Crit}(W), \beta)]^{\mathrm{vir}}$ in the case of a single log vertex. Moreover, there is the formula

$$[\mathcal{U}_{g,c=n}(\mathfrak{P}, \beta)]^{\mathrm{red}} = [\mathcal{U}_{g,c=n}(\mathfrak{P}, \beta)]^{\mathrm{vir}} - \tilde{r}[\Delta]^{\mathrm{red}},$$

where $\tilde{r}$ is the pole order of W along ∞.

7.4. **Tropical decomposition and finite generation.** From now on, we will consider a simpler setup where X is a smooth projetive variety and V is a line bundle on X. In this case, we will assume that $Z = (s = 0)$ is smooth of codimension 1. Moreover, we will assume that Z is a Calabi-Yau threefold. We will consider the invariants

$$N_{g,\beta} \coloneqq \int_{[\bar{M}_{g,0}(Z,\beta)]^{\mathrm{vir}}} 1 \in \mathbb{Q}$$

and the generating series

$$F_g = \sum_{\beta \geqslant 0} N_{g,\beta} Q^\beta \in \Lambda \coloneqq \mathbb{C}[\![Q]\!].$$

Conjecture 7.19 (Yamaguchi-Yau, Alim-Länge)**.** *Suppose $g = 0$ mirror symmetry holds for Z. Then there exists a finitely-generated $\mathbb{Q}$-algebra $R \subset \Lambda'$ such that for all $g \geqslant 2$, $F_g \in (I_0(q))^{2g-2}R$.*

Our main goal is the following.

Theorem 7.20. *Let Z be a Calabi-Yau threefold hypersurface in a Fano fourfold X with no odd cohomology. Let $\{\phi_j\}$ be a basis of $H^*(X)$ and $\left\{\phi^j\right\}$ be the dual basis. Then for all $g \geqslant 0$, we have*

$$F_g = (-1)^{1-g} \sum_{\tau' \in G'_g} \frac{(-1)^{|\mathbb{V}_\infty|}}{\mathrm{Aut}(\tau')} \cdot \prod_{E \in \mathbb{E}} c(E)$$

$$\cdot \sum_{j_E | E = \left\{E_{\log}, E_\infty\right\} \in \mathbb{E}} \prod_{\substack{v \in \mathbb{V}_\infty \\ g=1 \\ \mathbb{H}_v = \{E_\infty\}}} c_1^{\mathrm{eff}}(\phi^{j_E}) \cdot \prod_{\substack{v \in \mathbb{V}_\infty \\ g(v) \geqslant 2}} \tilde{Q}^{\beta(v)} \cdot \exp\left(\int_{\beta(v)} K(Q)\right)$$

$$\cdot \left(\prod_{E_\infty \in \mathbb{H}'_v} \int \beta(v) \phi^{j_E}\right) \cdot c^{\mathrm{eff}}_{g(v), \beta(v)} \cdot \prod_{v \in \mathbb{V}_{\log}} \Omega_{g(v), |\mathbb{H}_1(v)|, |\mathbb{H}_2(v)|}(\phi_{j_E} : E_{\log} \in \mathbb{H}_1(v)).$$

Example 7.21. Let Y be a toric variety with its toric log structure. Then $\Sigma(Y)$ is the toric fan treated as a cone complex, and we can consider its boundary $\Delta_Y = \sum_{\rho \in \Sigma(Y)^1} \Delta_\rho$.

If we consider $[\Delta]^{\mathrm{red}}$, we should obtain the similar formula

$$[\Delta]^{\mathrm{red}} = \sum_{\rho \in \Sigma(\Delta)(1)} [\Delta_\rho]^{\mathrm{red}}.$$

The question now is what the ρ represent. In fact, they parameterize rigid tropical curves. In the tropicalization

$$\begin{tikzcd} \Sigma(C)_\rho \arrow[r] \arrow[d] & \Sigma(\mathcal{C}) \arrow[r, "f"] \arrow[d] & \mathbb{R}_{\geqslant 0} \\ \rho \arrow[r, hook] & \Sigma(\mathcal{U}), \end{tikzcd}$$

we see that f_ρ is rigid if and only if the source tropical curve is a bipartite graph with one set of vertives going to $e_{\max}$ and the other set of vertices going to 0.

Example 7.22. Let X be a point and $Z = \emptyset$. Also consider $g = 2, n = 0$. Then the possible graphs are given in Figure 5.

Theorem 7.23 (Tropical quantum Lefschetz)**.** *Let X and Z be as before. For any $\beta \in H_2(X)$, any $\alpha_1, \ldots, \alpha_n \in H^*(X)$, and any $k_1, \ldots, k_n \in \mathbb{N}$, we have*

$$\int_{[\bar{M}_{g,n}(Z,\beta)]^{\mathrm{vir}}} \prod_{j=1}^n \mathrm{ev}_j^*(\alpha_j)\psi^{k_j} = (-1)^{1-g+\beta\cdot Z} \cdot \prod_{\tau \vdash (g,n,\beta)} \frac{(-1)^{|\mathbb{V}_\infty|}}{|\mathrm{Aut}(\tau)|} \cdot \prod_{E \in \mathbb{E}} c(E)$$

$$\cdot \sum_{\left\{j_E | E = \left\{E_{\log}, E_\infty\right\}\right\}} \prod_{v \in \mathbb{V}_\infty} \int_{[\mathcal{U}_v]^{\mathrm{red}}} \prod_{E_\infty \in \mathbb{H}_v} \mathrm{ev}^*_{E_\infty} \phi^{j_E}$$

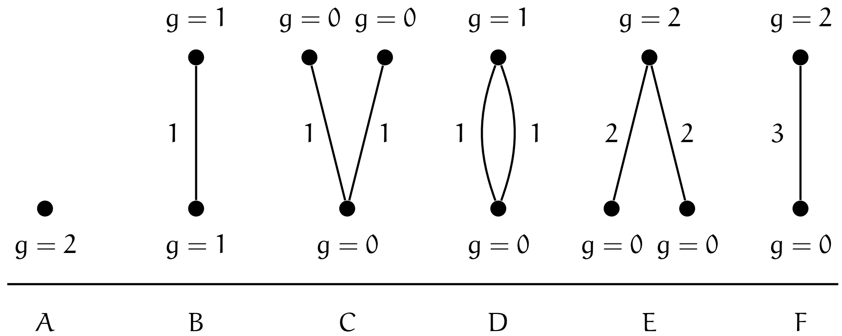


FIGURE 5. Graphs of tropical types for $X = \mathrm{pt}$ and $Z = \emptyset$ when $g = 2$.

$$\cdot \prod_{v \in \mathbb{V}_{\log}} \int_{[\mathcal{R}_v]^{\mathrm{vir}}} \prod_{E_{\log}} \mathrm{ev}^*_{E_{\log}}(\phi_{j_E}) \cdot \prod_{h \in \mathbb{L}_v} \mathrm{ev}^*_h \alpha_h \cdot \psi_h^{k_n}.$$

Remark 7.24. There is also a virtual localization formula, which can be applied to either the reduced theory (with only the R-torus action) or the canonical theory (which can be applied to the full automorphisms of the ambient space).

Now consider an ∞-vertex. Set $\tau_v = (g, c, \beta)$ and set $\mathcal{U}_v = \mathcal{U}_{g,c}(\infty, \beta)$. Then the first computation is that

$$\begin{aligned} \operatorname{red}\dim \mathcal{U}_v &= \chi(f^* T_{\mathfrak{P}/B\mathbb{C}^\times_\omega}) + 1 + \operatorname{trop}\dim \\ &= \operatorname{vir}\dim \bar{M}_{g,n}(Z, \beta) + \sum_{h \in \mathbb{H}} (c(h) + 1), \end{aligned}$$

where the tropical dimension is $3g - 4 + n$. Note that all contact orders are negative, so the second term is at most zero. In particular, we have proved that

Corollary 7.25. *If* $\operatorname{vir}\dim \bar{M}_{g,n}(Z, \beta) < 0$, *then* $[\mathcal{U}_v]^{\mathrm{red}} = 0$.

We also have an *balancing condition*, which states that

$$\sum_{h \in \mathbb{H}} c(h) = \deg f^* \mathcal{O}(\infty) = \int_\beta c_1(V) - (2g - 2 + n). \tag{1}$$

This can be rewritten as

$$0 \geqslant \sum_{h \in \mathbb{H}} (c(h) + 1) = \int_\beta c_1(V) - (2g - 2).$$

Proposition 7.26. *We see that ∞-vertices do not exist in the following cases:*

(1) We have $g = 0$ *and* V *is nef;*
(2) We have $g = 1$ *and* V *is ample, unless* $\beta = 0$ *and all* $c(h) = -1$.

Definition 7.27. A marking is a *unit* if its contact order is -1.

Notation 7.28. We will denote the unit sector by $\mathbf{1}$ and denote $\tau_{v+\mathbf{1}} = (g, c + \mathbf{1}, \beta)$, $\mathcal{U}_{v+\mathbf{1}} = U_{g,c+\mathbf{1}}(\infty, \beta)$, and $\mathcal{R}_{v+\mathbf{1}} = \mathcal{R}_{g,c+\mathbf{1}}(\infty, \beta)$.

Theorem 7.29.

(1) There is a map $F_{\mathbf{1}} \colon \mathcal{U}_{v+\mathbf{1}} \xrightarrow{s} \mathcal{C}^\circ \xrightarrow{\pi} \mathcal{U}_v$, *where* s *is birational;*

(2) *We have the equality* $s_*[\mathcal{U}_{v+\mathbf{1}}]^{\mathrm{red}} = \pi^*[\mathcal{U}_v]^{\mathrm{red}}$ *of reduced virtual cycles;*
(3) *If* D *is a divisor on* X*, then* $F_{\mathbf{1}*}(\mathrm{ev}_{\mathbf{1}}^* D \cap [\mathcal{U}_{v+\mathbf{1}}]^{\mathrm{red}}) = \left(\int_\beta D\right)[\mathcal{U}_v]^{\mathrm{red}}$*;*
(4) *Consider the diagram*

$$\begin{array}{ccc} [\mathrm{U}]_{v+\mathbf{1}} & \xrightarrow{F} & \bar{M}_{g,n+1} \\ {\scriptstyle F_{\mathbf{1}}}\downarrow & & {\scriptstyle p}\downarrow \\ \mathcal{U}_v & \xrightarrow{F} & \bar{M}_{g,n}. \end{array}$$

Then $F_*[\mathcal{U}_{v+\mathbf{1}}]^{\mathrm{red}} = p^*F_*[\mathcal{U}_v]^{\mathrm{red}}$.

Example 7.30. Consider the setting where $g = 1$ and $Z_d \subset X = \mathbb{P}^N$ is a hypersurface. We only need to consider $\beta = 0$ and $c(h) = -1$ for all $h \in H$. Then we see that

$$\prod_h \mathrm{ev}_h^* \alpha_h \cdot [\mathcal{U}_{1,c}(\infty,\beta)]^{\mathrm{red}} = \mathrm{ev}_{h^\star}^*\left(\prod_h \alpha_h\right) \cap [\mathcal{U}_{1,\mathbf{1}}(\infty,\beta)]^{\mathrm{red}}.$$

Proposition 7.31. *We in fact have*

$$\mathcal{U}_{1,\mathbf{1}}(\infty,0) \simeq \mathcal{R}_{1,\mathbf{1}}(\infty,0)$$

and the equality

$$[\mathcal{U}_{1,\mathbf{1}}(\infty,0)]^{\mathrm{red}} = c_{\dim X}\left(\mathcal{H}^\vee \boxtimes T_{\mathfrak{P}/B\mathbb{C}_\omega^\times} + \mathcal{O} - V\right).$$

Example 7.32. Now consider the quintic threefold $Z_5 \subset \mathbb{P}^4$. Then we see that $\operatorname{red}\dim \mathcal{U}_v = \sum_{h \in \mathbb{H}}(c(h)+2)$. The unit axiom reduces us to the case when $c(h) \leqslant -2$. We must now determine the number of new invariants we need to consider if all $c(h) = -2$, which is equivalent to finding the number of possible β. The balancing condition states that

$$\begin{aligned}\sum_{h\in\mathbb{H}} c(h) &= \int_\beta c_1(V) - (2g-2+n) \\ &= 5\beta - (2g-2+n),\end{aligned}$$

which implies that $0 \geqslant \sum(c(h)+1) = 5\beta - (2g-2)$ and therefore that $\beta \leqslant \frac{2g-2}{5}$.[13]

Conjecture 7.33 (Janda)**.**

(1) *We have* $K(Q) = \log I_0(q) \cdot c_1(K_X^\vee) - \frac{I_1}{I_0}$
(2) *The canonical theory satisfies* $\Omega \in I_0(q)^{2g-2+2n_1+3n_2} \cdot R$.

Corollary 7.34. *The above conjecture for* (X, V) *implies Yamaguchi-Yau finite generation for* Z.

[13] This is exactly the number of unknowns predicted if we assume finite generation, the holomorphic anomaly equations, orbifold regularity, and the conifold gap condition.

8. Structure of higher-genus GW invariants of the quintic (Shuai Guo)

8.1. **The problem.** We will consider the quintic threefold $X = X_5 \subset \mathbb{P}^4$ and ask how many curves of genus g and degree d it contains. In genus zero, the answers begin

$$2875, \qquad 609250, \qquad 317206375, \qquad \ldots$$

Because X is Calabi-Yau, the moduli space $\overline{\mathcal{M}}_{g,n}(X, \beta)$ has virtual dimension zero, and the string equation gives $\langle \mathbf{1}, \gamma_2, \ldots, \gamma_n \rangle_{g,n,\beta} = 0$ unless $(g, n, \beta) = (0, 3, 0)$. Therefore the primary theory is carried by the numbers

$$N_{g,\beta} \coloneqq \langle\rangle_{g,0,\beta} = \int_{[\overline{\mathcal{M}}_g(X,\beta)]^{\mathrm{vir}}} 1$$

and we would like to compute the generating series

$$F_g(Q) \coloneqq \sum_{\beta \text{ effective}} N_{g,\beta} Q^\beta.$$

These are rational numbers, and the integers we actually want are the BPS invariants $n_{g,\beta}$, which are defined from the positive-degree part F_g^+ by the multicover transformation

$$\sum_{g \geqslant 0} F_g^+(Q) \lambda^{2g-2} = \sum_{\substack{g \geqslant 0 \\ \beta > 0}} n_{g,\beta} \sum_{k \geqslant 1} \frac{1}{k} \left(2 \sin \frac{k\lambda}{2} \right)^{2g-2} Q^{k\beta}.$$

Note that only the $n_{g,\beta}$ are expected to be integral, and that the numbers above are the $n_{0,d}$.

Table 2. Leading BPS invariants $n_{g,d}$ of the quintic threefold

g	$d=1$	$d=2$	$d=3$	$d=4$	$d=5$
0	2875	609250	317206375	242467530000	229305888887625
1	0	0	609250	3721431625	12129909700200
2	0	0	0	534750	75478987900
3	0	0	0	8625	−15663750
4	0	0	0	0	49250
5	0	0	0	0	1100
6	0	0	0	0	10
7	0	0	0	0	0

Every column of Table 2 eventually vanishes. This is the Castelnuovo bound, which for the quintic states that $n_{g,d} = 0$ once $g > \frac{d^2+5d+10}{10}$ and was proved by Liu-Ruan. Integrality of the entries is the Gopakumar-Vafa conjecture, which was proved by Ionel-Parker.

In genus zero, Candelas-de la Ossa-Green-Parkes computed F_0 using period integrals on the mirror, which we discussed in Section 3.2. In higher genus, Bershadsky-Cecotti-Ooguri-Vafa gave a recursion which produces F_g from the F_h with $h < g$ together with finitely many unknown constants. Determining those constants is the hard part, and these lectures explain how log GLSM does it.

8.2. **Genus zero and the polynomial structure of the B-model.** Recall from Section 3.2 that for the quintic we consider the I-function

$$I(q,z) = z\sum_{d\geqslant 0} q^{H/z+d}\frac{\prod_{m=1}^{5d}(5H+mz)}{\prod_{m=1}^{d}(H+mz)^5} = I_0 z + I_1 H + I_2\frac{H^2}{z} + I_3\frac{H^3}{z^2},$$

whose components are annihilated by the Picard-Fuchs operator

$$D^4 - 5q(5D+1)(5D+2)(5D+3)(5D+4), \qquad D = q\frac{d}{dq}.$$

Writing $J_k = I_k/I_0$ and $\frac{I_1}{I_0} = \log q + \tau(q)$, so that $Q = e^{I_1/I_0} = qe^{\tau(q)} = q + O(q^2)$, the genus zero mirror theorem states that

$$F_0(Q) = \frac{5}{2}(J_1J_2 - J_3).$$

More generally, let $\mathcal{X} = \{(X,\omega)\}$ be a family of Calabi-Yau threefolds with Kähler moduli $\mathcal{M} \ni Q = e^{2\pi i\omega}$ and let $\mathcal{X}^\vee = \left\{(X^\vee, J)\right\}$ be a mirror family with complex moduli $\mathcal{M}^\vee \ni q$. The solutions of the Picard-Fuchs equation are periods on $X^\vee$ and assemble into

$$I(z) = zI_0\varphi_0 + I_1 + I_2z^{-1} + I_3z^{-2},$$

which lies in the span of $z\varphi_0, \varphi_\alpha, \varphi^\alpha z^{-1}, \varphi^0 z^{-2}$. Here $I_1 = \sum_\alpha I_1^\alpha\varphi_\alpha$, $I_2 = \sum_\alpha I_{2,\alpha}\varphi^\alpha$, and $I_3 = I_{3,0}\varphi^0$, and the mirror map is again $Q = e^{I_1/I_0} = qe^{\tau(q)}$.

The complex moduli line of the quintic has three special points, namely the large complex structure point $q = 0$, the conifold point $q = 5^{-5}$, and the orbifold (or Landau-Ginzburg) point $q = \infty$. The mirror map is defined at $q = 0$ and the expansion there produces Gromov-Witten invariants, while in higher genus the boundary conditions will be imposed at the other two.

We now restate the mirror theorem in terms of three-point functions. Pick local coordinates q_i on $\mathcal{M}^\vee$, let $\{\phi_i\}$ be the corresponding basis on the A-side, and define

$$C^A_{ijk}(Q) \coloneqq I_0^2\langle\phi_i,\phi_j,\phi_k\rangle^X_{0,3}, \qquad C^B_{ijk}(q) \coloneqq \int_{X^\vee}\partial_i\,\partial_j\,\partial_k\Omega\wedge\Omega.$$

Theorem 8.1 (Mirror theorem)**.** *We have $C^A_{ijk}(Q(q)) = C^B_{ijk}(q)$. Moreover $C^B_{ijk} \in \Delta^{-1}\mathbb{Q}[q]$, so it is a globally defined meromorphic function on $\mathcal{M}^\vee$ with poles only along the discriminant.*

Note that after multiplying by the correct power of I_0 and applying the mirror map, a genus-zero generating function becomes a rational function of q with poles only along the discriminant, and is therefore determined by finitely many coefficients.

Example 8.2. For the quintic in the hyperplane basis,

$$C_{111} = \frac{5}{1-5^5q}.$$

Example 8.3. For $X = X_{3,3} \subset \mathbb{P}^2\times\mathbb{P}^2$, the Yukawa couplings are

$$C_{111} = \frac{3^4q_1(2+3^3q_1+3^3q_2)}{\Delta}, \qquad C_{222} = \frac{3^4q_2(2+3^3q_1+3^3q_2)}{\Delta},$$

$$C_{112} = \frac{(1-3^3q_1-3^3q_2)(1+2\cdot 3^3q_1-3^3q_2)}{\Delta},$$
$$C_{122} = \frac{(1-3^3q_1-3^3q_2)(1-3^3q_1+2\cdot 3^3q_2)}{\Delta},$$

where

$$\Delta = 1-3\cdot 3^3(q_1+q_2)+3\cdot 3^6(q_1^2-7q_1q_2+q_2^2)-3^9(q_1+q_2)^3.$$

In higher genus, we expect normalized potentials $f^A_{g,i}$ satisfying

$$f^A_{g,i}(Q(q)) = f^B_{g,i}(q) \in \Delta^{-(2g-2+n)}\mathbb{Q}[q],$$

where the right hand side is computed by a sum over graphs.

8.3. **The BCOV Feynman rule.** The BCOV recursion is a Feynman rule, and it comes from a finite-dimensional model for the quantization of a symplectic transformation as in Section 3.6. Let $\mathcal{H} = \mathcal{H}^- \oplus \mathcal{H}^+ \subset \mathcal{H}_\infty$ be symplectic with $d+1 = \dim \mathcal{H}^+ = \dim \mathcal{H}^-$ and set

$$R = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \operatorname{End}\mathcal{H}, \qquad \mathcal{Q}(\mathbf{x},\mathbf{p}) = (\mathbf{p}, D^{-1}\mathbf{x}) - \frac{1}{2}(\mathbf{p}, D^{-1}C\mathbf{p}).$$

Given a total potential $F(\hbar,\mathbf{x})$, the transformed potential is defined formally by the Gaussian integral

$$f(\hbar,\mathbf{x}) \coloneqq (\widehat{R}F)(\hbar,\mathbf{x}) = \ln \int e^{\hbar^{-1}(\mathcal{Q}(\mathbf{x},\mathbf{p}')-\mathbf{x}'\mathbf{p}')+F(\hbar,\mathbf{x}')}\, d\mathbf{x}'\, d\mathbf{p}'.$$

Expanding by stationary phase gives a sum over graphs, where $D^{-1}C$ supplies the propagator on each edge and F supplies the vertices.

For a Calabi-Yau threefold the propagators are written $S^{\varphi\varphi}$, S^{φ}, and S according to the number of indices, and the vertices are the normalized A-model correlators

$$P^A_{g,n}(Q) = I_0^{2-2g}\langle \phi_1,\ldots,\phi_n\rangle^X_{g,n}.$$

Note that these have nothing to do with the operator $S_\tau(z)$ solving the quantum differential equation, which will appear later.

Example 8.4. For the quintic in genus two with no markings, the Feynman rule reads

$$\begin{aligned} f^B_2 &= P_2 + \frac{1}{2}S^{\varphi\varphi}P_{1,1}^2 + \frac{1}{2}S^{\varphi\varphi}P_{1,2} + (S^{\varphi\varphi})^2P_{1,1} + \frac{1}{8}(S^{\varphi\varphi})^2P_{0,4} \\ &\quad + \frac{1}{8}(S^{\varphi\varphi})^3 + \frac{1}{12}(S^{\varphi\varphi})^3 + \frac{\chi}{24}S^{\varphi}P_{1,1} + \frac{1}{2}\frac{\chi}{24}S^{\varphi}S^{\varphi\varphi} \\ &\quad + \frac{1}{2}\frac{\chi}{24}\Big(\frac{\chi}{24}-1\Big)S, \end{aligned}$$

where $\chi = \chi(X_5) = -200$. This is a cubic polynomial in the propagators.

The graphs here are stable graphs, as in Section 3.4. A vertex ν contributes the lower-genus correlator $P_{g_\nu,\mathrm{val}(\nu)}$, an internal edge contracts two of the insertions against a propagator, the legs record the remaining insertions, and the graph is weighted by $\frac{1}{|\operatorname{Aut}\Gamma|}$. As usual

$$g(\Gamma) = h^1(\Gamma) + \sum_\nu g_\nu, \qquad 2g_\nu - 2 + \mathrm{val}(\nu) > 0,$$

and f^B_g is the sum of the contributions of all stable graphs of genus g.

8.4. **The holomorphic anomaly equation.** The recursion can also be written as a differential equation. The B-model amplitudes $\mathcal{F}_g(z, \bar{z})$ are not holomorphic, and for $g \geqslant 2$ their failure to be holomorphic is given by

$$\partial_{\bar{\imath}} \mathcal{F}_g = \frac{1}{2} \bar{C}_{\bar{\imath}}^{jk} \left(D_j D_k \mathcal{F}_{g-1} + \sum_{r=1}^{g-1} D_j \mathcal{F}_r D_k \mathcal{F}_{g-r} \right).$$

Here C_{ijk} is the Yukawa coupling, $\bar{C}_{\bar{\imath}}^{jk}$ is obtained from $\bar{C}_{\bar{\imath}\bar{\jmath}\bar{k}}$ by raising two indices with the Weil-Petersson metric, and D_j is the covariant derivative on the appropriate power of the Hodge line bundle. The two terms on the right are the two boundary strata of $\overline{\mathcal{M}}_g$, namely a non-separating and a separating node, and the $\frac{1}{2}$ forgets the ordering of the two branches.

Choosing local $\bar{\partial}$-primitives

$$\partial_{\bar{\imath}} S^{jk} = \bar{C}_{\bar{\imath}}^{jk}, \qquad \partial_{\bar{\imath}} S^j = G_{\bar{\imath}\ell} S^{\ell j}, \qquad \partial_{\bar{\imath}} S = G_{\bar{\imath}\ell} S^{\ell},$$

the anomaly equation becomes, modulo a holomorphic term, the polynomial PDE

$$\frac{\partial \mathcal{F}_g}{\partial S^{jk}} = \frac{1}{2} \left(D_j D_k \mathcal{F}_{g-1} + \sum_{r=1}^{g-1} D_j \mathcal{F}_r D_k \mathcal{F}_{g-r} \right).$$

These propagators are only defined up to holomorphic tensors, and this freedom is the source of the holomorphic ambiguity.

Remark 8.5. In the one-parameter normalized variables we will use for the quintic, this reads

$$-\partial_0 P_g = \frac{1}{2} P_{g-1,2} + \frac{1}{2} \sum_{r=1}^{g-1} P_{r,1} P_{g-r,1}, \qquad -\partial_1 P_g = 0.$$

The graph sum satisfies this equation because in an unmarked potential the non-holomorphic generators occur only in the edge factors. A derivative in a propagator direction therefore picks out a single edge and cuts it, the two half-edges become markings on the resulting lower-genus vertices, and the covariant derivatives correspond to those markings by the divisor equation. Cutting a loop produces $P_{g-1,2}$, while cutting a separating edge produces $P_{g_1,1} P_{g_2,1}$.

Remark 8.6. The implication only goes one way. The BCOV graph rule implies the anomaly equation, but the anomaly equation by itself does not determine the graph rule, because to recover the same solution one also needs the propagator convention, the lower-genus vertex data, and the boundary conditions.

If all lower-genus vertices are known, then the right hand side is known, and integrating with respect to the non-holomorphic generators gives

$$P_g = P_g^{\text{particular}} + f_g(Y),$$

where finite generation gives the holomorphic ambiguity

$$f_g(Y) = P_g|_{A=B_1=B_2=B_3=0} = \sum_{i=0}^{3g-3} a_{i,g} Y^i, \qquad Y = (1 - 5^5 q)^{-1}.$$

Therefore the anomaly equation leaves $3g - 2$ unknown coefficients in each genus.

Remark 8.7. For the quintic, the degree-zero data together with orbifold regularity remove $\lceil \frac{3g-3}{5} \rceil + 1$ of these freedoms and the conifold gap supplies $2g-2$ further conditions, so that

$$3g - 2 - \left(\left\lceil \frac{3g-3}{5} \right\rceil + 1 \right) - (2g-2) = \left\lfloor \frac{2g-2}{5} \right\rfloor$$

parameters remain and are genuinely new effective invariants. This is exactly the number of ∞-vertex unknowns predicted by Equation (1), namely the number of β with $\beta \leqslant \frac{2g-2}{5}$.

The conjectural algorithm of the physicists therefore consists of the BCOV Feynman rule, orbifold regularity at the Landau-Ginzburg point (a form of the LG/CY correspondence, which for the quintic is work in progress of Guo-Ruan-Janda), and the conifold gap condition. To these we may add the Gopakumar-Vafa integrality and the Castelnuovo bound.

8.5. **Four theories.** Recall from Section 7 that the log GLSM carries two virtual cycles. The reduced cycle recovers quintic Gromov-Witten theory, while the canonical equivariant cycle admits a root of unity specialization which is semisimple and therefore computable by Givental-Teleman. The bipartite graph sum

$$\Omega^{\mathrm{GW}}_{g,n} = \sum_{\Gamma \text{ bipartite}} \frac{1}{|\mathrm{Aut}\,\Gamma|} \left(\prod_{v \in V_\infty(\Gamma)} c^{\mathrm{eff}}_{g_v, \beta_v} \right) \mathrm{Cont}^{\mathrm{can}}_{\Gamma}$$

expresses the former in terms of the latter, at the cost of introducing the effective invariants $c^{\mathrm{eff}}_{g,\beta}$ at the ∞-vertices. Replacing these by formal parameters $\mathbf{C} = \{c_{g,\beta}\}$ gives the *extended* $\mathbf{C}$*-family*, whose vertex data is supplied by the meromorphic and rubber log GLSM theories.

There are four theories in play.

- The t-*twisted geometric theory* sets $\lambda_i = 0$ and keeps the auxiliary weight t. It has both stable-map and stable-quotient versions, and the quotient version gives the genus two localization formula below.
- The *formal quintic*, or formal λ-twisted theory, sets $t = 0$ and $\lambda_i = \zeta_5^i \lambda$. It is a semisimple CohFT, so it has a universal and computable R-matrix.
- The *extended* $\mathbf{C}$*-family* interpolates between these, and is where bipartite graphs with both formal and meromorphic vertices live.
- *Quintic Gromov-Witten theory* is obtained by setting $\mathbf{C} = \mathbf{C}^{\mathrm{eff}}$ and then letting $\lambda \to 0$, and is what we want.

Warning 8.8. In higher genus the t^0 coefficient of the t-twisted stable quotient theory is not the λ^0 coefficient of the formal λ-twisted theory.

8.6. **The formal quintic.** Let $E = \mathcal{O}_{\mathbb{P}^4}(5)$ and let $\pi\colon \mathcal{C} \to \overline{\mathcal{M}}_{g,n}(\mathbb{P}^4, d)$ be the universal curve. The t-twisted cycle is

$$e(R\pi_* f^* E \otimes [1]) \cap [\overline{\mathcal{M}}_{g,n}(\mathbb{P}^4, d)]^{\mathrm{vir}},$$

where $[1]$ has equivariant first Chern class t. Starting instead from the fully $(\mathbb{C}^\times)^5$-equivariant cycle, the formal quintic is the root of unity specialization

$$[\overline{\mathcal{M}}_{g,n}(\mathbb{P}^4, d)]^{\mathrm{vir},\lambda} := \Big(e(R\pi_* f^* E \otimes [1]) \cap [\overline{\mathcal{M}}_{g,n}(\mathbb{P}^4, d)]^{\mathrm{vir}}_{(\mathbb{C}^\times)^5} \Big)\Big|_{t=0,\, \lambda_i = \zeta_5^i \lambda},$$

where $\zeta_5 = e^{2\pi i/5}$. The relation $H^5 = \lambda^5$ makes the quantum product generically semisimple while retaining λ as a parameter.

The state space, unit, and pairing are

$$\mathcal{H}_\lambda = \mathbb{Q}(\lambda)[H]/(H^5-\lambda^5), \quad \mathbf{1} = 1, \quad (\gamma_1,\gamma_2)^\lambda = \int_{\mathbb{P}^4,(\mathbb{C}^\times)^5} e(\mathcal{O}(5))\gamma_1\gamma_2\Big|_{\lambda_i=\zeta_5^i\lambda},$$

and the CohFT classes are

$$\Omega^\lambda_{g,n}(\gamma_1,\ldots,\gamma_n) = \sum_{d\geqslant 0} q^d \rho_*\left(\prod_{i=1}^n \mathrm{ev}_i^*(\gamma_i)\cap[\overline{\mathcal{M}}_{g,n}(\mathbb{P}^4,d)]^{\mathrm{vir},\lambda}\right),$$

where ρ forgets the map and stabilizes the domain. After shifting by $\tau(q)H$ with $\tau(q) = I_1^\lambda(q)/I_0^\lambda(q)$, the shifted theory $\Omega^{\lambda,\tau}$ is semisimple, and after extending scalars to $\mathbb{Q}(\zeta_5,\lambda)$ the idempotent basis satisfies

$$\omega^\lambda_{g,n}(e_{\alpha_1},\ldots,e_{\alpha_n}) = \sum_\alpha \Delta_\alpha^{g-1}\prod_i \delta_{\alpha,\alpha_i}.$$

Therefore Theorem 3.24 applies and gives

$$\Omega^{\lambda,\tau}_{g,n} = \sum_{\Gamma\in G_{g,n}} \frac{1}{|\mathrm{Aut}\,\Gamma|}\iota_*\,\mathrm{Cont}_\Gamma$$

with the decorations of Section 3.4. A leg carrying γ gets $R^{-1}(\bar\psi)\gamma$, an edge gets

$$V_R(z_1,z_2) = \frac{\sum_\alpha e_\alpha\otimes e^\alpha - \sum_\alpha R^{-1}(z_1)e_\alpha\otimes R^{-1}(z_2)e^\alpha}{z_1+z_2},$$

and a vertex gets the topological field theory ω, with the missing unit-preserving part inserted at forgotten markings via $T(z) = z(\mathbf{1}-R^{-1}(z)\mathbf{1})$.

Remark 8.9. The edge factor has no pole. Setting $z_2=-z_1$ and using the symplectic condition in the form $R^{-1}(z) = R^*(-z)$, the numerator vanishes, so it is divisible by z_1+z_2 and V_R is a power series in the two cotangent line variables. The coefficient of $z_1^p z_2^q$ is the class placed at a node with $\bar\psi_1^p\bar\psi_2^q$, and contracting it against the two adjacent topological field theory tensors glues the branches.

8.7. **How to compute the** R**-matrix.** The genus zero input is the formal equivariant I-function

$$I^\lambda(q,z) = z\sum_{d\geqslant 0} q^d\frac{\prod_{m=1}^{5d}(5H+mz)}{\prod_{m=1}^{d}((H+mz)^5-\lambda^5)} = zI_0 + I_1^\lambda H + z^{-1}I_2^\lambda H^2+\cdots.$$

The coefficients through H are independent of λ, so $I_1^\lambda = I_1 - I_0\log q$ and the mirror shift $\tau(q) = I_1^\lambda(q)/I_0(q)$ is free of logarithms. Set

$$D = q\frac{d}{dq}, \qquad Y = (1-5^5q)^{-1}, \qquad L = Y^{1/5}$$

and introduce the bases

$$\phi_k = I_{1,1}\cdots I_{k,k}H^k, \qquad \bar\phi_k = I_0L^{-k-1}\phi_k,$$

where the diagonal Birkhoff factors are defined recursively by

$$I_{p,0} = I_p, \qquad I_{p,r} = \frac{I_{p-1,r-1}}{I_{r-1,r-1}} + D\left(\frac{I_{p,r-1}}{I_{r-1,r-1}}\right), \quad 1\leqslant r\leqslant p,$$

so that in particular $I_{1,1} = 1 + D\tau$. The idempotents and canonical coordinates are

$$e_\alpha = \frac{1}{5}\sum_{i=0}^{4} h_\alpha^{-i}\frac{\phi_i}{L^i}, \qquad u_\alpha = h_\alpha \int_0^q (L(s)-1)\frac{ds}{s}, \qquad h_\alpha = \zeta_5^\alpha \lambda.$$

The q-dependence of the R-matrix is forced by the quantum differential equation. The fundamental solution satisfies

$$zDS_\tau(z) + S_\tau(z)H = M_\tau S_\tau(z), \qquad M_\tau = \dot{\tau}\star, \quad \dot{\tau} := (1 + D\tau)H = I_{1,1}H,$$

and in the normalized canonical direction α the stationary phase asymptotics give

$$(S_\tau)_{i\bar{\alpha}}(z)C_\alpha(z)^{-1} = e^{u_\alpha/z}R_{i\bar{\alpha}}(z),$$

so that

$$(zD + h_\alpha L)R_{i\bar{\alpha}} = \sum_j (M_\tau)_i^j R_{j\bar{\alpha}}.$$

Here $C_\alpha(z)$ is the $q = 0$ quantum Riemann-Roch (equivalently, Gamma function) factor, and together with the symplectic condition it fixes the constants that the differential equation cannot see. As at the end of Section 3.7, the differential equation gives a recursion and an oscillating integral gives the initial conditions.

Writing $R(z) = \mathrm{Id} + R_1 z + R_2 z^2 + \cdots$, the equation above determines each R_{k+1} from the preceding ones, so the series is computed one layer at a time. Two features make the quintic tractable. First, the root of unity basis has five sectors and a mod 5 charge rule permits only one residue pattern in each row, so most matrix entries vanish before any computation starts. Second, every coefficient lies in the finitely generated ring

$$\mathcal{R} = \mathbb{Q}[A, B_1, B_2, B_3, Y], \qquad A = \frac{DI_{1,1}}{I_{1,1}}, \quad B_k = \frac{D^k I_0}{I_0},$$

so the eventual stable graph sum is finite algebra rather than an uncontrolled analytic expansion.

Remark 8.10. A fixed genus only needs finitely much of R for dimensional reasons. On $\overline{\mathcal{M}}_{g,n}$ only classes of total degree at most $3g-3+n$ contribute and everything else integrates to zero, so for an unmarked potential we need only $R_0, \ldots, R_{3g-3}$. Genus 4 stops at z^9 and genus 5 at z^{12}.

8.8. **Genus two.** We will now carry out both routes in genus two. The geometric route uses the t-twisted stable quotient theory, applies virtual localization to get five contributions, and then applies stable quotient wall crossing to land on quintic Gromov-Witten theory. The formal route uses the semisimple λ-twisted theory and the R-matrix to compute a stable graph sum, and gives only the formal leading part, so the effective and meromorphic rest graphs must still be added.

Notation 8.11. Write

$$\langle \alpha_1, \ldots, \alpha_n \rangle_{g,n}^{t,SQ} := \sum_{d \geqslant 0} q^d \int_{[\overline{\mathcal{M}}_{g,n}(\mathbb{P}^4, d)]^{\mathrm{vir}}} e(R\pi_* \mathcal{L}^{\otimes 5} \otimes [-1]) \prod_i \mathrm{ev}_i^*(\alpha_i),$$

where $\mathcal{L}$ is the universal stable quotient line bundle and $c_1^{\mathbb{C}^\times}([-1]) = -t$. This is the stable quotient sign convention, and is not the same as the [1] appearing in the stable map cycle above.

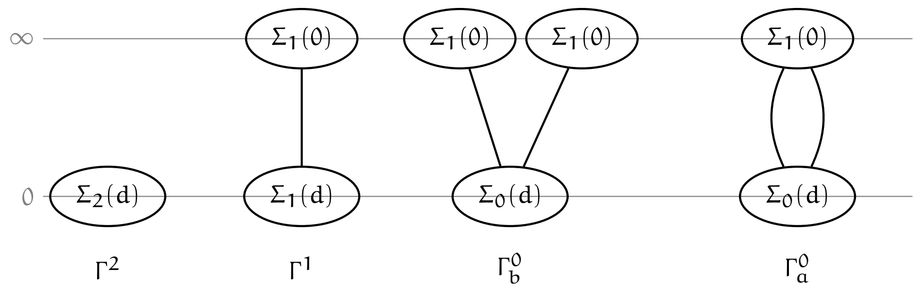


FIGURE 6. The four non-constant localization graphs in genus two, with each vertex drawn as its Riemann surface component.

The four non-constant graphs are drawn in Figure 6, and their contributions are

$$\mathrm{Cont}_{\Gamma^2} = \langle\rangle_{2,0}^{t,SQ},$$

$$\mathrm{Cont}_{\Gamma^1} = -\left\langle \frac{-\frac{5}{2}H^3 + \frac{5}{24}H^4 t^{-1}}{(t-5H)(t-5H-\psi)} \right\rangle_{1,1}^{t,SQ},$$

$$\mathrm{Cont}_{\Gamma^0_b} = \left\langle \frac{-\frac{5}{2}H^3 + \frac{5}{24}H^4 t^{-1}}{(t-5H)(t-5H-\psi_1)}, \frac{-\frac{5}{2}H^3 + \frac{5}{24}H^4 t^{-1}}{(t-5H)(t-5H-\psi_2)} \right\rangle_{0,2}^{t,SQ},$$

and, writing $H_1 = H \otimes 1$ and $H_2 = 1 \otimes H$ for the double edge graph,

$$\mathrm{Cont}_{\Gamma^0_a} = \left\langle \frac{\frac{5}{3}(H^3 \otimes H^4 + H^4 \otimes H^3)t^{-1} + \frac{65}{24}H^4 \otimes H^4 t^{-2}}{(t-5H_1)(t-5H_1-\psi_1)(t-5H_2)(t-5H_2-\psi_2)} \right\rangle_{0,2}^{t,SQ}.$$

Note that these correlators live on $\mathbb{P}^4$, so H^4 is present and nonzero.

There is a fifth contribution from the degree zero graph, which is the constant

$$-2c_{2,0} = -\frac{5}{144}, \qquad c_{2,0} = -\frac{1}{2}N_{2,0} = \frac{5}{288},$$

and altogether the localization identity is

$$F_2^{SQ}(q) = \mathrm{Cont}_{\Gamma^2} + \mathrm{Cont}_{\Gamma^1} + \frac{1}{2}\mathrm{Cont}_{\Gamma^0_a} + \frac{1}{2}\mathrm{Cont}_{\Gamma^0_b} - 2c_{2,0},$$

where the two factors of $\frac{1}{2}$ are the automorphisms of Γ^0_a and Γ^0_b.

Using the ordinary mirror shift $\tau(q) = I_1(q)/I_0(q) - \log q$ and setting $Q = qe^{\tau(q)}$ and $L = (1-5^5 q)^{-1/5}$, stable quotient wall crossing gives the exact identity

$$F_2^{GW}(Q) = I_0(q)^2 F_2^{SQ}(q).$$

Introducing $D_u = L^{-1} q\frac{d}{dq}$ and the generators

$$\mathcal{X}_k = D_u^k \log\frac{I_0}{L}, \qquad \mathcal{Y}_k = D_u^k \log\frac{I_0 I_{1,1}}{L^2}, \qquad I_{1,1} = 1 + q\frac{d\tau}{dq}, \quad D_u L = \frac{1}{5}(L^5 - 1),$$

the individual twisted graph contributions involve extra generators, but these cancel in the above linear combination and leave precisely the finite generation variables. Writing $X = \mathcal{X}_1$ and $Y_1 = \mathcal{Y}_1$, we obtain

$$
\begin{aligned}
F_2^{GW}(Q) = \frac{I_0^2}{L^2}\Big(&\frac{70}{9}\mathcal{X}_3 + \frac{575}{18}X\mathcal{X}_2 + \frac{557}{72}X^3 + \frac{5}{6}Y_1\mathcal{X}_2 \\
&- \frac{629}{72}Y_1X^2 - \frac{23}{24}Y_1^2X - \frac{1}{24}Y_1^3 \\
&+ \frac{125}{36}\mathcal{X}_2L^4 - \frac{5}{24}X^2L^4 - \frac{35}{9}Y_1XL^4 - \frac{5}{24}Y_1^2L^4 \\
&- \frac{1441}{300}XL^3 - \frac{41}{300}Y_1L^3 + \frac{31459}{7200}XL^8 + \frac{2141}{7200}Y_1L^8 \\
&+ \frac{29621}{12000}L^{12} - \frac{116369}{36000}L^7 + \frac{547}{750}L^2\Big).
\end{aligned}
$$

Expanding at large radius gives

$$
F_2^{GW}(Q) = -\frac{5}{144} + \frac{575}{48}Q + \frac{5125}{2}Q^2 + \frac{7930375}{6}Q^3 + O(Q^4),
$$

and these coefficients are the Gromov-Witten invariants $N_{2,d}$, not the BPS integers of Table 2.

For the formal route, the Givental-Teleman formula for $\Omega^{\lambda}_{2,0}$ is a sum over the seven stable graphs of genus two with no legs, which are drawn in Figure 7. Only $R^{-1}(z)$ through z^3 is needed, and the translations expand the seven topologies into 17 decorated strata.

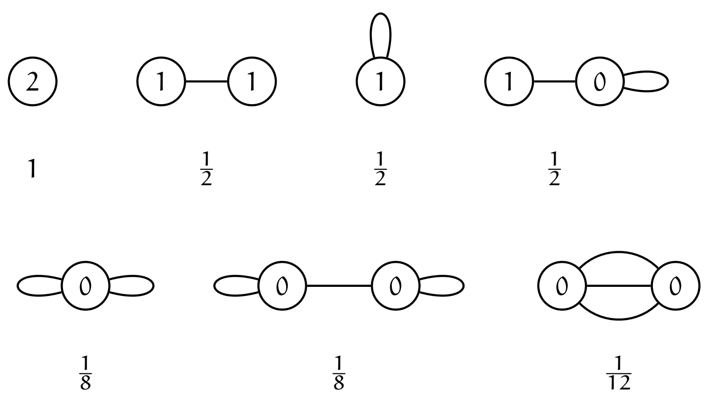


FIGURE 7. The seven stable graphs of genus two with no legs, labelled by the genera of the vertices and weighted by $\frac{1}{|\mathrm{Aut}\,\Gamma|}$. On the slides each vertex was drawn as its Riemann surface component.

Remark 8.12. These are dual graphs of stable curves and are not the graphs of Figure 5, which are the rigid tropical types obtained by applying the tropical decomposition formula to the reduced theory of the log GLSM. The two lists of topologies happen to coincide here, but a vertex means something different in each: here it is a component of the domain curve carrying a topological field theory tensor, while there it is a level of a bipartite tropical curve carrying either a log or an ∞-vertex moduli problem.

Defining the normalized formal potential intrinsically by

$$P_2^\lambda \coloneqq \frac{Y}{I_0^2} F_2^\lambda, \qquad Y = L^5,$$

the complete graph sum lies in $\mathbb{Q}[L, \mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3, \mathcal{Y}_1]$ and equals

$$\begin{aligned} P_2^\lambda = &-\frac{1}{128}L^{12} - \frac{23}{960}L^8\mathcal{X}_1 - \frac{9}{640}L^8\mathcal{Y}_1 + \frac{109}{16000}L^7 \\ &- \frac{3}{80}L^4\mathcal{X}_1^2 - \frac{19}{320}L^4\mathcal{X}_1\mathcal{Y}_1 - \frac{3}{80}L^4\mathcal{Y}_1^2 - \frac{1}{64}L^4\mathcal{X}_2 \\ &+ \frac{19}{1920}L^3\mathcal{X}_1 - \frac{47}{48000}L^2 \\ &- \frac{67}{1152}\mathcal{X}_1^3 - \frac{101}{1152}\mathcal{X}_1^2\mathcal{Y}_1 - \frac{5}{48}\mathcal{X}_1\mathcal{Y}_1^2 - \frac{1}{24}\mathcal{Y}_1^3 \\ &- \frac{25}{288}\mathcal{X}_1\mathcal{X}_2 - \frac{1}{48}\mathcal{X}_2\mathcal{Y}_1 - \frac{19}{1152}\mathcal{X}_3. \end{aligned}$$

This is the exact formal λ-twisted answer and is not F_2^{GW}. Note also that $Y = L^5$ is the discriminant generator while $\mathcal{Y}_1$ is a logarithmic derivative generator.

8.9. **Finite generation and the extended family.** Recall the finitely generated ring $\mathcal{R} = \mathbb{Q}[A, B_1, B_2, B_3, Y]$ from above, which we will grade by

$$\deg A = \deg B_1 = 1, \qquad \deg B_2 = 2, \qquad \deg B_3 = 3, \qquad \deg Y = 0.$$

For $\phi_k = I_{1,1} \cdots I_{k,k} H^k$ as before, set

$$\widetilde{\Omega}_{g,n}^\lambda \coloneqq 5^{g-1} \left(\frac{Y}{I_0^2} \right)^{g-1} \Omega_{g,n}^{\lambda,\tau}.$$

Theorem 8.13 (Graded finite generation)**.** *We have*

$$\widetilde{\Omega}_{g,n}^\lambda(\phi_{a_1}, \ldots, \phi_{a_n}) \in \bigoplus_{k \geqslant 0} \lambda^{5k} H^{2(3g-3+\sum_i a_i - 5k)}(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \otimes \mathcal{R}_{3g-3+\sum_i a_i}.$$

Consequently, after assembling the bipartite rest graphs, the non-equivariant component of the extended family satisfies

$$P_{g,n}^{\mathbf{C}} = 5^{1-g} \int_{\overline{\mathcal{M}}_{g,n}} \widetilde{\Omega}_{g,n}^{\mathbf{C}}(\phi_1^{\otimes n}) = \left(\frac{Y}{I_0^2} \right)^{g-1} \int_{\overline{\mathcal{M}}_{g,n}} \Omega_{g,n}^{\mathbf{C}}(\phi_1^{\otimes n}) \in \mathcal{R}_{3g-3+n}.$$

Therefore at each fixed (g, n) only finitely many graded monomials can appear, and only R-matrix orders up to $3g - 3 + n$ can contribute.

Everything is organized by the bipartite graph sum

$$\Omega_{g,n}^{\mathbf{C},\lambda} = \sum_{\Gamma \text{ bipartite}} \frac{1}{|\mathrm{Aut}\,\Gamma|} \prod_{v \in V_\infty(\Gamma)} C^\infty_{g_v, \mu_v}(\mathbf{C}; \vec{a}_v) \prod_{v \in V_0(\Gamma)} C^\lambda_{g_v, \vec{\delta}_v}(\vec{a}_v) \prod_{e \in E(\Gamma)} V_e,$$

where C^∞ is an effective or meromorphic vertex factor, C^λ is a formal zero-vertex correlator, the μ_v and $\vec{\delta}_v$ record incident degrees, and V_e is the edge factor. The limit $\lambda \to 0$ is taken only after the complete graph sum. Setting $\mathbf{C} = 0$ leaves the formal λ-twisted leading sector, setting $\mathbf{C} = \mathbf{C}^{\mathrm{eff}}$ and $\lambda \to 0$ recovers quintic Gromov-Witten theory, and differentiating a universal edge cuts it and produces the holomorphic anomaly equation while leaving the effective vertices untouched.

For the last of these, write $P^{\mathbf{C}}_g := P^{\mathbf{C}}_{g,0}$ and introduce

$$\mathcal{U} = B_1, \qquad \mathcal{V} = A + 2B_1, \qquad \mathcal{V}_2 = D\mathcal{U} + \mathcal{U}^2 - \mathcal{U}\mathcal{V}, \qquad \mathcal{V}_3 = (D + \mathcal{V})\mathcal{V}_2,$$

together with the two non-holomorphic directions

$$\partial_1 = \partial_{\mathcal{U}}, \qquad \partial_0 = \partial_{\mathcal{V}} - \mathcal{U}\,\partial_{\mathcal{V}_2} - \mathcal{U}^2\,\partial_{\mathcal{V}_3}.$$

Theorem 8.14 (Holomorphic anomaly equation for the extended family)**.** *We have*

$$-\partial_0 P^{\mathbf{C}}_g = \frac{1}{2} P^{\mathbf{C}}_{g-1,2} + \frac{1}{2} \sum_{\substack{g_1+g_2=g \\ g_1,g_2 \geqslant 1}} P^{\mathbf{C}}_{g_1,1} P^{\mathbf{C}}_{g_2,1}, \qquad -\partial_1 P^{\mathbf{C}}_g = 0.$$

The proof is the cutting argument of Section 8.4, which works because the derivative of the R-matrix propagator is simply

$$5\mathcal{B}_{\partial_0} = \bar{\phi}_1 \otimes \bar{\phi}_1, \qquad 5\mathcal{B}_{\partial_1} = \bar{\phi}_0 \otimes \bar{\phi}_2 + \bar{\phi}_2 \otimes \bar{\phi}_0.$$

Derivatives of R^{-1} and of every edge remain inside the same graded ring, so cutting a differentiated edge closes up on lower-genus correlators in $\mathcal{R}$. Setting $\mathbf{C} = 0$ recovers the same equations for the formal λ-twisted theory.

8.10. **Boundary conditions.** There are two sources of conditions on the holomorphic ambiguity.

The first is regularity at the orbifold or Landau-Ginzburg point $q = \infty$, equivalently $L \to 0$. For either the formal or an assembled extended potential $F^\star_g$, set

$$\mathcal{P}^{\mathrm{orb},\star}_g := \left(\frac{L}{I_0}\right)^{2g-2} F^\star_g = \Phi^\star_g(\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3, \mathcal{Y}_1, L^{-1}, L^5).$$

Theorem 8.13 bounds the possible Laurent powers of L, and the formal LG/CY identification says that after analytic continuation and identification of the normalized CY and LG bases, the same formal R-matrix polynomial describes both phases. Therefore

$$\lim_{L \to 0} \Phi^\star_g(\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3, \mathcal{Y}_1, L^{-1}, L^5) \text{ exists,}$$

and for $f_g(Y) = \sum_{i=0}^{3g-3} a_{i,g} Y^i$ with $Y = L^5$ the quintic bound is

$$a_{i,g} = 0 \qquad \text{for } i \leqslant \left\lceil \frac{3(g-1)}{5} \right\rceil.$$

Note that this is a statement about the assembled normalized graph contribution, and not about individual bare meromorphic or rubber vertex correlators.

The second source is the conifold. Set

$$q = 5^{-5}(1 + t_c)^{-1}, \qquad T_c = \frac{I^{co}_1(t_c)}{I^{co}_0(t_c)},$$

where I^{co}_0, I^{co}_1 is a local conifold period pair normalized so that the vanishing flat coordinate satisfies $T_c = 0$ at the conifold. Given a rational expression f in the finite generation data, its conifold value f^{co} is obtained by substituting the Laurent expansions of all generators as series in T_c. Theorem 8.13 gives the pole bound

$$F^{\mathbf{C},co}_g = O\left(T_c^{-(2g-2)}\right),$$

but what we need is the stronger *gap* condition

$$F_g^{co} = \frac{c_g}{T_c^{2g-2}} + O(1),$$

namely that every intermediate negative power $T_c^{-(2g-3)}, \dots, T_c^{-1}$ vanishes. This is $2g-2$ conditions and is not implied by finite generation.

Conjecture 8.15 (Formal conifold gap)**.** *The formal λ-twisted potential has the universal leading term*

$$F_g^{\lambda,co} = \frac{(-1)^{g-1}B_{2g}}{2g(2g-2)T_c^{2g-2}} + O(1),$$

where B_{2g} is a Bernoulli number.

Example 8.16. In genus two the computed expansion begins

$$F_2^{\lambda,co} = \frac{1}{240T_c^2} + \frac{31}{6000} - \frac{7169}{3600000}T_c + \frac{22951}{45000000}T_c^2 + O(T_c^3),$$

and the$2^{\lambda,co} = \frac{1}{240T_c^2} + c31\ 000$ - two gap and not merely the pole order bound.

The rest graphs also involve formal correlators with rational descendant insertions, namely

$$C^{\lambda}_{g,\vec{\delta}}(\vec{a}) := \left\langle \frac{H^{3-a_1}}{-5H\left(-\frac{5H}{\delta_1} - \psi_1\right)}, \dots, \frac{H^{3-a_n}}{-5H\left(-\frac{5H}{\delta_n} - \psi_n\right)} \right\rangle^{\lambda}_{g,n}$$

with $\delta_i \geqslant 1$ and $a_i \in \{0,1,2,3,4\}$.

Conjecture 8.17 (Regularity with special insertions)**.** *We have* $\left(C^{\lambda}_{g,\vec{\delta}}(\vec{a})\right)^{co} = O(1)$.

For fixed $\vec{a}$ and $\vec{\delta}$, individual terms in the ψ-expansion may have poles. The assertion is about the complete rational descendant insertion, whose negative powers cancel before the rest graph sum over $\vec{a}$ is taken, and it is a statement about these specific formal correlators rather than about every unassembled meromorphic correlator.

Conjectures 8.15 and 8.17 transfer the gap to quintic Gromov-Witten theory. A rest graph contribution has the shape

$$\sum_{\vec{a}} \prod_{v \in V_\infty(\Gamma)} I_0^{2g_v - 2 + \ell(\mu_v)} C^{\infty}_{g_v,\mu_v}(\vec{a}_v) \prod_{v \in V_0(\Gamma)} C^{\lambda}_{g_v,\vec{\delta}_v}(\vec{a}_v),$$

and the C^∞ factors are finite polynomials in q, hence conifold regular. Therefore the formal leading gap together with regularity of the special correlators implies the gap for the extended family, and hence for the quintic.

Remark 8.18. Conjectures 8.15 and 8.17 are open in general, but have been verified for the quintic through $g \leqslant 5$ by Guo-Janda-Ruan. The pole bound for the extended family is a theorem, and the gap follows from the two conjectures and is proved for $g \leqslant 5$ by explicit computation. The all-genus gap for quintic Gromov-Witten theory is therefore conditional on them and unconditional for $g \leqslant 5$.

Altogether, the computation runs from Picard-Fuchs data to the Givental R-matrix and the operator $S_\tau(z)$, then to formal stable graphs and meromorphic rest graphs, then to the holomorphic anomaly equation and finite generation, and finally to the orbifold and conifold boundary conditions, which determine F_g^{GW}.

9. Structure of higher-genus GW theory of local $\mathbb{P}^1$ (Shuai Guo)

This lecture is based on joint work with Jingyi Xu and Qingsheng Zhang.

9.1. **The geometry of X_p.** We will consider the total space

$$X_p := \mathrm{Tot}(\mathcal{O}_{\mathbb{P}^1}(p-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-p-1)),$$

which is a non-compact toric Calabi-Yau threefold. For $p = 0$ this is the resolved conifold.

Let $N \simeq \mathbb{Z}^3$ be a lattice of rank 3 and let $M := \mathrm{Hom}(N, \mathbb{Z})$ be the dual lattice, which we identify with the characters of $\mathbb{T} \simeq (\mathbb{C}^\times)^3$. The one-dimensional cones of the fan of X_p are generated by

$$b_1 = (1-p, -1, 1), \qquad b_2 = (0, 1, 1), \qquad b_3 = (1, 0, 1), \qquad b_4 = (0, 0, 1),$$

and X_p contains $\mathbb{T}$ as an open dense subset. Using the dual vector $e_3^\vee \in M$ we obtain the *Calabi-Yau subtorus*

$$\mathbb{T}' := \ker\left(\chi^{e_3^\vee} : \mathbb{T} \to \mathbb{C}^\times\right) \simeq (\mathbb{C}^\times)^2,$$

whose action preserves the Calabi-Yau form. The toric graph consists of two triangles, which correspond to the two fixed points of the torus action. The weights of the $\mathbb{T}'$-action at the first fixed point are

$$(0, 1), \qquad (1, 0), \qquad (-1, -1),$$

and at the second fixed point they are

$$(0, -1), \qquad (1, 1-p), \qquad (-1, p).$$

Localizing at these two fixed points expresses the invariants in terms of equivariant cohomology, where $H^*_{\mathbb{T}'}(\mathrm{pt}) = \mathbb{C}[\lambda_1, \lambda_2]$.

Note that for $p \geqslant 2$ the toric graph of X_p is not convex, so X_p is not semiprojective. This is the reason the case at hand is not covered by the existing literature, as we will explain in Section 9.5.

9.2. **The conjecture of CGMPS.** In 2007, Caporaso, Griguolo, Mariño, Pasquetti, and Seminara studied the Gromov-Witten theory of X_p. Writing

$$\check{F}^{X_p}(w, \hbar) := \sum_{g \geqslant 0} \check{F}_g^{X_p}(w) \hbar^{2g-2}$$

for the total free energy, they computed the genus-one part exactly and proposed the ansatz

$$\check{F}_g^{X_p} = \frac{\widetilde{\mathcal{P}}_g(w, p)}{(w - w_c)^{5g-5}}, \qquad \widetilde{\mathcal{P}}_g(w, p) = \sum_{i=1}^{5g-5} a_{g,i}(p)(w-1)^i \tag{2}$$

for $g \geqslant 2$. Here the variable w is related to the Kähler parameter q by $w = 1 - q$ and $w_c = 1 - \frac{1}{p^2}$. Our convention will differ by $F_g^{X_p} = (-1)^{g-1} \check{F}_g^{X_p}$ for $g \geqslant 2$.

Analyzing the asymptotics of the genus-zero invariants, CGMPS defined the scaled string coupling by

$$z^{5/2} = \hbar^{-2}\frac{p^8}{4w_c^5}(w-w_c)^5$$

and introduced the *double-scaling limit* $F_{ds}(z)$, which is obtained from $\check{F}^{X_p}(w,\hbar)$ by letting $w \to w_c$ and $\hbar \to 0$ with z fixed. Combining their genus-zero and genus-one results with the ansatz gives

$$F_{ds}(z) = -\frac{4}{15}z^{5/2} - \frac{1}{48}\log z + \sum_{g\geqslant 2} b_g z^{-5(g-1)/2},$$

where $b_g = \left(\frac{p^8}{4w_c^5}\right)^{g-1}\widetilde{\mathcal{P}}_g(w_c, p)$.

Conjecture 9.1 (CGMPS)**.** *The double-scaling limit* $F_{ds}(z)$ *coincides with the free energy* $F_{(3,2)}(z)$ *of the* $(3,2)$ *minimal model.*

The free energy of the $(3,2)$ minimal model has the form

$$F_{(3,2)}(z) = -\frac{4}{15}z^{5/2} - \frac{1}{48}\log z + \sum_{g\geqslant 2} a_g z^{-5(g-1)/2}, \quad a_g = \frac{4^{1-g}}{(3g-3)!}\left\langle \tau_2^{3g-3}\right\rangle_g,$$

where

$$\left\langle \tau_2^{3g-3}\right\rangle_g \coloneqq \int_{\overline{\mathcal{M}}_{g,3g-3}} \psi_1^2\cdots\psi_{3g-3}^2,$$

so Conjecture 9.1 is equivalent to the statement that $b_g = a_g$ for all $g \geqslant 2$.

9.3. **The main results.** The global mirror symmetry picture of Section 8 applies here as well, and the three main theorems describe the behavior of $F_g^{X_p}$ at each of the boundary points of the one-dimensional B-model moduli space. The limit $q \to 0$ is the large complex structure limit and gives the Gromov-Witten expansion in the A-model variable Q. At $q \to 1$ the framed R-matrix shows that F_g remains regular. The interesting point is $q \to \frac{1}{p^2}$, which is a higher-order critical point: unlike an ordinary conifold singularity, whose genus-g leading pole has order $2g-2$, this degeneration has pole order at most $5g-5$, and its leading coefficient is governed by the $(3,2)$ minimal model.

Theorem 9.2. *For any integer* $p \geqslant 2$ *and any* $g \geqslant 2$*, the Gromov-Witten potential of* X_p *is computed by the Eynard-Orantin topological recursion of a spectral curve* $\mathcal{C}$*, namely*

$$F_g^{X_p}(Q) = (-1)^{g-1}\mathcal{F}_g^{X_p}(q).$$

This gives a non-semiprojective toric Calabi-Yau example of the remodeling conjecture.

Theorem 9.3. *For* $g \geqslant 2$*, the potential* $F_g^{X_p}$ *is the expansion of a rational function*

$$F_g^{X_p} = (-1)^{g-1}\frac{\mathcal{P}_g(\Delta, p^2)}{\Delta^{5g-5}}, \qquad \Delta \coloneqq \frac{1-p^2q}{1-q} = \frac{p^2(w-w_c)}{w},$$

where $\mathcal{P}_g \in \frac{1}{(p^2-1)^{2g-2}}\mathbb{Q}[p^2][\Delta]_{\leqslant 5g-5}$.

This validates the structural ansatz Equation (2).

Theorem 9.4. *The constant term of $\mathcal{P}_g$ is*

$$\mathcal{P}_g(0,p^2) = \frac{p^{6g-6}}{(p^2-1)^{2g-2}} \frac{1}{(3g-3)!} \left\langle \tau_2^{3g-3} \right\rangle_g.$$

Because $\widetilde{\mathcal{P}}_g(w_c, p) = \mathcal{P}_g(0,p^2)\left(\frac{1}{p^2} - \frac{1}{p^4}\right)^{5g-5}$, Theorem 9.4 gives

$$b_g = \frac{4^{1-g}}{(3g-3)!} \left\langle \tau_2^{3g-3} \right\rangle_g = a_g,$$

which proves Conjecture 9.1.

9.4. **Gromov-Witten theory of X_p.** Because X_p is noncompact, the moduli space $\overline{\mathcal{M}}_{g,0}(X_p, d)$ is not proper and the degree of the virtual class is not defined a priori. The standard remedy is to introduce a torus action and define the invariants by virtual localization, and we will do this by passing to a twisted theory on $\mathbb{P}^1$.

Choose a framing $f \in \mathbb{Z}$ and let $\mathbb{T}_f := \ker(\lambda_2 - f\lambda_1)$. The two localized weight rows, each of which includes one zero weight, are

$$\begin{bmatrix} 0 & f & 1 & -f-1 \end{bmatrix}, \qquad \begin{bmatrix} -f & 0 & 1-fp+f & -1+fp \end{bmatrix}.$$

For $f \neq 0$, rescaling the character gives the equivalent symmetric weight vector

$$\begin{bmatrix} 1 & -1 & -\mathfrak{f} & \mathfrak{f} \end{bmatrix}, \qquad \mathfrak{f} = 1 - p + \frac{2}{f},$$

and we denote the adjusted torus by $\mathbb{T}_{\mathfrak{f}}$. If $H^*_{\mathbb{T}_{\mathfrak{f}}}(\mathrm{pt}) = \mathbb{C}[\lambda]$, this normalization is

$$\lambda_1 = -\frac{2}{\mathfrak{f}}\lambda, \qquad \lambda_2 = -2\lambda.$$

Let $\pi \colon \overline{\mathcal{M}}_{g,1}(\mathbb{P}^1, d) \to \overline{\mathcal{M}}_{g,0}(\mathbb{P}^1, d)$ be the forgetful map and let ev be the evaluation at the marking. Then we define

$$N^{X_p}_{g,d} := \int_{[\overline{\mathcal{M}}_{g,0}(\mathbb{P}^1,d)]^{\mathrm{vir}}_{\mathbb{T}_{\mathfrak{f}}}} e^{-1}_{\mathbb{T}_{\mathfrak{f}}}(R^{\bullet}\pi_* \mathrm{ev}^*(\mathcal{O}(p-1) \oplus \mathcal{O}(-p-1))),$$

where $e_{\mathbb{T}_{\mathfrak{f}}}$ is the equivariant Euler class. For $p = 0$ this specializes to the usual resolved conifold formula.

This theory is a CohFT $\Omega^{\lambda,\mathfrak{f}}$ with state space the equivariant cohomology ring

$$H^*_{\mathbb{T}_{\mathfrak{f}}}(\mathbb{P}^1) = \frac{\mathbb{C}[H,\lambda]}{(H+\lambda)(H-\lambda)}$$

and pairing

$$(a,b) := \int_{\mathbb{P}^1} a \cup b \cup e^{-1}_{\mathbb{T}_{\mathfrak{f}}}(\mathcal{O}(p-1) \oplus \mathcal{O}(-p-1)).$$

In the flat basis $\{\mathbf{1}, H\}$ the Gram matrix is

$$\frac{1}{((p-\mathfrak{f})^2-1)((p+\mathfrak{f})^2-1)} \begin{pmatrix} 2\mathfrak{f}p\lambda^{-3} & (1-\mathfrak{f}^2-p^2)\lambda^{-2} \\ (1-\mathfrak{f}^2-p^2)\lambda^{-2} & 2\mathfrak{f}p\lambda^{-1} \end{pmatrix}.$$

The correlators are

$$\Omega^{\lambda,\mathfrak{f}}_{g,n}(\alpha_1,\ldots,\alpha_n) := \sum_{d \geqslant 0} \mathrm{st}_* \left(\prod_{i=1}^{n} \mathrm{ev}_i^* \alpha_i \cap [\overline{\mathcal{M}}_{g,n}(\mathbb{P}^1,d)]^{\mathrm{vir},\lambda,\mathfrak{f}} \right) e^{-dt},$$

where st is the forgetful map to $\overline{\mathcal{M}}_{g,n}$ and the twisted virtual class is

$$[\overline{\mathcal{M}}_{g,n}(\mathbb{P}^1, d)]^{\mathrm{vir},\lambda,\mathfrak{f}} := e^{-1}_{\mathbb{T}_\mathfrak{f}}(R^\bullet\pi_* \operatorname{ev}^*(\mathcal{O}(p-1)\oplus\mathcal{O}(-p-1))) \cap [\overline{\mathcal{M}}_{g,n}(\mathbb{P}^1, d)]^{\mathrm{vir}}_{\mathbb{T}_\mathfrak{f}}.$$

Setting $Q = (-1)^{p+1}e^{-t}$, the genus-g potential is

$$F_g^{X_p}(t) = \int_{\overline{\mathcal{M}}_g} \Omega^{\lambda,\mathfrak{f}}_{g,0}.$$

Remark 9.5. The degree-positive invariants $N^{X_p}_{g,d\geqslant 1}$ are independent of $\mathfrak{f}$ by the topological vertex, and their λ-independence also follows from Theorem 9.8. We will use the $\mathfrak{f}$-independence in the proof of Theorem 9.3.

In genus zero the invariants are given by

$$N^{X_p}_{0,d} = \frac{(-1)^{(p+1)d-1}}{d^3}\binom{p^2d-1}{d-1}, \qquad d \geqslant 1,$$

which allows us to compute the Yukawa coupling exactly:

$$Y = -\partial_t^3 F_0^{X_p}(t) = \frac{1-\mathfrak{f}^2-p^2}{((p-\mathfrak{f})^2-1)((p+\mathfrak{f})^2-1)} + \frac{q}{p^2q-1}.$$

Here q and Q are related by the mirror map $Q = q(1-q)^{p^2-1}$. Evaluating the quantum product using $(v_1 \star v_2, v_3) = \Omega^{\lambda,\mathfrak{f}}_{0,3}(v_1, v_2, v_3)$, we obtain

$$H\star H = -\frac{(\mathfrak{f}^2-1)q+1}{p^2q-1}\lambda^2\mathbf{1} - \frac{2\mathfrak{f}pq}{p^2q-1}\lambda H,$$

or in matrix form

$$H\star = \begin{pmatrix} 0 & -\frac{(\mathfrak{f}^2-1)q+1}{p^2q-1}\lambda^2 \\ 1 & -\frac{2\mathfrak{f}pq}{p^2q-1}\lambda \end{pmatrix}.$$

The eigenvalues L_α of this matrix are generically distinct, so $\Omega^{\lambda,\mathfrak{f}}$ is semisimple. We write $\{e_\alpha\}$ for the idempotent basis, $\Delta_\alpha^{-1} = (e_\alpha, e_\alpha)$ for the lengths, and $\bar{e}_\alpha := \Delta_\alpha^{1/2}e_\alpha$ for the normalized canonical basis.

By Theorem 3.24, we have

$$\Omega^{\lambda,\mathfrak{f}} = R.T_R.\left(\bigoplus_{\beta=0}^{1}\Omega^{\mathrm{KW}}_\beta\right),$$

where $\Omega^{\mathrm{KW}}_\beta$ is a one-dimensional trivial CohFT and $T_R(z) = z(\widetilde{\mathbf{1}} - R^{-1}(z)\mathbf{1})$ with $\widetilde{\mathbf{1}} := \sum_\beta \bar{e}_\beta$. The R-matrix is obtained from the quantum differential equation

$$z\, dS_\tau(z) = d\tau \star_\tau S_\tau(z).$$

Equivariant localization and quantum Riemann-Roch relate the fundamental solution to the R-matrix by

$$(S^*_\tau(z)\mathbf{1}, \bar{e}_\alpha)\big|_{q=0}\, C_\alpha(z) = (\mathbf{1}, R(z)\bar{e}_\alpha)e^{u^\alpha(q)/z},$$

where the initial condition $C_\alpha(z)$ is an exponential involving Bernoulli numbers. Writing $R_{i\bar{\alpha}}(z) := (R^*(z)H^i, \bar{e}_\alpha)$, the quantum differential equation becomes the second-order equation

$$\left((zD+L_\alpha)^2+\frac{2\mathfrak{f}pq}{p^2q-1}\lambda(zD+L_\alpha)+\frac{(\mathfrak{f}^2-1)q+1}{p^2q-1}\lambda^2\right)R_{0\bar{\alpha}}=0,$$

where $D=-\partial_t$. This generates the R-matrix order by order in z.

9.5. **Topological recursion and the remodeling conjecture.** The Eynard-Orantin topological recursion takes as input a *spectral curve* (Σ, x, y, B), where Σ is a Riemann surface, x is locally defined with meromorphic differential dx having finitely many simple zeros, y is holomorphic near those zeros with $dy \neq 0$ there, and B is a fundamental bidifferential on $\Sigma\times\Sigma$. Near a zero z^β of dx, let $\bar{z}\neq z$ be the local conjugate point with $x(\bar{z})=x(z)$ and define the recursion kernel

$$K_\beta(z_0,z)\coloneqq\frac{\int_{\bar{z}}^{z}B(z_0,z')}{2(y(z)-y(\bar{z}))\,dx(z)}.$$

Setting $\omega_{0,0}=\omega_{1,0}=0$, $\omega_{0,1}(z)=y(z)\,dx(z)$, and $\omega_{0,2}(z_1,z_2)=B(z_1,z_2)$, the multi-differentials are defined recursively for $2g-2+n+1>0$ by

$$\begin{aligned}\omega_{g,n+1}(z_0,z_{[n]})=\sum_\beta &\operatorname{Res}_{z=z^\beta}K_\beta(z_0,z)\\ &\cdot\Bigg(\omega_{g-1,n+2}(z,\bar{z},z_{[n]})\\ &\qquad+\sum_{\substack{g_1+g_2=g\\ I\sqcup J=[n]}}{}^{'}\ \omega_{g_1,|I|+1}(z,z_I)\,\omega_{g_2,|J|+1}(\bar{z},z_J)\Bigg),\end{aligned}$$

where the primed sum excludes $\omega_{0,1}$. For $g\geqslant 2$ the free energy is

$$\mathcal{F}_g=\omega_{g,0}\coloneqq\frac{1}{2-2g}\sum_\beta\operatorname{Res}_{z=z^\beta}\omega_{g,1}(z)\int_{z'=z^\beta}^{z}\omega_{0,1}(z').$$

The remodeling conjecture of Bouchard, Klemm, Mariño, and Pasquetti asserts that the all-genus open and closed Gromov-Witten invariants of a toric Calabi-Yau threefold X coincide with the invariants $\omega_{g,n}$ produced by topological recursion on its mirror curve. It was proved for $\mathbb{C}^3$, the resolved conifold, and local curves by Eynard-Orantin, Zhou, and Chen-Li-Liu, and then in general by Fang-Liu-Zong for all semiprojective toric Calabi-Yau threefolds. Because X_p is not semiprojective for $p\geqslant 2$, it falls outside the scope of that proof.

To compare the two sides we use the reconstruction of Dunin-Barkowski, Orantin, Shadrin, and Spitz, which identifies the stable topological recursion correlators with the Givental reconstruction of a semisimple CohFT. Around each critical point z^β, define the local Airy coordinate $\eta^\beta(z)$ by

$$x(z)=x(z^\beta)+\frac{1}{2}(\eta^\beta(z))^2$$

and the globally defined meromorphic reference form

$$d\zeta^{\bar{\beta}}(z)\coloneqq-\operatorname{Res}_{z'=z^\beta}\frac{B(z',z)}{\eta^\beta(z')}.$$

The state space V of the associated CohFT is spanned by the basis $\{\bar{e}_\beta\}$ corresponding to the critical points, and the pairing is diagonal with $\eta(\bar{e}_\beta, \bar{e}_\gamma) = \delta_{\beta,\gamma}$. The curve R-matrix $\check{R}$ is defined by the asymptotic Laplace transform

$$\frac{u}{\sqrt{2\pi u}} \int_{\mathcal{L}_\gamma} e^{-x(z)/u}\, d\zeta^{\bar{\beta}}(z) \asymp e^{-x^\gamma/u} \eta\big(\bar{e}_\gamma, \check{R}^*(-u)\bar{e}_\beta\big),$$

and the analogous transform of dy defines $T(u)$, which equals $u(\widetilde{\mathbf{1}} - \check{R}^{-1}(u)\mathbf{1})$ when the resulting CohFT has flat unit. Then $\check{R}$ is symplectic, the curve CohFT is $\check{\Omega} = \check{R}.T.\left(\bigoplus_\beta \Omega^{KW}_\beta\right)$, and writing $d\zeta^{\bar{\beta}}_k := \left(-d \circ \frac{1}{dx}\right)^k d\zeta^{\bar{\beta}}$, the correspondence reads

$$\begin{aligned}\omega_{g,n}(z_1, \ldots, z_n) = &\sum_{\substack{k_1,\ldots,k_n \geqslant 0 \\ \beta_1,\ldots,\beta_n \in [N]}} \left(\int_{\overline{\mathcal{M}}_{g,n}} \check{\Omega}_{g,n}(\bar{e}_{\beta_1}\psi_1^{k_1}, \ldots, \bar{e}_{\beta_n}\psi_n^{k_n})\right) \\ &\cdot \prod_{i=1}^n d\zeta^{\beta_i}_{k_i}(z_i).\end{aligned}$$

For X_p we consider the spectral curve

$$\mathcal{C}_{\mathfrak{f}} = \left(\mathbb{P}^1, x_{\mathfrak{f}}(z), y(z), B(z_1, z_2) = \frac{dz_1\, dz_2}{(z_1 - z_2)^2}\right), \qquad x_{\mathfrak{f}}(z) = x(z) + 2\mathfrak{f}\lambda^2 y(z),$$

where

$$\begin{aligned} x(z) &= t^0 + \lambda \log q + \lambda(p-1)\log z + \lambda(-p-1)\log\frac{q-z}{1-z}, \\ y(z) &= \frac{p+1}{2\lambda}\log(1-q) + \frac{1}{2\lambda}\log z - \frac{1}{2\lambda}\log(1-z) - \frac{1}{2\lambda}\log(q-z).\end{aligned}$$

Solving $dx_{\mathfrak{f}} = 0$, the critical points are

$$z^\alpha = \frac{1 - L_\alpha}{1 + \mathfrak{f} + pL_\alpha}, \qquad L_\alpha = \lambda^{-1}L_\alpha, \qquad q = \frac{L_\alpha^2 - 1}{(pL_\alpha + 1)^2 - 1},$$

and computing the inner products $\eta(e_\alpha, e_\beta) = \delta_{\alpha,\beta}\frac{y'(z^\alpha)^2}{x_{\mathfrak{f}}''(z^\alpha)}$ yields lengths $\check{\Delta}_\alpha$ which satisfy $\check{\Delta}_\alpha = \Delta_\alpha$. Therefore the Frobenius structure of the spectral curve is the quantum product of X_p.

It remains to match the two R-matrices. Integration by parts on the Lefschetz thimbles gives

$$\begin{aligned} u\,\partial_{t^0}\left(e^{-x^\gamma/u}\check{R}^*(-u)^{\bar{\gamma}}_{\mathbf{1}}\right) &= -e^{-x^\gamma/u}\check{R}^*(-u)^{\bar{\gamma}}_{H}, \\ u\,\partial_{t^0}\left(e^{-x^\gamma/u}\check{R}^*(-u)^{\bar{\gamma}}_{H}\right) &= -\lambda b(q) e^{-x^\gamma/u}\check{R}^*(-u)^{\bar{\gamma}}_{H} - \lambda^2 c(q) e^{-x^\gamma/u}\check{R}^*(-u)^{\bar{\gamma}}_{\mathbf{1}},\end{aligned}$$

where

$$c(q) = -\frac{(\mathfrak{f}^2 - 1)q + 1}{p^2 q - 1}, \qquad b(q) = -\frac{2\mathfrak{f}pq}{p^2 q - 1}$$

are exactly the entries of the quantum product computed above, so these agree with the quantum differential equation for the Gromov-Witten R-matrix. For the initial conditions, taking the limit $q \to 0$ of the Laplace transforms along the thimbles produces Beta functions, hence products of Gamma functions, and Stirling's expansion matches the Bernoulli-number initial condition $C_\alpha(z)$.

Proposition 9.6. *The two* R*-matrices coincide, namely* $R(z) = \check{R}(u)|_{u=-z}$.

In Proposition 9.6 the canonical lengths agree, and after the substitution $u = -z$ the genus-g stable graph sums differ by the overall parity $(-1)^{3g-3} = (-1)^{g-1}$, which gives Theorem 9.2.

9.6. **Rationality.** Because the higher-genus potential is independent of $\mathfrak{f}$, we may set $\mathfrak{f} = 0$, which simplifies both the canonical basis and the quantum differential equation. The latter becomes

$$\left((zD + L_\alpha)^2 - \frac{1-q}{1-p^2q}\lambda^2\right)R_{0\bar{\alpha}} = 0, \qquad L_\alpha = (-1)^\alpha \lambda L, \quad L = \left(\frac{1-q}{1-p^2q}\right)^{1/2},$$

and we denote the entries of the R-matrix by $(R_k)_i^\alpha$. Set

$$\Delta \coloneqq \frac{1}{L^2} = \frac{1-p^2q}{1-q}, \qquad \mathcal{R} \coloneqq \mathbb{Q}[p^2, \Delta], \qquad \mathcal{R}_{\leqslant d} \coloneqq \left\{P \in \mathcal{R} \mid \deg_\Delta P \leqslant d\right\}.$$

Induction on the coefficients of the quantum differential equation then gives a hidden polynomial structure.

Proposition 9.7. *For* $k \in \mathbb{Z}_{\geqslant 0}$ *and* $i = 0, 1$, *we have*

$$(R_k)_i^\alpha \in \frac{1}{(p^2-1)^k \lambda^{k-i} \Delta^{(3k+i)/2}} \mathcal{R}_{\leqslant 2k}.$$

Analyzing the degree contributions of the stable graph sum in the Givental-Teleman reconstruction, and tracking the powers of λ, Δ, and $p^2 - 1$, we obtain the following.

Theorem 9.8. *Writing* $[\cdot]_d$ *for the codimension-*d *component, we have*

$$\left[\Omega_{g,n}^\lambda(H^{a_1}, \ldots, H^{a_n})\right]_d \in \frac{\lambda^{3g-3-d+\sum_i a_i}}{(p^2-1)^{d-(g-1)} \Delta^{\frac{1}{2}(g-1+3d+\sum_i a_i)}} \mathcal{R}_{\leqslant 2d} \otimes H^{2d}(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}).$$

Taking $n = 0$ and $d = 3g - 3 = \dim \overline{\mathcal{M}}_g$ in Theorem 9.8 gives

$$F_g^{X_p} \in \frac{1}{(p^2-1)^{2g-2} \Delta^{5g-5}} \mathcal{R}_{\leqslant 6g-6},$$

so Theorem 9.3 will follow once we know that the degree of the numerator in Δ is bounded by $5g - 5$. Since Δ has a simple pole at $q = 1$, it suffices to prove that $F_g^{X_p}$ is regular at $q = 1$, because the numerator must then cancel the highest powers of Δ. Regularity at $q = 1$ follows from the $\mathfrak{f}$-independence of $F_g^{X_p}$ together with the existence of the limit $q \to 1$ of the R-matrix for $\mathfrak{f} \neq 0$.

9.7. **The limit** $\Delta = 0$**.** Evaluating $\mathcal{P}_g(\Delta, p^2)$ at $\Delta = 0$ is precisely the double-scaling limit. Rescale the canonical lengths by

$$\delta_\alpha \coloneqq \Delta^{1/2}\Delta_\alpha = (-1)^{\alpha+1} 2(p^2-1)\lambda^3$$

and define the modified R-matrix entries by isolating the leading terms at $\Delta = 0$ in Proposition 9.7, namely

$$(\bar{R}_k)_i^\alpha \coloneqq \left[(-1)^k (R_k)_i^\alpha \Delta^{(3k+i)/2}\right]\Big|_{\Delta=0}, \qquad i = 0, 1.$$

Then the graph sum for $F_g^{X_p}$ becomes a formula for $\mathcal{P}_g(0, p^2)$ after replacing Δ_α by δ_α and $(R_k)_i^\alpha$ by $(\bar{R}_k)_i^\alpha$.

Lemma 9.9. *Solving the leading-order quantum differential equation at $\Delta = 0$ gives the closed forms*

$$(\bar{R}_k)_0^\alpha = \frac{p^{2k}}{(p^2-1)^k \lambda^k}(-1)^{k\alpha} \prod_{n=1}^{k} \frac{(6n-7)(6n+1)}{48n},$$
$$(\bar{R}_k)_1^\alpha = \frac{p^{2k}}{(p^2-1)^k \lambda^{k-1}}(-1)^{(k-1)\alpha} \frac{1-6k}{1+6k} \prod_{n=1}^{k} \frac{(6n-7)(6n+1)}{48n}.$$

The p-dependent factors in Lemma 9.9 now factor out and leave purely rational coefficients. Construct the topological field theory Ω^{top} on $V = \text{span}\{\phi_0, \phi_1\}$ with unit ϕ_0, where

$$(\phi_0, \phi_0) = (\phi_1, \phi_1) = 0, \qquad (\phi_0, \phi_1) = -1, \qquad \phi_1 \star \phi_1 = \phi_0.$$

Reconstruction by $R(z)$ together with its flat-unit translation produces a constant C_g with

$$\mathcal{P}_g(0, p^2) = \frac{p^{6g-6}}{(p^2-1)^{2g-2}} C_g,$$

and it remains to identify the reconstructed CohFT.

The $(2, 3)$ minimal model has spectral curve

$$x(z) = \frac{1}{3}(z^3 - 3\varepsilon^2 z) + \tau^0, \qquad y(z) = \frac{1}{2}(z^2 - 2),$$

with flat coordinate $\tau^1 = \varepsilon^4/2$. Setting $\tau^0 = 0$ and $\varepsilon = 1$ and identifying $\phi_0 = \partial_{\tau^0}$ and $\phi_1 = \partial_{\tau^1}$, the R-matrix obtained from X_p at $\Delta = 0$ coincides with the R-matrix of this curve. Therefore C_g is exactly the free energy of the $(2, 3)$ minimal model.

Finally, apply the symplectic swap $(x, y) \mapsto (y, -x)$, under which the curve becomes

$$x(z) = \frac{1}{2}(z^2 - 2), \qquad y(z) = -\frac{1}{3}(z^3 - 3z).$$

The swapped curve has one-dimensional state space, trivial R-matrix, and translation $T(u) = u^2 \bar{e}$, so that

$$C_g = \frac{1}{(3g-3)!}\left\langle \tau_2^{3g-3} \right\rangle_g,$$

which is Theorem 9.4. Note that this swap is the correspondence between the $(2, 3)$ and the $(3, 2)$ minimal models, which is why the intersection numbers appearing here are those of Conjecture 9.1.